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1. INTRODUCTION

The problem under consideration is the dynamic optimal control problem in the space-time domain

$$\begin{aligned}
 (P) \quad & \inf_{u \in L^2(t_0, T, \mathbb{R}^{m_u})} J(u; t_0, y) := \int_{t_0}^T l(y(t)) + \beta |u(t)|^2 dt, \quad \beta > 0 \\
 & \text{s.t.} \\
 & \dot{y} = f(y) + g(y)u, \\
 & y(t_0) = x,
 \end{aligned}$$

where we assume $y(t) \in \mathbb{R}^d$, $u(t) \in \mathbb{R}^{m_u}$, $l \in C^1(\mathbb{R}^d, \mathbb{R})$, $f \in C^1(\mathbb{R}^d, \mathbb{R}^d)$, $g \in C^1(\mathbb{R}^d, \mathbb{R}^{d \times m_u})$. We assume throughout that a minimizer of (P) exists.

The first method to solve the optimization problem involves computing a state-adjoint-control triple originating from the initial condition x , known as the open-loop control. The Pontryagin Maximum Principle (PMP) yields the first-order optimality condition in the form of a coupled forward-backward two-point boundary value problem:

$$(TPBVP) \quad \begin{cases} \frac{d}{dt} y^* = f(y^*) + g(y^*)u^*, \\ y^*(t_0) = x, \\ -\frac{d}{dt} p^* = \partial_y \left(f(y^*) + g(y^*)u^* \right)^T p^* + \partial_y l(y^*), \\ p^*(T) = 0. \end{cases}$$

By the maximality condition of the Hamiltonian at the optimal control we get the optimality condition

$$(1) \quad u^* = -\frac{1}{2\beta} g(y^*)^T p^* \quad \forall t \in (t_0, T).$$

The open-loop control is not robust with respect to disturbances of the state [1]: it is computed for a single initial condition, and deviations from the optimal trajectory are not corrected. In addition, numerical methods for the two-point boundary value problem are sensitive to the initial guess and may fail to converge. An alternative is based on the Dynamic Programming principle, we introduce the value function

$$V(t, y) := \inf_u J(u; t, y) \quad \text{subject to the dynamic system,}$$

which satisfies a nonlinear hyperbolic differential equation, the Hamilton-Jacobi-Bellman (HJB) equation:

$$(HJB) \quad \begin{cases} \partial_t V - \frac{1}{4\beta} \nabla V(t, x)^T g(x) g^T(x) \nabla V(t, x) + \nabla V(t, x)^T f(x) + l(x) = 0 \\ V(T, x) = 0. \end{cases}$$

The value function encodes the optimal cost from every state at once; the feedback law synthesized from it corrects deviations along the trajectory, which is the robustness the open-loop control lacks.

For most of the nonlinear dynamic control problems, one desires a feedback control law, which involves solving the optimal value function of the HJB equation. However, solving the HJB function is notoriously difficult since it suffers from the curse-of-dimensionality. There is an extensive literature on finding the numerical solutions. In addition to the classical methods, the data-driven methods have become increasingly popular. One line of work, introduced in [19], generates the training data with (TPBVP) and approximates the optimal value function from it. The optimal feedback law is then synthesized from the fitted value function after validation. The method effectively overcomes the curse-of-dimensionality, since the data generated is causality-free (not depending on the nearby grid). Moreover, Azmi, Kalise and Kunisch [1] enrich the dataset with the gradient of the value function: a state-adjoint-control triple is generated by solving the open-loop problem, and the adjoint supplies the gradient. The optimal value function is in turn approximated by hyperbolic cross basis functions with l^1 penalty to enhance the sparsity. By the gradient training we get better approximation in the H^1 space. In terms of different approximation methods besides the hyperbolic basis functions, there is a growing literature on solving the nonlinear optimal control problem using neural networks [17, 26].

In this study the approximating function is a two-layer network,

$$(2) \quad \mathcal{N}_{\omega, c}(x) := \sum_{n=1}^N c_n \rho(a_n \cdot x + b_n) \in \mathbb{R},$$

where ρ is a sufficiently smooth activation function and N the width of the network. And the gradient is given by:

$$\nabla_x \mathcal{N}_{\omega, c}(x) = \sum_{n=1}^N c_n \nabla_x \rho(a_n \cdot x + b_n) \in \mathbb{R}^d.$$

The training data is generated by solving (TPBVP). The adjoint state p^* is obtained as a byproduct and provides the gradient information of the optimal value function: $p^*(t) = \nabla V(t, y^*(t))$ wherever $V(t, \cdot)$ is differentiable at $y^*(t)$ ([8] [6] [7]). Where V is not differentiable — on the switching sets studied in Section 6 — competing optimal trajectories carry distinct branchwise costates, and the sampled p^* is the one-sided gradient of its own branch. Instead

of state-value pair we get the dataset of the triple $\{x^i, V(x^i), \nabla V(x^i)\}_{i=1}^M$ that is enriched with gradient information. We have the minimization problem

$$(3) \quad \min_{N \in \mathbb{N}, (a_n, b_n, c_n) \in \mathbb{R}^d \times \mathbb{R} \times \mathbb{R}} l_M(\mathcal{N}_{\omega, c}; V),$$

with

$$l_M(\mathcal{N}_{\omega, c}; V) := \frac{1}{2M} \left(\sum_{m=1}^M |V(x^m) - \mathcal{N}_{\omega, c}(x^m)|^2 + \sum_{m=1}^M \sum_{i=1}^d |\partial_{x_i} V(x^m) - \partial_{x_i} \mathcal{N}_{\omega, c}(x^m)|^2 \right),$$

where the x^m are independent samples of a random variable on domain D with distribution ν . The loss function can be understood as the empirical estimation of the H^1 loss

$$l_\nu(\mathcal{N}_{\omega, c}; V) = \frac{1}{2} \int_D |V(x) - \mathcal{N}_{\omega, c}(x)|^2 + \sum_{i=1}^d |\partial_{x_i} V(x) - \partial_{x_i} \mathcal{N}_{\omega, c}(x)|^2 d\nu.$$

Enhancing Sparsity with a Nonconvex Penalty. Notice that N , the number of neurons, is also an optimization parameter. Although the number of neurons does not act as the inductive bias of the model in general [20], sparsity is crucial for computational efficiency in many problems. Furthermore, sparsity is necessary for certain applications, such as deconvolution of point-like objects. A sparse learning procedure usually adopts convex reformulation of the problem and employ a sparse promoting convex penalty term on the outer weights, a popular choice is the l^1 function (LASSO). However, Pieper and Petrosyan [23] showed that convex penalties cannot effectively eliminate the redundancy of a shallow network when neurons cluster in a small neighborhood, and proposed a nonconvex sparse learning method as the remedy. Their principal application is the ReLU activation; the abstract framework is broader. A finite ReLU network is Lipschitz and lies in H^1 , so the gradient-augmented loss itself is well defined for it; what ReLU lacks is smoothness in the parameters — the candidate search and the second-order training steps of Section 5 differentiate the atom in (a, b) , which ReLU permits only distributionally. In this thesis, we combine gradient-augmented training with nonconvex sparse regularization for a class of activation functions that extends well beyond ReLU, and we analyze the resulting variational problem over measures. The contributions are the following.

- (1) *Existence theory on the unbounded parameter domain.* For localized ridge activations (Assumption 3.2) we prove existence of global minimizers of the gradient-augmented, nonconvex-penalized problem over measures on $\Omega = \mathbb{R}^{d+1}$ (Theorem 3.18). The proof combines weak-* to weak continuity of the value operator (Lemma 3.3) with weak compactness and distributional identification of the gradients — the gradient map itself is not weak-* to weak continuous. Counterexamples show the localization cannot be dropped: for saturating or growing activations the problem degenerates at infinity, and a compact parameter set becomes necessary (Example 3.1).
- (2) *The dual variable at infinity.* On the unbounded parameter domain the gradient kernel grows without bound, so the dual variable $\bar{p}$ is defined only pointwise. For H^2 -regular residuals, Green's identity yields a boundary representation through the bounded kernel σ (Lemma 3.6), which makes $\bar{p}$ bounded and continuous; if moreover no positive- $\mathcal{H}^{d-1}$ -measure piece of ∂D lies in a hyperplane, $\bar{p}$ vanishes at infinity (Lemma 3.7). The failure mode is characterized completely: a non-vanishing escaping sequence must align with the normal of a flat boundary face (Corollary 3.22), a mechanism realized by an explicit example on the cube (Remark 3.21) and quantified as a smallness condition on the boundary flux of the residual (Remark 3.8).
- (3) *Optimality conditions and finite support.* We derive the necessary first-order conditions by directional derivatives and truncated perturbations, without predual structure or moment assumptions on the minimizer (Theorem 3.19); prove that local minimizers are finitely supported under an explicit no-escape condition on the dual variable at infinity —

automatic, by the previous item, on domains without flat faces (Theorem 3.20); and give a sufficient condition for local optimality of finitely supported candidates (Theorem 3.23).

- (4) *The k -homogeneous formulation.* For positively k -homogeneous activations, an equivalent sphere-constrained problem with the fractional-exponent penalty $|c|^{2/(k+1)}$ is derived (Propositions 4.3, 4.5), and the optimality and existence theory is extended to this singular penalty class (Corollaries 4.6, 4.8).

Two node-insertion algorithms with a semismooth Newton inner solver implement the two penalty classes. On a smooth benchmark (Van der Pol) the nonconvex penalties reach the same gradient accuracy as convex ℓ^1 training with a fraction of the neurons (Figure 1, left). On the pendulum swing-up, whose value function has a gradient discontinuity along a switching set, only the ReLU² atom yields a feedback law that makes the correct branch decision across it in our closed-loop tests (Figure 1, right).

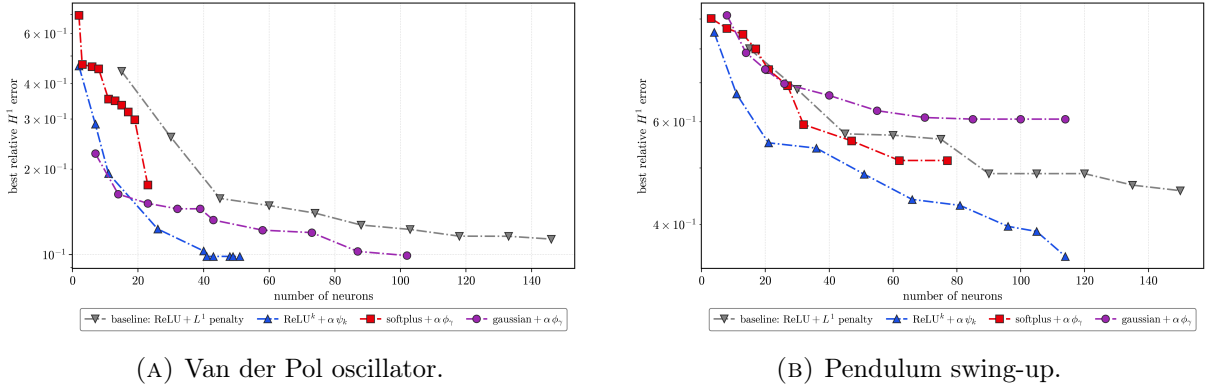


FIGURE 1. Number of neurons against the best relative H^1 validation error on the two benchmark problems, at fixed $\alpha = 10^{-5}$; each curve is the running best error as neurons are inserted. Four series per panel: the convex baseline $\text{ReLU} + \ell^1$; ReLU^k with the fractional-exponent penalty $\psi_k(t) = |t|^{2/(k+1)}$, here at $k = 2$; and softplus and Gaussian with the nonconvex ℓ^1 -log blend ϕ_γ of Section 5 ($\theta = \frac{1}{2}$) at $\gamma = 10$.

The thesis is organized as follows. Section 2 reviews the space of finite Radon measures and the Barron spaces attached to the activation classes. Section 3 introduces the sparse learning problem over measures and develops the existence, optimality, and finite-support theory. Section 4 treats the k -homogeneous reformulation and the fractional-exponent penalty. Section 5 describes the data generation and the two insertion algorithms, and Section 6 reports the numerical studies.

The overall pipeline has four steps:

- (1) **data generation:** initially, a dataset consisting of causality-free state-value-gradient triple must be generated. Many methods are discussed in the literature [13, 10]. The process is highly parallelizable and adaptive: dense samples can be placed in regions of interest.
- (2) **model selection:** depending on the underlying system, different activation function and penalty can be chosen. The choice of activation function and regularization impacts the model performance significantly. We can also incorporate other prior information such as semiconcavity of the value function or the physics-informed loss [9, 18, 24, 27, 16].
- (3) **training and validation:** this step follows the standard machine-learning pipeline. A neural network is trained to approximate the optimal value function, and the performance of the model is subsequently evaluated on a separate validation set. The validation set can be chosen to probe specific regions, such as the switching set.

- (4) **feedback control**: once the value function is trained, the feedback law follows from the Hamiltonian minimization. Sparsity keeps the evaluation of the law cheap.

2. PRELIMINARIES

This section reviews the two function space ingredients of the framework: the space of finite Radon measures, over which the learning problem is posed and whose duality furnishes the optimality conditions; and the Barron spaces, which quantify what the corresponding networks can approximate.

2.1. The space of finite Radon measures. Throughout the paper we denote Ω as $\mathbb{R}^{d+1}$. A finite Radon measure μ is a σ -additive $\mathbb{R}^m$ -valued set function defined on the Borel sets of Ω with $\mu(\emptyset) = 0$. The *total-variation measure* of a finite Radon measure μ is the positive measure defined by

$$|\mu|(E) = \sup \left\{ \sum_{n=1}^{\infty} |\mu(E_n)| : \bigcup_{n=1}^{\infty} E_n = E, \{E_n\} \text{ disjoint and measurable} \right\}.$$

We denote the space of finite Radon measures by $\mathcal{M}(\Omega, \mathbb{R}^m)$ (and $\mathcal{M}(\Omega)$ if $m = 1$). The following spaces will be considered

- $C_0(\Omega; \mathbb{R}^m)$, the space of all continuous functions $u : \Omega \rightarrow \mathbb{R}^m$ *vanishing at infinity*, that is, for every $\varepsilon > 0$ there exists a compact set $K_\varepsilon \subset \Omega$ such that $|u(x)| < \varepsilon$ for all $x \in \Omega \setminus K_\varepsilon$;
- $\mathcal{M}(\Omega; \mathbb{R}^m)$, the space of all vector-valued measures $\mu : \mathcal{B} \rightarrow \mathbb{R}^m$ with finite variation on Ω .

By the Riesz representation theorem, $\mathcal{M}(\Omega; \mathbb{R}^m)$ is isometrically the dual space of $C_0(\Omega; \mathbb{R}^m)$ under the pairing

$$\langle u, \mu \rangle = \sum_{i=1}^m \int_{\Omega} u_i d\mu_i, \quad \mu \in \mathcal{M}(\Omega; \mathbb{R}^m), \quad u \in C_0(\Omega; \mathbb{R}^m);$$

in particular, $\mathcal{M}(\Omega; \mathbb{R}^m)$ equipped with the total variation norm is a Banach space. Without further specification, we denote by $\langle \cdot, \cdot \rangle$ the dual bracket $\langle \cdot, \cdot \rangle_{C_0(\Omega), \mathcal{M}(\Omega)}$.

The following spaces of measures will also be considered:

- $\mathcal{M}^0(\Omega)$: the space of all nonatomic measures on Ω .
- $\mathcal{M}^\sharp(\Omega)$: the space of all purely atomic measures on Ω .

For $\mu \in \mathcal{M}(\Omega)$ we define the *atom set* $\text{atom } \mu := \{\omega \in \Omega : |\mu|(\{\omega\}) > 0\}$, which is at most countable, the purely atomic part $\mu_{\text{atom}} := \sum_{\omega \in \text{atom } \mu} \mu(\{\omega\}) \delta_\omega \in \mathcal{M}^\sharp(\Omega)$, and the nonatomic part $\mu_{\text{cont}} := \mu - \mu_{\text{atom}} \in \mathcal{M}^0(\Omega)$ [5]. This gives the decomposition

$$\mathcal{M}(\Omega) = \mathcal{M}^0(\Omega) \oplus \mathcal{M}^\sharp(\Omega).$$

Note that $\text{atom } \mu$ need not be closed, and it can be a proper subset of $\text{supp } \mu_{\text{atom}}$ (atoms may accumulate at a point carrying no mass).

2.2. Barron spaces. In view of the approximation by neural networks, the seminal paper [3] develops a universal approximation theorem on function approximation by feedforward neural networks with sigmoidal activation functions. Later, due to the popularity of ReLU, the approximation properties of networks with ReLU activation are investigated in the subsequent studies [21, 25, 4, 22]. In particular, it is argued that generalization is achieved not by limiting the size of the network but by controlling the magnitude of the weights. Let $\mathcal{W}(D)$ be the space of functions f on D that can be represented by $f(x) = \int_{\mathbb{S}^d} (a \cdot x + b)_+ d\mu(a, b)$ for every $x \in D$ and some $\mu \in \mathcal{M}(\mathbb{S}^d)$. For $f \in \mathcal{W}(D)$, we define the norm

$$(4) \quad \|f\|_{\mathcal{W}(D)} = \inf \left\{ \|\mu\|_{\mathcal{M}(\mathbb{S}^d)} : f(x) = \int_{\mathbb{S}^d} (a \cdot x + b)_+ d\mu(a, b) \text{ for every } x \in D \right\}.$$

It is shown that this norm is closely related to a fractional Laplacian of f [21, 25].

For the gradient-augmented training, higher regularity of the activation is required, and it is natural to consider ReLU^k with $k \geq 2$. We follow the extended Barron spaces of [15]. There, a function of the form

$$f(x) = \int c(a \cdot x + b)_+^k d\varrho(a, b, c),$$

with ϱ a *probability* measure over the parameters $(a, b) \in \mathbb{R}^{d+1}$ and the outer weight $c \in \mathbb{R}$, is assigned the representation cost

$$\|f\|_{B_{1,\varrho}^k} := \int |c| (\|a\|_1 + |b|)^k d\varrho(a, b, c),$$

and the Barron norm is the infimum over all representing probabilities,

$$\|f\|_{B_1^k} := \inf_{\varrho} \|f\|_{B_{1,\varrho}^k}.$$

The space $B_1^k(D)$ of functions on D with finite Barron norm is a normed vector space [15, Theorem 6]. The two pictures are compatible: a finite signed measure $\mu \in \mathcal{M}(\mathbb{S}^d)$ representing f through ReLU^k atoms on the sphere induces, after normalization, a representing probability whose cost is bounded by a dimensional constant times $\|\mu\|_{\mathcal{M}(\mathbb{S}^d)}$, so that $\|f\|_{B_1^k} \leq C(d, k) \|\mu\|_{\mathcal{M}(\mathbb{S}^d)}$. The following result links the Barron space to the standard Sobolev space.

Proposition 2.1 (Barron–Sobolev embedding, [15, Theorem 7]). *Given any fixed integer k , there holds $B_1^k(D) \subset H^k(D)$; moreover there is a constant $C(D, d, k)$ such that*

$$\|f\|_{H^m(D)} \leq C(D, d, k) \|f\|_{B_1^k}, \quad 0 \leq m \leq k,$$

for all $f \in B_1^k(D)$.

The proposition shows both the power and the limitation of the Barron space: every function that the ReLU^k dictionary represents with finite total weight is automatically H^k -regular. For smooth value functions this yields good approximation together with control of the derivatives. Conversely, for $k \geq 2$, a value function whose gradient jumps across a switching set does not belong to H^k near the jump, so its B_1^k norm cannot be finite. The embedding turns this into a quantitative obstruction at every finite budget: networks with variation norm at most R are bounded by $C(D, d, k) R$ in $H^k(D)$, and any $H^1(D)$ -limit of an H^k -bounded sequence lies in $H^k(D)$ (weak compactness in the Hilbert space $H^k(D)$ and uniqueness of limits), so for $V \notin H^k(D)$

$$\inf \{ \|\mathcal{N}\mu - V\|_{H^1(D)} : \|\mu\|_{\mathcal{M}(\mathbb{S}^d)} \leq R \} > 0 \quad \text{for every } R < \infty.$$

The floor can only vanish as $R \rightarrow \infty$ — and a bounded variation budget is precisely what the sparsity regularizer enforces; the experiments make the resulting cost visible at the switching set (see Figure 1). For $k = 1$ there is no such obstruction: membership of V in $H^1(D)$ is compatible with the ReLU class, and a single ReLU ridge represents a flat gradient jump exactly.

Moreover, the approximation properties of other sufficiently smooth activation functions are covered by embedding them into the ReLU^k Barron scale. For a Lipschitz activation σ , let B_σ denote the Barron space defined as above with atoms $\sigma(a \cdot x + b)$.

Proposition 2.2 (Embeddings into ReLU^k Barron spaces, [11, Theorem 1]). *Let σ be a Lipschitz activation function and let D be bounded.*

- (1) *If $\sigma \in C^1(\mathbb{R})$ and $\partial^2 \sigma \in L^1(\mathbb{R})$, then $B_\sigma \hookrightarrow B_{\text{ReLU}}$.*
- (2) *If $k \geq 1$, $\sigma \in C^k(\mathbb{R})$, and the one-sided (Caputo) derivatives of order $k + 1$ satisfy $\partial_+^{k+1} \sigma \in L^1((0, \infty))$ and $\partial_-^{k+1} \sigma \in L^1((-\infty, 0))$ — for which it suffices that the $(k + 1)$ -st distributional derivative of σ exists as a finite measure — then*

$$B_\sigma \hookrightarrow B_{\text{ReLU}^k}.$$

Notation. Throughout the thesis: $D \subset \mathbb{R}^d$ is the bounded Lipschitz domain and ν the data distribution on D ; M denotes the number of samples. In the analysis of Sections 3–4, ν is taken to be the Lebesgue measure on D and all norms are the unweighted ones, $L^2(D)$ and $H^1(D)$; the sampling distribution reappears in the empirical losses of Sections 5–6. $\mathcal{H}^{d-1}$ denotes $(d-1)$ -dimensional Hausdorff measure, which coincides with surface measure on Lipschitz hypersurfaces such as ∂D . $\Omega = \mathbb{R}^{d+1}$ is the parameter domain with generic element $\omega = (a, b)$ (inner weight and bias).

3. THE LEARNING PROBLEM

We define the *value operator* of a measure $\mu \in \mathcal{M}(\Omega)$ by

$$[\mathcal{N}\mu](x) = \int_{\Omega} \sigma(x, \omega) d\mu(\omega).$$

The integral converges absolutely for every $\mu \in \mathcal{M}(\Omega)$ whenever the value dictionary is bounded, i.e. $S_{\infty} := \sup_{\omega \in \Omega} \|\sigma(\cdot, \omega)\|_{L^{\infty}(D)} < \infty$, with $\|\mathcal{N}\mu\|_{L^{\infty}(D)} \leq S_{\infty} \|\mu\|_{\mathcal{M}(\Omega)}$. This holds in each of the three settings considered in this thesis: (i) localized ridge activations on the unbounded domain $\Omega = \mathbb{R}^{d+1}$ (Assumption 3.2 below), (ii) arbitrary continuous activations on a compact parameter set $\Omega_R \subset \mathbb{R}^{d+1}$, and (iii) k -homogeneous activations on the sphere (Section 4).

The gradient of the operator is understood distributionally; whenever

$$(B) \quad \int_{\Omega} \|\nabla_x \sigma(\cdot, \omega)\|_{L^2(D)} d|\mu|(\omega) < \infty,$$

the map $\omega \mapsto \partial_{x_i} \sigma(\cdot, \omega)$ is Bochner integrable into $L^2(D)$. The gradient operator is given by the classical formula

$$\nabla(\mathcal{N}\mu)(x) = \int_{\Omega} \nabla_x \sigma(x, \omega) d\mu(\omega),$$

by Fubini's theorem against test functions, and we write $\nabla \mathcal{N}\mu := \nabla(\mathcal{N}\mu)$ in this case.

The fidelity term is the gradient-augmented loss with the distributional gradient convention,

$$L(\mu) := \begin{cases} \frac{1}{2} \|\mathcal{N}\mu - V\|_{L^2(D)}^2 + \frac{1}{2} \|\nabla(\mathcal{N}\mu) - \nabla V\|_{L^2(D)}^2, & \mathcal{N}\mu \in H^1(D), \\ +\infty, & \text{otherwise.} \end{cases}$$

3.1. Predual representation of the value operator $\mathcal{N}$. For the existence of the minimizer we need the weak-* l.s.c. of the loss function, which includes the value operator $\mathcal{N}$. It is not obvious once we extend the dictionary to the unbounded parameter space. In particular, the operator $\mathcal{N}$ is not weak-*to-weak continuous in general.

Example 3.1 (loss of weak-* lower semicontinuity without localization). Let $\rho = \tanh$, $V \equiv 1$, and $\mu_n = \delta_{(a, b_n)}$ with $b_n \rightarrow \infty$. Then $\mu_n \xrightarrow{*} 0$, yet

$$\mathcal{N}\mu_n = \tanh(a \cdot x + b_n) \rightarrow 1$$

strongly in $H^1(D)$. Hence

$$\liminf_{n \rightarrow \infty} L(\mu_n) = 0 < L(0),$$

so the fidelity term is not weak-* lower semicontinuous. The issue is that mass escaping to infinity can carry a non-vanishing value contribution unless the dictionary atoms decay at infinity.

For the profile $\rho(z) = \log(1 + e^z)$ (softplus) the measures $\mu_n = \rho(b_n)^{-1} \delta_{(0, b_n)}$ fit $V \equiv 1$ exactly for every n at penalty $\alpha \phi(1/\rho(b_n)) \rightarrow 0$, so $\inf J = 0$ is not attained; moreover $n^{-1} \rho(n(e \cdot x - t)) \rightarrow (e \cdot x - t)_+$ in $H^1(D)$, so every ReLU network is reached at vanishing penalty and the regularization degenerates entirely (It might be possible to leverage the homogeneity to control the domain like the ReLU^k case).

We need the decay condition to prevent this escape and make the limit attained. For the unbounded-parameter ridge analysis in this section we use the following standing value-dictionary assumption.

Assumption 3.2 (localized ridge profile). The parameter domain is $\Omega = \mathbb{R}^{d+1}$ and the activation has the ridge form

$$\sigma(x, \omega) = \rho(a \cdot x + b), \quad \omega = (a, b) \in \Omega,$$

where $\rho \in C_0(\mathbb{R})$, i.e. ρ is continuous and $\rho(z) \rightarrow 0$ as $|z| \rightarrow \infty$.

Lemma 3.3 (weak-* to weak continuity of $\mathcal{N}$). *Under Assumption 3.2, set $\kappa(\omega) := \|\sigma(\cdot, \omega)\|_{L^2(D)}$, then $\kappa \in C_0(\Omega)$. Moreover, for every sequence $\mu_n \xrightarrow{*} \mu$ in $\mathcal{M}(\Omega)$ it holds $\mathcal{N}\mu_n \rightharpoonup \mathcal{N}\mu$ weakly in $L^2(D)$.*

Proof. (1) We first show that $\kappa \in C_0(\Omega)$. Continuity follows from dominated convergence, since $\rho \in C_0(\mathbb{R})$ is bounded. It remains to prove decay at infinity.

Let $R_D > 0$ satisfy $|x| \leq R_D$ on D , and let $|\omega_j| \rightarrow \infty$, where $\omega_j = (a_j, b_j)$. It is enough to show that every subsequence has a further subsequence along which $\kappa(\omega_j) \rightarrow 0$. Passing to such a further subsequence, either $|a_j|$ is bounded or $|a_j| \rightarrow \infty$.

If $|a_j|$ is bounded, then $|b_j| \rightarrow \infty$, and

$$\inf_{x \in D} |a_j \cdot x + b_j| \geq |b_j| - |a_j|R_D \rightarrow \infty.$$

Thus $\rho(a_j \cdot x + b_j) \rightarrow 0$ uniformly in $x \in D$, and hence $\kappa(\omega_j) \rightarrow 0$.

It remains to consider the case $|a_j| \rightarrow \infty$. For $\tau > 0$ set

$$B_\tau(\omega_j) := \{x \in D : |a_j \cdot x + b_j| \leq \tau\}, \quad \eta(\tau) := \sup_{|z| \geq \tau} |\rho(z)|.$$

Since $\rho \in C_0(\mathbb{R})$, $\eta(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$. The set $B_\tau(\omega_j)$ is the intersection of D with a slab of width $2\tau/|a_j|$, and therefore

$$|B_\tau(\omega_j)| \leq c_D \frac{\tau}{|a_j|}$$

with c_D depending only on D . Splitting D into $B_\tau(\omega_j)$ and its complement gives

$$\begin{aligned} \kappa(\omega_j)^2 &\leq \|\rho\|_\infty^2 |B_\tau(\omega_j)| + \eta(\tau)^2 |D| \\ &\leq \|\rho\|_\infty^2 c_D \frac{\tau}{|a_j|} + \eta(\tau)^2 |D|. \end{aligned}$$

Hence $\limsup_j \kappa(\omega_j)^2 \leq \eta(\tau)^2 |D|$ for every $\tau > 0$. Letting $\tau \rightarrow \infty$ gives $\kappa(\omega_j) \rightarrow 0$. Therefore $\kappa(\omega) \rightarrow 0$ as $|\omega| \rightarrow \infty$.

(2) Given $\mu_n \xrightarrow{*} \mu$ in $\mathcal{M}(\Omega)$, for every $\varphi \in L^2(D)$ we have

$$\langle \mathcal{N}\mu_n, \varphi \rangle_{L^2(D)} = \int_\Omega \langle \sigma(\cdot, \omega), \varphi \rangle_{L^2(D)} d\mu_n(\omega)$$

by Fubini's theorem, since $|\langle \sigma(\cdot, \omega), \varphi \rangle_{L^2(D)}| \leq \|\rho\|_\infty |D|^{1/2} \|\varphi\|_{L^2(D)}$ and μ_n is finite. The function $f : \omega \mapsto \langle \sigma(\cdot, \omega), \varphi \rangle_{L^2(D)}$ is continuous by dominated convergence. By Cauchy-Schwarz we have

$$|\langle \sigma(\cdot, \omega), \varphi \rangle_{L^2(D)}| \leq \kappa(\omega) \|\varphi\|_{L^2(D)},$$

which goes to 0 as $|\omega|$ goes to infinity by the previous step. Consequently, $f \in C_0(\Omega)$, and by weak-* convergence,

$$\lim_{n \rightarrow \infty} \langle \mathcal{N}\mu_n, \varphi \rangle_{L^2(D)} = \lim_{n \rightarrow \infty} \langle f, \mu_n \rangle = \langle f, \mu \rangle = \langle \mathcal{N}\mu, \varphi \rangle_{L^2(D)}.$$

□

Remark 3.4 (The gradient map is not needed as a weak- $*$ -to-weak continuous operator). Lemma 3.3 gives weak- $*$ -to-weak continuity of the value operator

$$\mathcal{N} : \mathcal{M}(\Omega) \rightarrow L^2(D).$$

Since we are doing the gradient training, we need to have a similar result for the gradient operator. However on the unbounded parameter domain, for ridge atoms

$$\sigma_{a,b}(x) = \rho(a \cdot x + b),$$

we have

$$\nabla_x \sigma_{a,b}(x) = a \rho'(a \cdot x + b).$$

When the ridge intersects D , the $L^2(D)$ -norm of this gradient can grow without bound as $|a| \rightarrow \infty$ and

$$\sup_{\omega \in \Omega} \|\nabla_x \sigma(\cdot, \omega)\|_{L^2(D)} = \infty.$$

Hence, $\mu \mapsto \nabla(\mathcal{N}\mu)$ is not a bounded operator from $\mathcal{M}(\Omega)$ into $L^2(D)^d$ and cannot be weak- $*$ -to-weak continuous on all of $\mathcal{M}(\Omega)$.

Fortunately, it does not obstruct the direct method of calculus variation. Along a minimizing sequence (μ_n) , the loss itself gives a bound on $\|\nabla(\mathcal{N}\mu_n)\|_{L^2(D)}$. Therefore, after passing to a subsequence,

$$\nabla(\mathcal{N}\mu_n) \rightharpoonup g \quad \text{weakly in } L^2(D)^d.$$

The limit is identified distributionally using only the weak convergence of the value functions. Indeed, for every $\psi \in C_c^\infty(D)^d$,

$$\int_D g \cdot \psi \, dx = \lim_{n \rightarrow \infty} \int_D \nabla(\mathcal{N}\mu_n) \cdot \psi \, dx = - \lim_{n \rightarrow \infty} \int_D \mathcal{N}\mu_n \operatorname{div} \psi \, dx = - \int_D \mathcal{N}\mu \operatorname{div} \psi \, dx.$$

Thus $g = \nabla(\mathcal{N}\mu)$ in the sense of distributions, and $\mathcal{N}\mu \in H^1(D)$. In this way the existence proof uses weak compactness of bounded sequences in the Hilbert space and distributional identification, rather than weak- $*$ -to-weak continuity of the gradient map.

3.2. Predual representation of the gradient of the loss function L . Since the optimization problem is nonconvex, the global minimizer is difficult to find in general. Instead, we define the local minimum:

Definition 3.5 (local minimizer). $\bar{\mu} \in \mathcal{M}(\Omega)$ is a local minimizer if there exists $\epsilon > 0$ such that

$$J(\bar{\mu}) \leq J(\mu) \quad \text{for all } \mu \text{ with } \|\mu - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq \epsilon.$$

To investigate the optimization we need the derivative. In this section we discuss how to describe the derivative properly. As we have already seen in the last remark, in the case of gradient training we do not have a canonical bounded operator that maps the measure to the gradient if we assume the unbounded dictionary. Let $\bar{\mu}$ be such that $\mathcal{N}\bar{\mu} \in H^1(D)$, and set $r := \mathcal{N}\bar{\mu} - V$. Denote by $\partial L(\mu)[u]$ the Gâteaux derivative at μ for an admissible perturbation u one has

$$(5) \quad \partial L(\bar{\mu})[u] = \langle r, \mathcal{N}u \rangle_{L^2(D)} + \langle \nabla r, \nabla(\mathcal{N}u) \rangle_{L^2(D)}.$$

If the differentiation under the parameter integral is justified — for instance if the perturbation u itself satisfies (B), with $|\mu|$ replaced by $|u|$ — then the derivative can be written as

$$\partial L(\bar{\mu})[u] = \int_{\Omega} p_r(\omega) \, du(\omega),$$

where

$$(6) \quad p_r(\omega) := \langle r, \sigma(\cdot, \omega) \rangle_{L^2(D)} + \langle \nabla r, \nabla_x \sigma(\cdot, \omega) \rangle_{L^2(D)}.$$

On compact parameter sets, or whenever the atom gradients are uniformly bounded in $L^2(D)$, the formula defines a bounded continuous dual variable p_r on Ω . On the unbounded ridge parameter domain this boundedness fails in general, as we discussed in the last section. Hence p_r need not belong to the predual $C_0(\Omega)$ of $\mathcal{M}(\Omega)$ without additional assumptions.

We circumvent this barrier with integration by parts; the price is higher regularity of the residual. For $r \in H^2(D)$, Green's formula moves the derivative off the unbounded gradient kernel $\nabla_x \sigma$ and onto r , representing p_r through the bounded kernel σ alone — the following lemma makes this precise, and this representation is the one used for the analysis on the unbounded parameter domain.

Lemma 3.6 (boundary representation of the dual pairing). *Let $D \subset \mathbb{R}^d$ be a bounded Lipschitz domain. Assume that*

$$\sigma(x, \omega) = \rho(a \cdot x + b), \quad \omega = (a, b) \in \Omega,$$

with $\rho \in C^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$. Let $r \in H^2(D)$ and p_r defined as above, then:

(1) *For every $\omega \in \Omega$,*

$$p_r(\omega) = \langle r - \Delta r, \sigma(\cdot, \omega) \rangle_{L^2(D)} + \langle \partial_n r, \sigma(\cdot, \omega) \rangle_{L^2(\partial D)}.$$

In particular, p_r is continuous and bounded, with

$$\sup_{\omega \in \Omega} |p_r(\omega)| \leq C_D \|\rho\|_{L^\infty(\mathbb{R})} \|r\|_{H^2(D)}.$$

(2) *For every $\mu' \in \mathcal{M}(\Omega)$ such that $\mathcal{N}\mu' \in H^1(D)$,*

$$\langle r, \mathcal{N}\mu' \rangle_{L^2(D)} + \langle \nabla r, \nabla(\mathcal{N}\mu') \rangle_{L^2(D)} = \int_{\Omega} p_r(\omega) d\mu'(\omega).$$

Proof. (1) For fixed $\omega = (a, b)$, the function $\sigma(\cdot, \omega) = \rho(a \cdot x + b)$ belongs to $C^1(\overline{D}) \subset H^1(D)$. Moreover $|\sigma(x, \omega)| \leq \|\rho\|_{L^\infty(\mathbb{R})}$ on both D and ∂D . Let $u = \sigma(\cdot, \omega)$. Since $r \in H^2(D)$ and $u \in H^1(D)$, Green's formula gives

$$\langle \nabla r, \nabla u \rangle_{L^2(D)} = -\langle \Delta r, u \rangle_{L^2(D)} + \langle \partial_n r, u \rangle_{L^2(\partial D)}.$$

Adding $\langle r, u \rangle_{L^2(D)}$ yields

$$p_r(\omega) = \langle r - \Delta r, \sigma(\cdot, \omega) \rangle_{L^2(D)} + \langle \partial_n r, \sigma(\cdot, \omega) \rangle_{L^2(\partial D)}.$$

This proves the representation in part (1). The bound follows from Cauchy–Schwarz and the trace estimate for the normal derivative:

$$\begin{aligned} |p_r(\omega)| &\leq \|r - \Delta r\|_{L^2(D)} \|\sigma(\cdot, \omega)\|_{L^2(D)} + \|\partial_n r\|_{L^2(\partial D)} \|\sigma(\cdot, \omega)\|_{L^2(\partial D)} \\ &\leq C_D \|\rho\|_{L^\infty(\mathbb{R})} \|r\|_{H^2(D)}, \end{aligned}$$

with constant depending on the domain D . The right-hand side is independent of ω . Continuity of p_r follows from the representation above: if $\omega_j \rightarrow \omega$, then $\sigma(\cdot, \omega_j) \rightarrow \sigma(\cdot, \omega)$ in both $L^2(D)$ and $L^2(\partial D)$. Continuity follows by dominated convergence.

(2) Now let $\mu' \in \mathcal{M}(\Omega)$ with $\mathcal{N}\mu' \in H^1(D)$ and set $u = \mathcal{N}\mu'$. We use the trace theorem on bounded Lipschitz domains: there is a bounded linear trace operator

$$\text{Tr} : H^1(D) \rightarrow H^{1/2}(\partial D) \hookrightarrow L^2(\partial D),$$

and, since $r \in H^2(D)$, the normal trace $\partial_n r := \text{Tr}(\nabla r) \cdot n$ belongs to $L^2(\partial D)$. Green's formula gives

$$\langle r, u \rangle_{L^2(D)} + \langle \nabla r, \nabla u \rangle_{L^2(D)} = \langle r - \Delta r, u \rangle_{L^2(D)} + \langle \partial_n r, \text{Tr } u \rangle_{L^2(\partial D)}.$$

Since ρ is bounded and μ' is finite, Fubini's theorem gives

$$\langle r - \Delta r, \mathcal{N}\mu' \rangle_{L^2(D)} = \int_{\Omega} \langle r - \Delta r, \sigma(\cdot, \omega) \rangle_{L^2(D)} d\mu'(\omega).$$

Moreover, the boundedness and continuity of $\sigma(\cdot, \omega)$ imply by dominated convergence that $\mathcal{N}\mu'$ has a continuous representative on $\overline{D}$,

$$(\mathcal{N}\mu')(x) = \int_{\Omega} \sigma(x, \omega) d\mu'(\omega).$$

For functions in $H^1(D) \cap C(\overline{D})$, the Sobolev trace coincides with the classical boundary restriction. Hence, for $\mathcal{H}^{d-1}$ -almost every $x \in \partial D$,

$$\text{Tr}(\mathcal{N}\mu')(x) = (\mathcal{N}\mu')(x) = \int_{\Omega} \sigma(x, \omega) d\mu'(\omega).$$

Using this identity and applying Fubini on ∂D gives

$$\langle \partial_n r, \text{Tr}(\mathcal{N}\mu') \rangle_{L^2(\partial D)} = \int_{\Omega} \langle \partial_n r, \sigma(\cdot, \omega) \rangle_{L^2(\partial D)} d\mu'(\omega).$$

Combining the last three identities and using the representation from part (1), we obtain

$$\langle r, \mathcal{N}\mu' \rangle_{L^2(D)} + \langle \nabla r, \nabla(\mathcal{N}\mu') \rangle_{L^2(D)} = \int_{\Omega} p_r(\omega) d\mu'(\omega).$$

This proves part (2). $\square$

If the residual $r := \mathcal{N}\bar{\mu} - V$ belongs to $H^2(D)$, then $\bar{p} = p_r$ in the notation of Lemma 3.6, and the representation of Lemma 3.6(1) splits the Gâteaux derivative into an interior term and a boundary term. The interior term vanishes as $|\omega| \rightarrow \infty$ by Lemma 3.3. The boundary term pairs the normal derivative of the residual against the trace of the atom on the boundary,

$$\langle \partial_n r, \sigma(\cdot, \omega) \rangle_{L^2(\partial D)} = \int_{\partial D} \partial_n r(x) \rho(a \cdot x + b) d\mathcal{H}^{d-1}(x),$$

so everything depends on whether the trace $\rho(a \cdot x + b)|_{\partial D}$ becomes small. If ∂D meets every hyperplane in an $\mathcal{H}^{d-1}$ -null set — no flat boundary pieces, e.g. D smooth and strictly convex — then every sharpening ridge crosses the boundary transversally, its trace is supported on a band of vanishing surface measure, and the term vanishes (Lemma 3.7). In that case $\limsup_{|\omega| \rightarrow \infty} |p_r(\omega)| = 0$.

Lemma 3.7 (decay of the dual variable). *In the setting of Lemma 3.6, assume in addition that Assumption 3.2 holds and that $\mathcal{H}^{d-1}(\partial D \cap \Pi) = 0$ for every hyperplane $\Pi \subset \mathbb{R}^d$. Then $p_r \in C_0(\Omega)$.*

Proof. By Lemma 3.6(1), p_r is continuous, so $p_r \in C_0(\Omega)$ follows once $|p_r(\omega)| \rightarrow 0$ as $|\omega| \rightarrow \infty$. The representation there gives

$$|p_r(\omega)| \leq (\|r\|_{L^2(D)} + \|\Delta r\|_{L^2(D)}) \kappa(\omega) + \|\partial_n r\|_{L^2(\partial D)} \kappa_{\partial}(\omega),$$

with $\kappa(\omega) := \|\sigma(\cdot, \omega)\|_{L^2(D)}$ and $\kappa_{\partial}(\omega) := \|\sigma(\cdot, \omega)\|_{L^2(\partial D)}$. By Lemma 3.3, $\kappa(\omega) \rightarrow 0$ as $|\omega| \rightarrow \infty$; it remains to show $\kappa_{\partial}(\omega) \rightarrow 0$.

Let $R_D > 0$ satisfy $|x| \leq R_D$ on $\overline{D}$, and set $\eta(\tau) := \sup_{|z| \geq \tau} |\rho(z)|$, so $\eta(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$. Fix $\tau > 0$; as in Lemma 3.3, splitting ∂D into the band $B_{\tau}(\omega) := \{x \in \partial D : |a \cdot x + b| \leq \tau\}$, where $|\sigma| \leq \|\rho\|_{\infty}$, and its complement, where $|\sigma| \leq \eta(\tau)$,

$$\kappa_{\partial}(\omega)^2 \leq \|\rho\|_{\infty}^2 \mathcal{H}^{d-1}(B_{\tau}(\omega)) + \eta(\tau)^2 \mathcal{H}^{d-1}(\partial D).$$

We claim $\mathcal{H}^{d-1}(B_{\tau}(\omega)) \rightarrow 0$ as $|\omega| \rightarrow \infty$. Let $|\omega_j| \rightarrow \infty$. It is enough to show that every subsequence has a further subsequence along which $\mathcal{H}^{d-1}(B_{\tau}(\omega_j)) \rightarrow 0$. Passing to such a further subsequence, either $|a_j|$ is bounded or $|a_j| \rightarrow \infty$.

If $|a_j|$ is bounded, then $|b_j| \rightarrow \infty$, and

$$\inf_{x \in \partial D} |a_j \cdot x + b_j| \geq |b_j| - |a_j|R_D \rightarrow \infty.$$

Hence $B_{\tau}(\omega_j) = \emptyset$ eventually.

It remains to consider the case $|a_j| \rightarrow \infty$. If $B_\tau(\omega_j) = \emptyset$ along a further subsequence, there is nothing to prove. Otherwise choose $x_j \in B_\tau(\omega_j)$. After passing to a further subsequence, $a_j/|a_j| \rightarrow e$ with $|e| = 1$ and $x_j \rightarrow x_* \in \partial D$. Since $|a_j \cdot x_j + b_j| \leq \tau$,

$$\frac{b_j}{|a_j|} \rightarrow \beta := -e \cdot x_*.$$

For $x \in B_\tau(\omega_j)$,

$$|e \cdot x + \beta| \leq \frac{|a_j \cdot x + b_j|}{|a_j|} + \left| e - \frac{a_j}{|a_j|} \right| R_D + \left| \beta - \frac{b_j}{|a_j|} \right| \rightarrow 0$$

uniformly in x . Thus for every $\varepsilon > 0$, eventually

$$B_\tau(\omega_j) \subset \{x \in \partial D : |e \cdot x + \beta| \leq \varepsilon\}.$$

Therefore

$$\limsup_j \mathcal{H}^{d-1}(B_\tau(\omega_j)) \leq \mathcal{H}^{d-1}(\{x \in \partial D : |e \cdot x + \beta| \leq \varepsilon\}).$$

Letting $\varepsilon \downarrow 0$ and using continuity from above gives

$$\limsup_j \mathcal{H}^{d-1}(B_\tau(\omega_j)) \leq \mathcal{H}^{d-1}(\partial D \cap \Pi) = 0, \quad \Pi := \{x : e \cdot x = -\beta\},$$

by the no-flat-face hypothesis. Hence $\mathcal{H}^{d-1}(B_\tau(\omega)) \rightarrow 0$, and so, for every τ , $\limsup_{|\omega| \rightarrow \infty} \kappa_\partial(\omega)^2 \leq \eta(\tau)^2 \mathcal{H}^{d-1}(\partial D)$; letting $\tau \rightarrow \infty$ gives $\kappa_\partial(\omega) \rightarrow 0$. With $\kappa(\omega) \rightarrow 0$ this yields $|p_r(\omega)| \rightarrow 0$, hence $p_r \in C_0(\Omega)$. $\square$

Remark 3.8. If instead ∂D contains a flat face — the *entire* coplanar section $F = \partial D \cap \{n_F \cdot x = c_F\}$ up to an $\mathcal{H}^{d-1}$ -null set, as for a face of the boxes of Section 6 — a ridge can align with it: along $a_j = t_j n_F$, $b_j = -t_j c_F + s_0$ with $t_j \rightarrow \infty$, the argument $a_j \cdot x + b_j \equiv s_0$ is *constant on the face*, so the trace does not shrink — it converges to the constant $\rho(s_0)$ on F and to 0 on the rest of the boundary. In the limit only the flux of the residual through the face survives:

$$p_r(\omega_j) \rightarrow \rho(s_0) \int_F \partial_n r d\mathcal{H}^{d-1}.$$

Geometrically, these atoms are ever-thinner ridges pressed against the face: they occupy vanishing volume, but their gradients keep correcting the normal derivative of the fit in a boundary layer along F , which remains profitable exactly as long as the residual has nonzero flux through the face.

3.3. Activation functions. For the local analysis of Section 3.6 — the optimality conditions and the finite-support theorem, whose arguments are local in ω — we use the following differentiability of the ridge profile.

Assumption 3.9. The activation has the ridge form $\sigma(x, \omega) = \rho(a \cdot x + b)$, $\omega = (a, b) \in \Omega = \mathbb{R}^{d+1}$, with $\rho \in C^1(\mathbb{R})$ and ρ, ρ' Lipschitz continuous.

In particular, on every compact set $K \subset \Omega$ there is a constant Λ_K such that, for all $x \in D$ and $\omega_1, \omega_2 \in K$,

$$\begin{aligned} |\sigma(x, \omega_1) - \sigma(x, \omega_2)| &\leq \Lambda_K(1 + \|x\|)|\omega_1 - \omega_2|, \\ |\partial_{x_i} \sigma(x, \omega_1) - \partial_{x_i} \sigma(x, \omega_2)| &\leq \Lambda_K(1 + \|x\|)|\omega_1 - \omega_2|, \quad i = 1, \dots, d, \end{aligned}$$

the Lipschitz-in- ω regularity the local analysis relies on; the constant is local in ω because $\partial_{x_i} \sigma = a_i \rho'$ carries the factor a_i .

Which activation suits which target is a question about the derivative $\partial_x \sigma$: under the gradient-augmented loss each neuron contributes σ to the reconstruction of V and $\partial_x \sigma$ to that of ∇V , so the shape of $\partial_x \sigma$ governs the approximation economy. This discussion is deferred to Section 6, where it accompanies the numerical studies: smooth targets in Section 6.1, targets with a gradient discontinuity in Section 6.2.

3.4. The nonconvex penalty. We implement an explicit sparsity-promoting penalty term. The penalization does not enhance the generalization performance for large neural networks [28]. Nevertheless, the explicit regularization still plays a central role in the sparse learning. As introduced in the introduction, the l^1 penalty fails to detect the redundancy of the neurons clustering in a small neighborhood, which motivates the nonconvex penalty term. First we introduce the following assumption:

Assumption 3.10. The penalty term $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfies: ϕ is differentiable at 0 with $\phi'(0) = 1$.

Assumption 3.11. The penalty term $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is differentiable, concave, nondecreasing and coercive ($\phi(z) \rightarrow \infty$ as $z \rightarrow \infty$) with $\phi(0) = 0$.

The Assumption 3.11 describe the global property of the penalty functions. By introducing the nonconvexity the coercivity of the quadratic cost is compromised. With Assumption 3.10, 3.11 the optimization problem stays coercive such that we can apply the direct method of calculus of variations. The following lemma follows from the assumption.

Lemma 3.12. *Let Assumption 3.11 be satisfied. Then:*

- (1) (subadditivity) $\phi(z_1 + z_2) \leq \phi(z_1) + \phi(z_2)$ for all $z_1, z_2 \geq 0$;
- (2) if in addition Assumption 3.10 is satisfied, then $\phi(z) \leq z$ for all $z \geq 0$.

Proof. **1.** Using $\phi(0) = 0$ and the fundamental theorem of calculus,

$$\begin{aligned} \phi(z_1 + z_2) &= \int_0^{z_1} \phi'(s) ds + \int_0^{z_2} \phi'(z_1 + s) ds \\ (\phi' \text{ nonincreasing by concavity}) &\leq \int_0^{z_1} \phi'(s) ds + \int_0^{z_2} \phi'(s) ds \\ &= \phi(z_1) + \phi(z_2). \end{aligned}$$

2. For $z \geq 0$,

$$\phi(z) = \int_0^z \phi'(s) ds \leq \int_0^z \phi'(0) ds = z,$$

using again that ϕ' is nonincreasing and $\phi'(0) = 1$. $\square$

Assumptions 3.10 and 3.11 ensure that the nonconvex problem admits a global minimizer. To enhance the sparsity, we need to exert strong penalizing effect on the weight on 2 neighboring neurons such that merging them would in fact reduce the cost. We are mainly interested in the behavior of ϕ' in a neighborhood of 0:

Assumption 3.13 (Local concavity). The penalty term $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}$ satisfies:

- (1) There exists $\gamma_1 \geq 0, \hat{z}_1 > 0$, such that $-\gamma_1(z_2 - z_1) \leq \phi'(z_2) - \phi'(z_1) \leq 0$ for all $0 \leq z_1 \leq z_2 \leq \hat{z}_1$. (γ -convex)
- (2) There exists $\gamma_2 > 0, \hat{z}_2 > 0$, such that $\phi'(z_2) - \phi'(z_1) \leq -\gamma_2(z_2 - z_1) \leq 0$ for all $0 \leq z_1 \leq z_2 \leq \hat{z}_2$. (strongly concave)

Remark 3.14. The Assumption 3.13 can be interpreted as the concave function being strongly concave but exhibits also certain convexity locally at 0. It is worth noticing that it is entirely a local property that only needs to take effect at 0. Assumption 3.13(2) induces strict subadditivity that takes effect at small weights.

In general the global concavity in Assumption 3.11 does not imply the γ -convexity in Assumption 3.13(1): the curvature of a concave ϕ may be unbounded from below near 0. Observe, for example, $\phi(x) = x - \frac{2}{3}x^{\frac{3}{2}}$ on a neighborhood of 0 (modified outside, e.g. extended non-decreasing and coercive beyond $x = \frac{1}{4}$, so that Assumption 3.11 holds on all of $\mathbb{R}_+$): here $\phi''(x) = -\frac{1}{2}x^{-1/2} \rightarrow -\infty$ as $x \rightarrow 0^+$, so no constant γ_1 satisfies Assumption 3.13(1), while

the strong concavity of Assumption 3.13(2) does hold near 0. Since Assumption 3.13 is a local property, such penalties can be constructed piecewise, prescribing the behavior in a neighborhood of 0 first.

Proposition 3.15 (second-order envelopes). *Given ϕ satisfying Assumptions 3.10 and 3.11, if ϕ satisfies in addition Assumption 3.13, then it holds*

- (1) for all $z \in [0, \hat{z}_1] : \phi(z) \geq z - \frac{\gamma_1}{2}z^2$,
- (2) for all $z \in [0, \hat{z}_2] : \phi(z) \leq z - \frac{\gamma_2}{2}z^2$.

Proof. By $\phi(0) = 0$ and $\phi'(0) = 1$ (Assumptions 3.11, 3.10),

$$\phi(z) - z = \int_0^z (\phi'(s) - \phi'(0)) ds.$$

On $[0, \hat{z}_1]$, Assumption 3.13(1) gives $-\gamma_1 s \leq \phi'(s) - \phi'(0) \leq 0$, and integrating yields (1). On $[0, \hat{z}_2]$, Assumption 3.13(2) gives $\phi'(s) - \phi'(0) \leq -\gamma_2 s$, and integrating yields (2). $\square$

By the proposition above we know any penalty satisfies Assumption 3.10 and 3.13 admits such upper and lower bound. The function

$$(7) \quad \phi_\gamma(z) = \frac{1}{\gamma} \log(1 + \gamma z)$$

for $\gamma > 0$, and its convex combination with the ℓ^1 norm (the blend $\phi_{\gamma,\theta}$ made explicit in Section 5) will be the main choice for the numerical experiments.

Next we incorporate the nonconvex penalty in the integral representation. For the atom parts it is defined in the discrete setting. For the continuous part on Ω the measure is penalized with the weight $\phi'(0)$. Under Assumption 3.10 it is the L^1 norm of measure. Concretely, using the decomposition $\mu = \mu_{\text{cont}} + \mu_{\text{atom}}$ of Section 2.1, we define the penalty term

$$(8) \quad \Phi_1(\mu) = \|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega \in \text{atom } \mu} \phi(|\mu|(\{\omega\})).$$

We formulate the optimization problem as

$$(P_{\Phi_1}) \quad \min_{\mu \in \mathcal{M}(\Omega)} J(\mu) := L(\mu) + \alpha \Phi_1(\mu).$$

3.5. Existence of the global minimizer. In this section we prove the existence of the global minimizer. We leverage the direct method of the calculus of variations. Since we are working on the space of measures $\mathcal{M}(\Omega)$, we consider the weak-* convergence. As usual, the difficulty is to prove the weak-* lower semicontinuity (l.s.c.). Furthermore, the coercivity is compromised by introducing the nonconvex penalty term Φ . The next lemma addresses this problem: under Assumptions 3.10 and 3.11 on ϕ , the coercivity of Φ_1 is preserved.

Lemma 3.16 (On the coercivity of Φ_1). *Under Assumptions 3.10 and 3.11, for any $\mu \in \mathcal{M}(\Omega)$ it holds*

$$\phi(\|\mu\|_{\mathcal{M}(\Omega)}) \leq \Phi_1(\mu) \leq \|\mu\|_{\mathcal{M}(\Omega)}.$$

In particular, the functional Φ_1 is coercive.

Proof. The proof relies on the subadditivity and $\phi(z) \leq z$ from Lemma 3.12; note that the subadditivity extends from two summands to countably many by induction and the continuity of ϕ .

$$\begin{aligned} \phi(\|\mu\|_{\mathcal{M}(\Omega)}) &= \phi\left(|\mu|(\Omega \setminus \text{atom } \mu) + \sum_{\omega \in \text{atom } \mu} |\mu|(\{\omega\})\right), \\ (\text{subadditivity}) \quad &\leq \phi(|\mu|(\Omega \setminus \text{atom } \mu)) + \sum_{\omega \in \text{atom } \mu} \phi(|\mu|(\{\omega\})), \end{aligned}$$

$$\begin{aligned}
(\phi(z) \leq z) &\leq |\mu|(\Omega \setminus \text{atom } \mu) + \sum_{\omega \in \text{atom } \mu} \phi(|\mu|(\{\omega\})), \\
&= \Phi_1(\mu).
\end{aligned}$$

In particular, by the coercivity of ϕ it holds that Φ_1 is also coercive.

For the second inequality, we apply $\phi(z) \leq z$ to every atom and use that the continuous part enters Φ_1 with weight one; this gives $\Phi_1(\mu) \leq \|\mu\|_{\mathcal{M}(\Omega)}$. $\square$

The next lemma proves the weak-* l.s.c. of Φ_1 . It is a special case of the lower-semicontinuity theory of [5] for nonconvex functionals on measures, which is developed on locally compact domains and so applies to $\Omega = \mathbb{R}^{d+1}$. Since Φ_1 only depends on the atomic and continuous parts of the measure, no activation assumption enters. Only Assumptions 3.10 and 3.11 are needed — in particular not the local concavity of Assumption 3.13.

Lemma 3.17 (weak-* lower semicontinuity of Φ_1). *Under Assumptions 3.10 and 3.11, the functional*

$$\Phi_1(\mu) = \|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega \in \text{atom } \mu} \phi(|\mu|(\{\omega\}))$$

is sequentially weak- l.s.c. on $\mathcal{M}(\Omega)$.*

Proof. Let $\mu_h \xrightarrow{*} \mu$; then $M := \sup_h \|\mu_h\|_{\mathcal{M}(\Omega)} < \infty$ by the uniform boundedness principle. Passing to a subsequence we may assume $\lim_h \Phi_1(\mu_h) = \liminf_h \Phi_1(\mu_h) =: \ell < \infty$ (otherwise there is nothing to prove). Fix $\epsilon > 0$. We decompose each element of the sequence — not the limit — into its ϵ -heavy atomic part ζ_h and the rest η_h :

$$\mu_h = \zeta_h + \eta_h, \quad \zeta_h := \mu_h|_{A_h}, \quad A_h := \{\omega \in \Omega : |\mu_h|(\{\omega\}) \geq \epsilon\},$$

so that η_h collects the atoms of mass below ϵ together with the continuous part, and, the two parts living on disjoint atom sets,

$$\Phi_1(\mu_h) = \Phi_1(\zeta_h) + \Phi_1(\eta_h).$$

Since $\|\mu_h\|_{\mathcal{M}(\Omega)} \leq M$, the set A_h contains at most $k := \lfloor M/\epsilon \rfloor$ points. Both parts are bounded by M in variation, and $C_0(\Omega)$ is separable for $\Omega = \mathbb{R}^{d+1}$; hence, after a further extraction,

$$\zeta_h \xrightarrow{*} \zeta, \quad \eta_h \xrightarrow{*} \eta, \quad \zeta + \eta = \mu.$$

We estimate the two parts separately.

Heavy part:

Each ζ_h is purely atomic with at most k atoms. By lemma 3.6 in [5], the set of measures with at most k atoms is sequentially weak-* closed — hence ζ is purely atomic with at most k atoms — and on this set the functional $\mu' \mapsto \sum_{\omega} \phi(|\mu'|(\{\omega\}))$ is sequentially weak-* l.s.c., since ϕ is continuous and subadditive (Lemma 3.12). Therefore

$$\liminf_h \Phi_1(\zeta_h) = \liminf_h \sum_{\omega \in \text{atom } \zeta_h} \phi(|\zeta_h|(\{\omega\})) \geq \sum_{\omega \in \text{atom } \zeta} \phi(|\zeta|(\{\omega\})) = \Phi_1(\zeta).$$

Light part:

Since ϕ is concave with $\phi(0) = 0$, it holds for all $z \in [0, \epsilon]$,

$$\phi(z) \geq \frac{\phi(\epsilon)}{\epsilon} z =: (1 - \delta_\epsilon) z, \quad \text{where } \delta_\epsilon = 1 - \frac{\phi(\epsilon)}{\epsilon} \rightarrow 1 - \phi'(0) = 0 \text{ as } \epsilon \rightarrow 0$$

by Assumption 3.10. Every atom of η_h has mass below ϵ , and the continuous part enters Φ_1 with weight one, so

$$\begin{aligned}\Phi_1(\eta_h) &= \|\eta_{h,\text{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega \in \text{atom } \eta_h} \phi(|\eta_h|(\{\omega\})) \\ &\geq (1 - \delta_\epsilon) \left(\|\eta_{h,\text{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega \in \text{atom } \eta_h} |\eta_h|(\{\omega\}) \right) = (1 - \delta_\epsilon) \|\eta_h\|_{\mathcal{M}(\Omega)}.\end{aligned}$$

By the weak-* l.s.c. of the norm $\|\cdot\|_{\mathcal{M}(\Omega)}$ and by $\Phi_1 \leq \|\cdot\|_{\mathcal{M}(\Omega)}$ (Lemma 3.16),

$$\liminf_h \Phi_1(\eta_h) \geq (1 - \delta_\epsilon) \|\eta\|_{\mathcal{M}(\Omega)} \geq (1 - \delta_\epsilon) \Phi_1(\eta).$$

The functional Φ_1 is subadditive: for $\mu = \zeta + \eta$ with ζ purely atomic we have $\mu_{\text{cont}} = \eta_{\text{cont}}$, every atom of μ is an atom of ζ or of η , and $\phi(|\zeta + \eta|(\{\omega\})) \leq \phi(|\zeta|(\{\omega\})) + \phi(|\eta|(\{\omega\}))$ by the monotonicity and subadditivity of ϕ ; hence $\Phi_1(\mu) \leq \Phi_1(\zeta) + \Phi_1(\eta)$. Combining the two parts,

$$\ell = \lim_h \Phi_1(\mu_h) \geq \liminf_h \Phi_1(\zeta_h) + \liminf_h \Phi_1(\eta_h) \geq \Phi_1(\zeta) + (1 - \delta_\epsilon) \Phi_1(\eta) \geq (1 - \delta_\epsilon) \Phi_1(\mu).$$

Since $\epsilon > 0$ was arbitrary and $\delta_\epsilon \rightarrow 0$ as $\epsilon \rightarrow 0$ — the selections above may depend on ϵ , but the conclusion does not — we obtain $\ell \geq \Phi_1(\mu)$. $\square$

On the unbounded parameter domain the difficulty beyond the lower semicontinuity of Φ_1 is the behavior of the fidelity term under mass escaping to infinity. Under Assumption 3.2 the value atoms of the dictionary vanish in $L^2(D)$ as the parameter tends to infinity (Lemma 3.3), and this is exactly what rules the escape out.

Theorem 3.18 (Existence on the unbounded parameter domain). *Let Assumption 3.2 and Assumptions 3.10, 3.11 hold. Then problem (P_{Φ_1}) admits at least one global minimizer in $\mathcal{M}(\Omega)$.*

Proof. Since $V \in H^1(D)$ we have $\inf J \leq J(0) < \infty$. Let (μ_n) be a minimizing sequence with $J(\mu_n) \leq C$. By Lemma 3.16, $\phi(\|\mu_n\|_{\mathcal{M}(\Omega)}) \leq \Phi_1(\mu_n) \leq C/\alpha$, and the coercivity of ϕ yields $\sup_n \|\mu_n\|_{\mathcal{M}(\Omega)} < \infty$. As $C_0(\Omega)$ is separable, bounded sets of $\mathcal{M}(\Omega)$ are sequentially weak-* compact; passing to a subsequence we have $\mu_n \xrightarrow{*} \bar{\mu}$.

By Lemma 3.3, $\mathcal{N}\mu_n \rightharpoonup \mathcal{N}\bar{\mu}$ weakly in $L^2(D)$. Moreover $J(\mu_n) \leq C$ implies $\mathcal{N}\mu_n \in H^1(D)$ with $\|\nabla(\mathcal{N}\mu_n)\|_{L^2(D)} \leq \|\nabla V\|_{L^2(D)} + \sqrt{2C}$, passing to a subsequence again we have $\nabla(\mathcal{N}\mu_n) \rightharpoonup g$ weakly for some g in $L^2(D)^d$. For every $\psi \in C_c^\infty(D)^d$,

$$\int_D g \cdot \psi \, dx = \lim_n \int_D \nabla(\mathcal{N}\mu_n) \cdot \psi \, dx = - \lim_n \int_D \mathcal{N}\mu_n \operatorname{div} \psi \, dx = - \int_D \mathcal{N}\bar{\mu} \operatorname{div} \psi \, dx,$$

hence $g = \nabla(\mathcal{N}\bar{\mu})$ in the sense of distributions, and $\mathcal{N}\bar{\mu} \in H^1(D)$. The two fidelity terms are weakly lower semicontinuous norms evaluated at the weak limits $\mathcal{N}\bar{\mu} - V$ and $\nabla(\mathcal{N}\bar{\mu}) - \nabla V$, and Φ_1 is weak-* lower semicontinuous by Lemma 3.17, whose proof requires only local compactness of Ω . Therefore $J(\bar{\mu}) \leq \liminf_n J(\mu_n) = \inf J$. $\square$

3.6. Optimality conditions and finite support.

Theorem 3.19 (the optimality condition). *Under Assumptions 3.9, 3.10, 3.11, if $\bar{\mu}$ is a local minimum of J with $L(\bar{\mu}) < \infty$, then, writing $\bar{p}(\omega) := \partial L(\bar{\mu})[\delta_\omega] = \langle r, \sigma(\cdot, \omega) \rangle_{L^2(D)} + \langle \nabla r, \nabla_x \sigma(\cdot, \omega) \rangle_{L^2(D)}$ with $r := \mathcal{N}\bar{\mu} - V$, it holds*

- (1) $|\bar{p}(\omega)| \leq \alpha$ for all $\omega \in \Omega$,
- (2) $\bar{p}(\omega) = -\alpha \phi'(|\bar{\mu}|(\{\omega\})) \operatorname{sign}(\bar{\mu})(\omega)$, for $|\bar{\mu}|$ -almost all $\omega \in \Omega$,

where the sign density $\operatorname{sign}(\bar{\mu}) : \Omega \rightarrow \{-1, 1\}$ is defined $|\bar{\mu}|$ -almost everywhere by the Hahn decomposition of $\bar{\mu}$.

Proof. We use the directional derivative $\partial L(\bar{\mu})[u]$, not a duality pairing — $\bar{p}$ need not lie in the predual $C_0(\Omega)$. For a single atom, $\partial L(\bar{\mu})[\delta_\omega] = \bar{p}(\omega)$ since $\mathcal{N}\delta_\omega = \sigma(\cdot, \omega)$. Since $\bar{\mu}$ is a local minimizer, for every fixed $u \in \mathcal{M}(\Omega)$ and all sufficiently small $\tau > 0$ (depending on u):

$$J(\bar{\mu} + \tau u) - J(\bar{\mu}) = L(\bar{\mu} + \tau u) - L(\bar{\mu}) + \alpha \Phi_1(\bar{\mu} + \tau u) - \alpha \Phi_1(\bar{\mu}) \geq 0.$$

Dividing by τ and letting $\tau \rightarrow 0^+$ along any direction u with $\mathcal{N}u \in H^1(D)$ (in particular the directions $u = \pm \delta_\omega$ and $u = \pm \bar{\mu}_{\text{cont}}^R$ used below), the fidelity difference converges to $\partial L(\bar{\mu})[u]$, so

$$-\partial L(\bar{\mu})[u] \leq \alpha \lim_{\tau \rightarrow 0^+} \frac{1}{\tau} (\Phi_1(\bar{\mu} + \tau u) - \Phi_1(\bar{\mu})),$$

whenever the right-hand side limit exists. Now we separate two cases.

1) For the atom part: let $\omega \in \text{atom } \bar{\mu}$ with $c := \bar{\mu}(\{\omega\}) \neq 0$. Taking $u = \delta_\omega$ and using that ϕ is differentiable at $|c| > 0$,

$$-\bar{p}(\omega) = -\partial L(\bar{\mu})[\delta_\omega] \leq \alpha \lim_{\tau \rightarrow 0^+} \frac{1}{\tau} (\phi(|c| + \tau) - \phi(|c|)) = \alpha \phi'(|c|) \text{sign}(c).$$

Repeating the argument with $u = -\delta_\omega$ yields the reverse inequality, hence $\bar{p}(\omega) = -\alpha \phi'(|c|) \text{sign}(c)$. Since $\phi'(0) = 1$ and ϕ' is nonincreasing and nonnegative, we have $\phi'(|c|) \in [0, 1]$, hence $|\bar{p}(\omega)| \leq \alpha$ at every atom.

2) For the remaining points: let $\omega \notin \text{atom } \bar{\mu}$ and perturb with $u = \delta_\omega$; then

$$-\bar{p}(\omega) = -\partial L(\bar{\mu})[\delta_\omega] \leq \alpha \lim_{\tau \rightarrow 0^+} \frac{1}{\tau} (\phi(\tau) - \phi(0)) = \alpha \phi'(0) = \alpha.$$

Taking $u = -\delta_\omega$, the mass at ω is again τ , so the penalty still contributes $\phi(\tau)$; this gives $\bar{p}(\omega) \leq \alpha$, and with the above $|\bar{p}(\omega)| \leq \alpha$ for all $\omega \notin \text{atom } \bar{\mu}$, which together with case 1) proves claim (1).

The continuous part cannot be inserted directly: $\int_\Omega \|\nabla_x \sigma(\cdot, \omega)\|_{L^2(D)} d|\bar{\mu}_{\text{cont}}|$ may be infinite, in which case $\mathcal{N}\bar{\mu}_{\text{cont}} \notin H^1(D)$, and this cannot be excluded a priori — the atomic part may carry infinitely many atoms, and $\mathcal{N}\bar{\mu} \in H^1(D)$ does not split over the summands. We therefore truncate. For $R > 0$ set $\bar{\mu}_{\text{cont}}^R := \bar{\mu}_{\text{cont}}|_{\{|\omega| \leq R\}}$. On the bounded set $\{|\omega| \leq R\}$ the gradient kernel is uniformly bounded, $\|\nabla_x \sigma(\cdot, \omega)\|_{L^2(D)} \leq C_R$, so $\int_\Omega \|\nabla_x \sigma(\cdot, \omega)\|_{L^2(D)} d|\bar{\mu}_{\text{cont}}^R| \leq C_R \|\bar{\mu}_{\text{cont}}^R\|_{\mathcal{M}(\Omega)} < \infty$. Hence $\omega \mapsto \nabla_x \sigma(\cdot, \omega)$ is Bochner integrable against $\bar{\mu}_{\text{cont}}^R$, so $\mathcal{N}\bar{\mu}_{\text{cont}}^R \in H^1(D)$ by itself; consequently $\mathcal{N}(\bar{\mu} \pm \tau \bar{\mu}_{\text{cont}}^R) \in H^1(D)$, the directions $\pm \bar{\mu}_{\text{cont}}^R$ are admissible, and $\partial L(\bar{\mu})[\bar{\mu}_{\text{cont}}^R] = \int_{\{|\omega| \leq R\}} \bar{p} d\bar{\mu}_{\text{cont}}$. Since perturbing by $\pm \bar{\mu}_{\text{cont}}^R$ only rescales the continuous part on $\{|\omega| \leq R\}$, we have $\Phi_1(\bar{\mu} \pm \tau \bar{\mu}_{\text{cont}}^R) - \Phi_1(\bar{\mu}) = \pm \tau \|\bar{\mu}_{\text{cont}}^R\|_{\mathcal{M}(\Omega)}$, and the variational inequality in both signs gives

$$\int_{\{|\omega| \leq R\}} \left(-\bar{p}(\omega) \text{sign}(\bar{\mu}_{\text{cont}})(\omega) - \alpha \right) d|\bar{\mu}_{\text{cont}}|(\omega) = 0.$$

The integrand is nonpositive by claim (1), hence it vanishes $|\bar{\mu}_{\text{cont}}|$ -almost everywhere on $\{|\omega| \leq R\}$; letting $R \rightarrow \infty$ gives $\bar{p} = -\alpha \text{sign}(\bar{\mu}_{\text{cont}})$ for $|\bar{\mu}_{\text{cont}}|$ -almost all ω . Together with case 1) this proves claim (2). $\square$

The theorem gives a necessary optimality condition for the local minimizer. But on the other hand, it lays the ground to prove that the local optimizer admits finite support. If the sample size is finite, then the optimal solution admits finite support by the Carathéodory's theorem. The following theorem proves that the optimal solution admits the finite support even though the dataset is infinitely large. In contrast to [23] the parameter domain is not assumed compact; the price is the explicit no-escape condition (H) on the dual variable at infinity, for which Remark 3.8 provides checkable sufficient conditions under the localization of Assumption 3.2. Note that Theorem 3.18 and the present theorem do not compose automatically: the existence theorem does not assert that its minimizer satisfies (H), and on domains with flat boundary faces — such as the boxes of Section 6 — the condition must be verified for the residual at hand (Remark 3.8).

Theorem 3.20 (finite support). *Let Assumptions 3.10, 3.11, 3.13 and Assumption 3.9 hold. Let $\bar{\mu}$ be a local minimizer of (P_{Φ_1}) with $L(\bar{\mu}) < \infty$ satisfying the no-escape condition*

$$(H) \quad \limsup_{|\omega| \rightarrow \infty} |\bar{p}(\omega)| < \alpha.$$

Then $\bar{\mu}$ is supported on finitely many points, $\bar{\mu} = \sum_{n=1}^N \bar{c}_n \delta_{\bar{\omega}_n}$. On a compact parameter set Ω_R the condition (H) is void and the conclusion holds unconditionally.

Proof. 1. $\text{cont } \bar{\mu} = \emptyset$: Argue by contradiction and assume $\bar{\mu}_{\text{cont}} \neq 0$; without loss of generality the positive part of the Hahn decomposition $\bar{\mu} = \bar{\mu}^+ - \bar{\mu}^-$ has a nontrivial continuous part $\bar{\mu}_{\text{cont}}^+ \neq 0$ (otherwise argue with $\bar{\mu}^-$ and reversed signs). By Theorem 3.19, $\bar{p} = -\alpha$ holds $\bar{\mu}_{\text{cont}}^+$ -almost everywhere, and the atoms of $\bar{\mu}$ form a $\bar{\mu}_{\text{cont}}^+$ -null set; we may therefore choose

$$\hat{\omega} \in \text{supp } \bar{\mu}_{\text{cont}}^+ \quad \text{with} \quad \bar{p}(\hat{\omega}) = -\alpha, \quad \bar{\mu}(\{\hat{\omega}\}) = 0.$$

For $\delta \in (0, 1]$ set $D_\delta := B_\delta(\hat{\omega})$ and $m_\delta := \bar{\mu}_{\text{cont}}^+(D_\delta)$; then $m_\delta > 0$ for every δ since $\hat{\omega} \in \text{supp } \bar{\mu}_{\text{cont}}^+$, and $m_\delta \rightarrow \bar{\mu}_{\text{cont}}^+(\{\hat{\omega}\}) = 0$ as $\delta \rightarrow 0$. We merge *only the continuous positive mass* of the ball onto its center:

$$\mu_\delta = \bar{\mu} - \bar{\mu}_{\text{cont}}^+|_{D_\delta} + m_\delta \delta_{\hat{\omega}}.$$

Note $\|\mu_\delta - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq 2m_\delta \rightarrow 0$, so μ_δ is an admissible competitor for the local minimizer once δ is small. We prove that it decreases the cost.

The perturbation $\mu_\delta - \bar{\mu}$ is supported in a compact subset of Ω , on which $\bar{p}$ is continuous (by Assumption 3.9 and dominated convergence) and hence bounded, so every $\int_\Omega \bar{p} d(\mu_\delta - \bar{\mu})$ below is an ordinary integral, not the $\mathcal{M}$ - C_0 duality pairing (which is unavailable, as $\bar{p}$ need not lie in $C_0(\Omega)$). The fidelity term is quadratic, therefore the second order Taylor expansion is exact:

$$L(\mu_\delta) = L(\bar{\mu}) + \int_\Omega \bar{p} d(\mu_\delta - \bar{\mu}) + \frac{1}{2} \|\mathcal{N}[\mu_\delta - \bar{\mu}]\|_{L^2(D)}^2 + \frac{1}{2} \sum_{i=1}^d \|\partial_{x_i} \mathcal{N}[\mu_\delta - \bar{\mu}]\|_{L^2(D)}^2.$$

The moved measure $\bar{\mu}_{\text{cont}}^+|_{D_\delta}$ is nonnegative with total mass m_δ , and $\sigma(x, \cdot)$, $\partial_{x_i} \sigma(x, \cdot)$ are Lipschitz on the compact ball $\overline{B_1(\hat{\omega})}$ with a constant of the form $\Lambda_{\hat{\omega}}(1 + \|x\|)$ (the compact- ω Lipschitz bounds of Assumption 3.9). Hence

$$|\mathcal{N}[\mu_\delta - \bar{\mu}](x)| = \left| \int_{D_\delta} (\sigma(x, \hat{\omega}) - \sigma(x, \omega)) d\bar{\mu}_{\text{cont}}^+(\omega) \right| \leq \delta \Lambda_{\hat{\omega}}(1 + \|x\|) m_\delta,$$

and analogously $|\partial_{x_i} \mathcal{N}[\mu_\delta - \bar{\mu}](x)| \leq \delta \Lambda_{\hat{\omega}}(1 + \|x\|) m_\delta$. Define

$$\Lambda_1^2(D, \Lambda_{\hat{\omega}}) = \Lambda_{\hat{\omega}}^2 \int_D (1 + \|x\|)^2 dx.$$

Hence

$$L(\mu_\delta) \leq L(\bar{\mu}) + \int_\Omega \bar{p} d(\mu_\delta - \bar{\mu}) + \frac{d+1}{2} \delta^2 \Lambda_1^2 m_\delta^2.$$

For the first order term, $\bar{p} = -\alpha$ holds $\bar{\mu}_{\text{cont}}^+$ -almost everywhere on D_δ and $\bar{p}(\hat{\omega}) = -\alpha$, so it vanishes exactly:

$$\int_\Omega \bar{p} d(\mu_\delta - \bar{\mu}) = m_\delta \bar{p}(\hat{\omega}) - \int_{D_\delta} \bar{p} d\bar{\mu}_{\text{cont}}^+ = -\alpha m_\delta + \alpha m_\delta = 0.$$

Concerning the penalty term, the moved mass is continuous — it enters Φ_1 with weight one — and creates the single new atom $m_\delta \delta_{\hat{\omega}}$ (recall $\bar{\mu}(\{\hat{\omega}\}) = 0$). For δ small enough that $m_\delta \leq \hat{z}_2$,

$$\begin{aligned} \Phi_1(\mu_\delta) &= \Phi_1(\bar{\mu}) - m_\delta + \phi(m_\delta), \\ &= \Phi_1(\bar{\mu}) + \int_0^{m_\delta} \phi'(\xi) - \phi'(0) d\xi, \end{aligned}$$

$$(\text{Assumption 3.10, 3.13(2)}) \leq \Phi_1(\bar{\mu}) - \frac{\gamma_2}{2} m_\delta^2.$$

Combining the two parts we get

$$J(\mu_\delta) \leq J(\bar{\mu}) - \left(\frac{\alpha\gamma_2 - (d+1)\delta^2\Lambda_1^2}{2} \right) m_\delta^2.$$

Choosing $\delta^2 < \frac{\alpha\gamma_2}{(d+1)\Lambda_1^2(D, \Lambda_\omega)}$ gives $J(\mu_\delta) < J(\bar{\mu})$, contradicting local minimality. Hence $\bar{\mu}_{\text{cont}} = 0$.

2. $|\text{atom } \bar{\mu}|$ is finite: First, the atoms cannot escape to infinity: if there were atoms ω_n with $|\omega_n| \rightarrow \infty$, then $|\bar{\mu}|(\{\omega_n\}) \rightarrow 0$ by the boundedness of the total variation, and Theorem 3.19 together with $\phi'(0) = 1$ would give

$$|\bar{p}(\omega_n)| = \alpha \phi'(|\bar{\mu}|(\{\omega_n\})) \rightarrow \alpha,$$

contradicting the no-escape condition (H). Hence atom $\bar{\mu}$ is contained in a compact subset of Ω .

Now assume by contradiction that atom $\bar{\mu}$ is infinite. Then there exists a sequence of distinct atoms with $\omega_n \rightarrow \hat{\omega}$ for some $\hat{\omega} \in \Omega$; after passing to a subsequence, infinitely many of the weights share one sign, w.l.o.g. $c_n := \bar{\mu}(\{\omega_n\}) > 0$. By the boundedness of the total variation it holds $\lim_{n \rightarrow \infty} c_n = 0$, and by Theorem 3.19 and the continuity of $\bar{p}$,

$$\bar{p}(\hat{\omega}) = \lim_{n \rightarrow \infty} \bar{p}(\omega_n) = -\alpha \lim_{n \rightarrow \infty} \phi'(c_n) = -\alpha.$$

Concentrate the tail of the sequence at the limit $\hat{\omega}$: for $N \in \mathbb{N}$ define

$$\mu_N := \bar{\mu} - \sum_{i=N}^{\infty} c_i \delta_{\omega_i} + C_N \delta_{\hat{\omega}}, \quad C_N := \sum_{i=N}^{\infty} c_i > 0, \quad \delta_N := \sup_{i \geq N} |\omega_i - \hat{\omega}| \rightarrow 0,$$

so that $\|\mu_N - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq 2C_N \rightarrow 0$. As in step 1 (with the local Lipschitz constant Λ_ω ; again $\mu_N - \bar{\mu}$ is compactly supported, since atom $\bar{\mu}$ lies in a compact set, so $\int_\Omega \bar{p} d(\mu_N - \bar{\mu})$ is an ordinary integral),

$$L(\mu_N) \leq L(\bar{\mu}) + \int_\Omega \bar{p} d(\mu_N - \bar{\mu}) + \frac{d+1}{2} \Lambda_1^2(D, \Lambda_\omega) \delta_N^2 C_N^2.$$

For the first order term we have by the optimality condition and the value $\bar{p}(\hat{\omega}) = -\alpha$:

$$\int_\Omega \bar{p} d(\mu_N - \bar{\mu}) = - \sum_{i=N}^{\infty} \bar{p}(\omega_i) c_i + C_N \bar{p}(\hat{\omega}) = \alpha \sum_{i=N}^{\infty} c_i \phi'(c_i) - \alpha C_N.$$

For the penalty term, by the monotonicity and subadditivity of ϕ (Lemma 3.12) — which also covers the case that $\hat{\omega}$ is itself an atom of $\bar{\mu}$, since $\phi(|\bar{\mu}(\{\hat{\omega}\})| + C_N) - \phi(|\bar{\mu}(\{\hat{\omega}\})|) \leq \phi(C_N)$ —

$$\Phi_1(\mu_N) \leq \Phi_1(\bar{\mu}) - \sum_{i=N}^{\infty} \phi(c_i) + \phi(C_N).$$

Combining the estimates, it holds

$$J(\mu_N) \leq J(\bar{\mu}) - \alpha \sum_{i=N}^{\infty} \left(\phi(c_i) - c_i \phi'(c_i) \right) - \alpha \left(C_N - \phi(C_N) \right) + \frac{d+1}{2} \Lambda_1^2(D, \Lambda_\omega) \delta_N^2 C_N^2.$$

By concavity of ϕ it holds

$$\phi(c_i) - c_i \phi'(c_i) \geq \phi(0) = 0.$$

For N large enough that $C_N \leq \hat{z}_2$, by Assumption 3.10 and 3.13(2) it holds

$$C_N - \phi(C_N) = C_N \phi'(0) - \left(\phi(0) + \int_0^{C_N} \phi'(\xi) d\xi \right) = - \int_0^{C_N} \phi'(\xi) - \phi'(0) d\xi \geq \frac{\gamma_2}{2} C_N^2.$$

Thus

$$J(\mu_N) \leq J(\bar{\mu}) - \frac{1}{2} \left(\alpha\gamma_2 - (d+1)\Lambda_1^2(D, \Lambda_\omega) \delta_N^2 \right) C_N^2.$$

Since $\delta_N \rightarrow 0$, choosing N large enough that additionally

$$\delta_N^2 < \frac{\alpha\gamma_2}{\Lambda_1^2(D, \Lambda_{\hat{\omega}})(d+1)}$$

gives $J(\mu_N) < J(\bar{\mu})$ with $\|\mu_N - \bar{\mu}\|_{\mathcal{M}(\Omega)}$ arbitrarily small, contradicting local minimality. Hence atom $\bar{\mu}$ is finite, and together with step 1 the claim follows. $\square$

Remark 3.21 (failure of no escape on a flat face). Condition (H) can fail because an escaping ridge remains visible on a flat boundary face. Let $D = (0, 1)^d$ with $d \geq 2$, let $F = \{x \in \partial D : x_1 = 0\}$, and take the localized profile $\rho(z) = e^{-z^2}$. Set

$$r(x) = \cosh(1 - x_1).$$

Then $r \in H^2(D)$ and $\Delta r = r$, so the interior term $\langle r - \Delta r, \sigma(\cdot, \omega) \rangle_{L^2(D)}$ vanishes. The boundary representation of Lemma 3.6 gives

$$p_r(a, b) = \int_{\partial D} \partial_n r(x) \rho(a \cdot x + b) d\mathcal{H}^{d-1}(x).$$

On F the outward normal is $-e_1$, so $\partial_n r = \sinh(1)$; on the other faces $\partial_n r = 0$. Along the escaping sequence

$$a_j = t_j e_1, \quad b_j = 0, \quad t_j \rightarrow \infty,$$

one has $a_j \cdot x + b_j = 0$ on F , and therefore

$$p_r(a_j, b_j) = \sinh(1) \rho(0) \mathcal{H}^{d-1}(F) = \sinh(1).$$

Thus the dual variable does not decay along this escaping sequence, and the quantity controlled by (H) equals $\sinh(1)$ for this dual field. This example does not assert that the displayed r is the residual of a local minimizer; it isolates the boundary-flux mechanism that (H) rules out. Corollary 3.22 shows that, in the H^2 regime, this normal-to-a-flat-face escape is the only way an escaping sequence can keep the dual variable from vanishing.

Corollary 3.22 (escaping directions in the H^2 regime). *Assume the hypotheses of Lemma 3.6 and Assumption 3.2. Let $\omega_j = (a_j, b_j)$ satisfy $|\omega_j| \rightarrow \infty$ and*

$$\limsup_{j \rightarrow \infty} |p_r(\omega_j)| > 0.$$

Then, after passing to a subsequence,

$$|a_j| \rightarrow \infty, \quad \frac{a_j}{|a_j|} \rightarrow e, \quad \frac{b_j}{|a_j|} \rightarrow -c$$

for some $|e| = 1$ and $c \in \mathbb{R}$ such that

$$\mathcal{H}^{d-1}(\partial D \cap \{x : e \cdot x = c\}) > 0.$$

In particular, if the positive-measure section above is a flat face $F \subset \{x : n_F \cdot x = c_F\}$, then the escaping sequence is asymptotically normal to that face: up to changing the sign of the normal,

$$\frac{a_j}{|a_j|} \rightarrow n_F, \quad \frac{b_j}{|a_j|} \rightarrow -c_F.$$

Proof. Choose a subsequence, not relabeled, and $\epsilon > 0$ such that $|p_r(\omega_j)| \geq \epsilon$. By Lemma 3.6(1),

$$|p_r(\omega_j)| \leq (\|r\|_{L^2(D)} + \|\Delta r\|_{L^2(D)}) \kappa(\omega_j) + \|\partial_n r\|_{L^2(\partial D)} \kappa_{\partial}(\omega_j),$$

where

$$\kappa(\omega) := \|\sigma(\cdot, \omega)\|_{L^2(D)}, \quad \kappa_{\partial}(\omega) := \|\sigma(\cdot, \omega)\|_{L^2(\partial D)}.$$

Lemma 3.3 gives $\kappa(\omega_j) \rightarrow 0$.

If $|a_j|$ stayed bounded along a further subsequence, then $|b_j| \rightarrow \infty$, and since ∂D is bounded, $|a_j \cdot x + b_j| \rightarrow \infty$ uniformly on ∂D . Hence $\kappa_{\partial}(\omega_j) \rightarrow 0$, contradicting $|p_r(\omega_j)| \geq \epsilon$. Therefore

$|a_j| \rightarrow \infty$ along the non-vanishing subsequence. Passing to a further subsequence, $a_j/|a_j| \rightarrow e$ with $|e| = 1$.

If $|b_j|/|a_j|$ were unbounded, then again $|a_j \cdot x + b_j| \rightarrow \infty$ uniformly on ∂D , so $\kappa_{\partial}(\omega_j) \rightarrow 0$, a contradiction. Thus, after passing to a further subsequence, $b_j/|a_j| \rightarrow -c$ for some $c \in \mathbb{R}$. Let $\Pi := \{x : e \cdot x = c\}$.

If $\mathcal{H}^{d-1}(\partial D \cap \Pi) = 0$, we show directly that $\kappa_{\partial}(\omega_j) \rightarrow 0$. For $\tau > 0$ set

$$B_j(\tau) := \{x \in \partial D : |a_j \cdot x + b_j| \leq \tau\}, \quad \eta(\tau) := \sup_{|z| \geq \tau} |\rho(z)|.$$

Then

$$\kappa_{\partial}(\omega_j)^2 \leq \|\rho\|_{\infty}^2 \mathcal{H}^{d-1}(B_j(\tau)) + \eta(\tau)^2 \mathcal{H}^{d-1}(\partial D).$$

For $x \in B_j(\tau)$,

$$|e \cdot x - c| \leq \frac{|a_j \cdot x + b_j|}{|a_j|} + \left|e - \frac{a_j}{|a_j|}\right| R_D + \left|c + \frac{b_j}{|a_j|}\right| \rightarrow 0$$

uniformly in x , where $R_D > 0$ satisfies $|x| \leq R_D$ on ∂D . Hence, for every $\varepsilon > 0$, eventually

$$B_j(\tau) \subset \{x \in \partial D : |e \cdot x - c| \leq \varepsilon\}.$$

By continuity from above of $\mathcal{H}^{d-1} \llcorner \partial D$,

$$\limsup_j \mathcal{H}^{d-1}(B_j(\tau)) \leq \mathcal{H}^{d-1}(\partial D \cap \Pi) = 0.$$

Therefore

$$\limsup_j \kappa_{\partial}(\omega_j)^2 \leq \eta(\tau)^2 \mathcal{H}^{d-1}(\partial D)$$

for every $\tau > 0$, and letting $\tau \rightarrow \infty$ gives $\kappa_{\partial}(\omega_j) \rightarrow 0$. Together with $\kappa(\omega_j) \rightarrow 0$, this would imply $p_r(\omega_j) \rightarrow 0$, contradicting the choice of the subsequence. Hence $\mathcal{H}^{d-1}(\partial D \cap \Pi) > 0$.

The last assertion follows because a hyperplane containing a flat face has normal $\pm n_F$ and offset $\pm c_F$, with the signs linked as above. $\square$

Now we have a necessary optimality condition. But it is not sufficient. More importantly, we cannot formulate an algorithm to determine the optimal value of N . To overcome this difficulty, we apply a heuristic approach: for fixed inner weights $\bar{\omega} = (\bar{\omega}_n)_{n=1}^N$, the finite-dimensional problem

$$(9) \quad J_{\bar{\omega}}(c) := L\left(\sum_{n=1}^N c_n \delta_{\bar{\omega}_n}\right) + \alpha \sum_{n=1}^N \phi(|c_n|), \quad c \in \mathbb{R}^N,$$

is amenable to gradient-based optimization. Now we need to find a way to heuristically determine the optimal $\bar{\omega}$ and the optimal number N . To this end, we introduce the following theorem as a sufficient condition.

For the analysis it is convenient to collect the two components of the fidelity term in the product space

$$\mathcal{Y} := L^2(D) \times L^2(D)^d, \quad \mathcal{T}\mu := (\mathcal{N}\mu, \nabla(\mathcal{N}\mu)), \quad \mathcal{V} := (V, \nabla V),$$

so that $L(\mu) = \frac{1}{2} \|\mathcal{T}\mu - \mathcal{V}\|_{\mathcal{Y}}^2$ whenever $\mathcal{N}\mu \in H^1(D)$, and the dual variable reads $\bar{p}(\omega) = \langle \mathcal{T}\bar{\mu} - \mathcal{V}, (\sigma(\cdot, \omega), \nabla_x \sigma(\cdot, \omega)) \rangle_{\mathcal{Y}}$.

Theorem 3.23 (sufficient condition for local optimality). *Let ϕ satisfy Assumptions 3.10, 3.11, 3.13, let σ satisfy Assumptions 3.2 and 3.9 with $\rho \in C^2(\mathbb{R})$, and let $V \in H^2(D)$. Let $\bar{\mu} := \sum_{n=1}^N \bar{c}_n \delta_{\bar{\omega}_n}$ with distinct $\bar{\omega}_n$ and $\bar{c}_n \neq 0$ be such that*

- (1) $\bar{c} \in (\mathbb{R} \setminus \{0\})^N$ is a local minimum of $J_{\bar{\omega}}$,
- (2) $|\bar{p}(\omega)| < \alpha$ for all $\omega \in \Omega \setminus \{\bar{\omega}_n\}_{n=1, \dots, N}$,
- (3) the no-escape condition (H) holds. (By Remark 3.8 and Lemma 3.7, this is automatic when ∂D meets every hyperplane in an $\mathcal{H}^{d-1}$ -null set, and void on a compact parameter set Ω_R .)

Then $\bar{\mu}$ is a local minimizer of J : there exists $\epsilon > 0$ such that $J(\bar{\mu}) \leq J(\mu)$ for all $\mu \in \mathcal{M}(\Omega)$ with $\|\mu - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq \epsilon$.

Proof. Throughout, $\bar{r} := \mathcal{N}\bar{\mu} - V \in H^2(D)$ (a finite sum of $C^2(\bar{D})$ atoms plus $V \in H^2(D)$), and $\bar{p} = p_{\bar{r}}$ in the notation of Lemma 3.6.

Step 1 (stationarity at the atoms). Since $\bar{c}_n \neq 0$ and ϕ is differentiable, $J_{\bar{\omega}}$ is differentiable at $\bar{c}$, and hypothesis (1) gives $\partial_{c_n} J_{\bar{\omega}}(\bar{c}) = 0$, i.e. $\bar{p}(\bar{\omega}_n) = -\alpha\phi'(|\bar{c}_n|) \text{sign}(\bar{c}_n)$. By Assumption 3.13(2) and the monotonicity of ϕ' it holds $\phi'(|\bar{c}_n|) < \phi'(0) = 1$, hence $|\bar{p}(\bar{\omega}_n)| < \alpha$.

Step 2 (uniform dual gap). By Step 1 and hypothesis (2), $|\bar{p}(\omega)| < \alpha$ for every $\omega \in \Omega$; $\bar{p}$ is continuous (Lemma 3.6(1)); and by (H) there is $R > 0$ with $\sup_{|\omega| > R} |\bar{p}(\omega)| < \alpha$. On the compact ball $\{|\omega| \leq R\}$ the continuous function $|\bar{p}|$ attains its maximum, which is $< \alpha$. Hence there exists $\delta \in (0, 1)$ with

$$\sup_{\omega \in \Omega} |\bar{p}(\omega)| \leq (1 - \delta)\alpha.$$

Step 3 (decomposition of a competitor). By hypothesis (1) there is $\epsilon_1 > 0$ with $J_{\bar{\omega}}(c) \geq J_{\bar{\omega}}(\bar{c})$ for all $\|c - \bar{c}\|_{\ell^1} \leq \epsilon_1$. Fix

$$\epsilon := \min \left\{ \epsilon_1, \hat{z}_1, \frac{\alpha\delta}{C_{\bar{\omega}} + \alpha\gamma_1/2} \right\}, \quad C_{\bar{\omega}} := C_{D,\rho} \max_{n=1,\dots,N} \|\sigma(\cdot, \bar{\omega}_n)\|_{H^2(D)},$$

with $C_{D,\rho}$ from Lemma 3.6(1). Let $\mu \in \mathcal{M}(\Omega)$ with $\|\mu - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq \epsilon$. If $\mathcal{N}\mu \notin H^1(D)$ then $J(\mu) = +\infty$ and there is nothing to prove; so assume $\mathcal{N}\mu \in H^1(D)$ and decompose

$$\mu = \mu_0 + \tilde{\mu}, \quad \mu_0 := \sum_{n=1}^N c_n \delta_{\bar{\omega}_n}, \quad c_n := \mu(\{\bar{\omega}_n\}),$$

so that $\tilde{\mu} := \mu - \mu_0$ carries no mass at the support points, and

$$\|c - \bar{c}\|_{\ell^1} + \|\tilde{\mu}\|_{\mathcal{M}(\Omega)} = \|\mu - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq \epsilon.$$

Note $\mathcal{N}\mu_0 \in C^2(\bar{D})$ and hence $\mathcal{N}\tilde{\mu} = \mathcal{N}\mu - \mathcal{N}\mu_0 \in H^1(D)$.

Step 4 (expansion of the fidelity term). Expanding the square in $\mathcal{Y}$ and discarding $\frac{1}{2}\|\mathcal{T}\tilde{\mu}\|_{\mathcal{Y}}^2 \geq 0$,

$$L(\mu) = L(\mu_0) + \langle \mathcal{T}\mu_0 - \mathcal{V}, \mathcal{T}\tilde{\mu} \rangle_{\mathcal{Y}} + \frac{1}{2}\|\mathcal{T}\tilde{\mu}\|_{\mathcal{Y}}^2 \geq L(\mu_0) + \langle \mathcal{T}\tilde{\mu} - \mathcal{V}, \mathcal{T}\tilde{\mu} \rangle_{\mathcal{Y}} + \langle \mathcal{T}(\mu_0 - \bar{\mu}), \mathcal{T}\tilde{\mu} \rangle_{\mathcal{Y}}.$$

By Lemma 3.6(2) with $r = \bar{r}$ and $\mu' = \tilde{\mu}$,

$$\langle \mathcal{T}\tilde{\mu} - \mathcal{V}, \mathcal{T}\tilde{\mu} \rangle_{\mathcal{Y}} = \int_{\Omega} \bar{p} d\tilde{\mu}.$$

For the second pairing, $r' := \mathcal{N}(\mu_0 - \bar{\mu}) = \sum_n (c_n - \bar{c}_n) \sigma(\cdot, \bar{\omega}_n) \in H^2(D)$ with $\|r'\|_{H^2(D)} \leq \max_n \|\sigma(\cdot, \bar{\omega}_n)\|_{H^2(D)} \|c - \bar{c}\|_{\ell^1}$, so Lemma 3.6(1)–(2) give

$$|\langle \mathcal{T}(\mu_0 - \bar{\mu}), \mathcal{T}\tilde{\mu} \rangle_{\mathcal{Y}}| = \left| \int_{\Omega} p_{r'} d\tilde{\mu} \right| \leq C_{D,\rho} \|r'\|_{H^2(D)} \|\tilde{\mu}\|_{\mathcal{M}(\Omega)} \leq C_{\bar{\omega}} \epsilon \|\tilde{\mu}\|_{\mathcal{M}(\Omega)}.$$

Step 5 (penalty split and lower bound). The atoms of μ_0 and $\tilde{\mu}$ are disjoint, so $\Phi_1(\mu) = \sum_{n=1}^N \phi(|c_n|) + \Phi_1(\tilde{\mu})$. Every atom of $\tilde{\mu}$ has mass at most $\|\tilde{\mu}\|_{\mathcal{M}(\Omega)} \leq \epsilon \leq \hat{z}_1$, so by Proposition 3.15(1),

$$\Phi_1(\tilde{\mu}) \geq \left(1 - \frac{\gamma_1 \epsilon}{2}\right) \|\tilde{\mu}\|_{\mathcal{M}(\Omega)},$$

while Step 2 gives $\int_{\Omega} \bar{p} d\tilde{\mu} \geq -(1 - \delta)\alpha \|\tilde{\mu}\|_{\mathcal{M}(\Omega)}$.

Step 6 (assembly). Combining Steps 4–5 and $L(\mu_0) + \alpha \sum_n \phi(|c_n|) = J_{\bar{\omega}}(c)$,

$$\begin{aligned} J(\mu) &\geq J_{\bar{\omega}}(c) + \int_{\Omega} \bar{p} d\tilde{\mu} + \alpha \Phi_1(\tilde{\mu}) - C_{\bar{\omega}} \epsilon \|\tilde{\mu}\|_{\mathcal{M}(\Omega)} \\ &\geq J_{\bar{\omega}}(c) + \left(\alpha\delta - \frac{\alpha\gamma_1 \epsilon}{2} - C_{\bar{\omega}} \epsilon \right) \|\tilde{\mu}\|_{\mathcal{M}(\Omega)} \geq J_{\bar{\omega}}(c), \end{aligned}$$

by the choice of ϵ . Since $\|c - \bar{c}\|_{\ell^1} \leq \epsilon \leq \epsilon_1$, hypothesis (1) gives $J_{\bar{\omega}}(c) \geq J_{\bar{\omega}}(\bar{c}) = J(\bar{\mu})$, which completes the proof. $\square$

4. THE k -HOMOGENEOUS FORMULATION

In this section, we discuss a specific class of activation function: the k -homogeneous activation function.

Assumption 4.1 (k -homogeneity). σ is positively k -homogeneous with respect to ω for some $k \geq 1$, i.e., for all $x \in D$, $\lambda > 0$ it holds

$$\sigma(x, \lambda\omega) = \lambda^k \sigma(x, \omega) \quad \text{for all } \omega \in \mathbb{R}^{d+1}.$$

Remark 4.2. For the gradient training we use $k \geq 2$: the ReLU^k atom is then C^1 in x , so the gradient-augmented loss is well defined, and for $k > 2$ the inner optimization is even twice continuously differentiable. The derivative of a ReLU^k ridge is, up to the factor k , a ReLU^{k-1} ridge: the value term of the loss sees atoms of order k while the gradient term sees atoms of order $k - 1$. In the borderline case $k = 2$ the problem is still well defined, but $\partial_{x_i} \sigma$ has a non-differentiable kink, as in the ReLU case. For $k < 2$ the Lipschitz continuity of the gradient of the activation fails. The algebraic results of this section (Propositions 4.3 and 4.5) hold for every $k \geq 1$.

Proposition 4.3 (Generalization of [23, Proposition 3] to k -homogeneous activations). *Let $p \geq 1$, $q \geq 1$, and let $r: \mathbb{R}^{d+1} \rightarrow \mathbb{R}_+$ be a positive-definite, positively 1-homogeneous functional (a gauge; any norm qualifies):*

$$r(\tau\omega) = |\tau| r(\omega) \quad \text{for all } \omega \in \mathbb{R}^{d+1}, \tau \in \mathbb{R}, \quad r(\omega) > 0 \quad \text{for } \omega \neq 0.$$

Suppose the activation function $\sigma(x, \omega)$ is positively k -homogeneous in ω , i.e.,

$$\sigma(x, \tau\omega) = \tau^k \sigma(x, \omega) \quad \text{for all } \tau > 0, \omega \in \mathbb{R}^{d+1}, x \in D.$$

Then the unconstrained problem

$$\min_{N \in \mathbb{N}, (c_n, \omega_n) \in \mathbb{R} \times \mathbb{R}^{d+1}} l(\mathcal{N}_{\omega, c}; V) + \alpha \sum_{n=1}^N \left[\frac{1}{p} |c_n|^p + \frac{1}{q} r(\omega_n)^q \right],$$

and the sphere-constrained problem

$$\min_{N \in \mathbb{N}, \{c_n\} \in \mathbb{R}, \{\tilde{\omega}_n: r(\tilde{\omega}_n)=1\}} l(\mathcal{N}_{\tilde{\omega}, c}; V) + \frac{2\alpha}{s} k^{-kp/(kp+q)} \sum_{n=1}^N |c_n|^{s/2},$$

where

$$s = \frac{2pq}{kp + q},$$

are equivalent.

Remark 4.4. For $p = q = 2$ we get $s = 4/(k+1)$, so the exponent is $s/2 = 2/(k+1)$ and the constant is $2/s \cdot k^{-k/(k+1)} = \frac{k+1}{2} k^{-k/(k+1)}$: the sphere penalty is

$$\alpha \frac{k+1}{2} k^{-k/(k+1)} \sum_{n=1}^N |c_n|^{2/(k+1)}.$$

For $k = 1$ this gives $\alpha \sum |c_n|$, recovering [23, Proposition 3]; for $k = 2$ it gives $\frac{3}{2} \cdot 2^{-2/3} \alpha \sum |c_n|^{2/3}$. In the numerical sections the penalty is implemented in the plain form $\alpha \sum_n |c_n|^{2/(k+1)}$: the constant $C_k := \frac{k+1}{2} k^{-k/(k+1)}$ is absorbed into an effective, k -dependent regularization strength $C_k \alpha$. A nominal α therefore does not represent the same effective regularization across different powers k , and the power sweeps of Section 6 compare at equal nominal α .

Proof. Since σ is positively k -homogeneous, for any $\tau > 0$ we have

$$c_n \sigma(x, \omega_n) = c_n \sigma(x, \tau_n \tilde{\omega}_n) = c_n \tau_n^k \sigma(x, \tilde{\omega}_n),$$

so $\mathcal{N}_{\omega, c} = \mathcal{N}_{\tilde{\omega}, \tilde{c}}$ where $\tilde{\omega}_n = \omega_n / \tau_n$, $\tilde{c}_n = c_n \tau_n^k$, for any choice of $\tau_n > 0$. In particular, the loss term $l(\mathcal{N}_{\omega, c}; V)$ is invariant under this reparametrization. Atoms with $\omega_n = 0$ contribute nothing to the network ($\sigma(\cdot, 0) = 0$ by k -homogeneity) and only pay penalty, so we may discard them without increasing the cost. For each remaining $\omega_n \neq 0$, write $\omega_n = \tau_n \tilde{\omega}_n$ with $\tau_n = r(\omega_n) > 0$ (positive definiteness of r) and $r(\tilde{\omega}_n) = 1$ (by 1-homogeneity). Then the unconstrained problem is equivalent to

$$\min_{\substack{N \in \mathbb{N}, \{c_n\} \in \mathbb{R}^N \\ \{\tilde{\omega}_n: r(\tilde{\omega}_n)=1\}, \tau_n > 0}} l(\mathcal{N}_{\tilde{\omega}, \tilde{c}}; V) + \alpha \sum_{n=1}^N \left[\frac{1}{p} \frac{|\tilde{c}_n|^p}{\tau_n^{kp}} + \frac{1}{q} \tau_n^q \right],$$

For fixed $\tilde{c}_n \neq 0$, define

$$g(\tau) = \frac{1}{p} |\tilde{c}_n|^p \tau^{-kp} + \frac{1}{q} \tau^q.$$

Since $g(\tau) \rightarrow +\infty$ as $\tau \rightarrow 0^+$ and $\tau \rightarrow +\infty$, the minimum exists and is unique. Differentiating:

$$g'(\tau) = -k |\tilde{c}_n|^p \tau^{-kp-1} + \tau^{q-1} = 0 \iff \tau^{kp+q} = k |\tilde{c}_n|^p.$$

Hence

$$\tau_n^* = (k |\tilde{c}_n|^p)^{1/(kp+q)}.$$

Substituting τ_n^* into g :

First term:

$$\frac{1}{p} |\tilde{c}_n|^p \cdot (k |\tilde{c}_n|^p)^{-kp/(kp+q)} = \frac{k^{-kp/(kp+q)}}{p} |\tilde{c}_n|^{p - kp^2/(kp+q)} = \frac{k^{-kp/(kp+q)}}{p} |\tilde{c}_n|^{pq/(kp+q)}.$$

Here we used $p - kp^2/(kp+q) = p(kp+q - kp)/(kp+q) = pq/(kp+q)$.

Second term:

$$\frac{1}{q} (k |\tilde{c}_n|^p)^{q/(kp+q)} = \frac{k^{q/(kp+q)}}{q} |\tilde{c}_n|^{pq/(kp+q)}.$$

Adding the two terms and noting that $pq/(kp+q) = s/2$:

$$g(\tau_n^*) = \left(\frac{k^{-kp/(kp+q)}}{p} + \frac{k^{q/(kp+q)}}{q} \right) |\tilde{c}_n|^{s/2}.$$

We simplify the coefficient. Factor out $k^{q/(kp+q)}$:

$$\frac{k^{-kp/(kp+q)}}{p} + \frac{k^{q/(kp+q)}}{q} = k^{q/(kp+q)} \left(\frac{k^{-1}}{p} + \frac{1}{q} \right) = k^{q/(kp+q)} \frac{q + kp}{kpq}.$$

Since $s = 2pq/(kp+q)$, we have $(kp+q)/(kpq) = 2/(ks)$, so

$$g(\tau_n^*) = \frac{2}{s} k^{q/(kp+q)} k^{-1} |\tilde{c}_n|^{s/2} = \frac{2}{s} k^{q/(kp+q)-1} |\tilde{c}_n|^{s/2} = \frac{2}{s} k^{-kp/(kp+q)} |\tilde{c}_n|^{s/2}.$$

(using $q/(kp+q) - 1 = -kp/(kp+q)$). $\square$

For the ReLU ^{k} activation function with $k \geq 2$, the reparametrization induces the fractional-exponent penalty

$$\phi_k(z) := |z|^{2/(k+1)}$$

(the constant C_k of Proposition 4.3 being absorbed into α , cf. the remark there). Since $2/(k+1) < 1$, this penalty fails Assumption 3.10: $\phi'_k(0^+) = \infty$. In this section we no longer use the measure

functional Φ_1 from Section 3. The fractional-exponent penalty is used only in the discrete, finitely atomic formulation

$$\mu = \sum_{n=1}^N c_n \delta_{\omega_n}, \quad N < \infty,$$

with penalty

$$\Phi_k(\mu) = \sum_{n=1}^N \phi_k(|c_n|).$$

Continuous measure components are therefore not part of the admissible class. The loss in (10) is the empirical l_M ; every statement of this section holds verbatim with the population fidelity L of Section 3 in its place (a finite network always satisfies (B)), and the two closing corollaries are stated in the population setting for consistency with Section 3. The optimization problem becomes:

$$(10) \quad \min_{N \in \mathbb{N}, \|a_n\|^2 + |b_n|^2 \leq 1, c_n \in \mathbb{R}, n=1, \dots, N} l_M(\mathcal{N}_{\omega, c}; V) + \alpha \sum_{n=1}^N \phi_k(|c_n|).$$

The following proposition states that any optimal solution of (10) can be projected onto the unit sphere without increasing the cost. Hence we may constrain the optimization problem to the unit sphere.

Proposition 4.5 (homogeneous reformulation). *Assume ϕ_k is nondecreasing with $\phi_k(0) = 0$ (in particular, Assumption 3.11 without the differentiability at zero suffices), assume (10) admits at least one global minimizer, and let σ be k -homogeneous with $k \geq 1$. Then (10) is equivalent to*

$$(P_{\phi_k}) \quad \min_{N \in \mathbb{N}, \|a_n\|^2 + |b_n|^2 = 1, c_n \in \mathbb{R}, n=1, \dots, N} l_M(\mathcal{N}_{\omega, c}; V) + \alpha \sum_{n=1}^N \phi_k(|c_n|).$$

Proof. By assumption the optimal solution exists; denote $(\hat{N}, (\hat{a}_n, \hat{b}_n, \hat{c}_n)_{n=1}^{\hat{N}})$ one of the global minimizers. The optimal value of the first part in the fidelity term is

$$\frac{1}{2M} \sum_{m=1}^M \left| \left(\sum_{n=1}^{\hat{N}} \hat{c}_n \sigma(\hat{a}_n \cdot x_m + \hat{b}_n) \right) - V(x_m) \right|^2.$$

Neurons with $(\hat{a}_n, \hat{b}_n) = 0$ satisfy $\sigma(x, 0) = 0$ by k -homogeneity and can be discarded without changing the network or increasing the penalty. For the remaining $(\hat{a}_n, \hat{b}_n)$ in the interior of the unit ball with $0 < \|\hat{a}_n\|^2 + |\hat{b}_n|^2 < 1$, denote $\lambda_n = \frac{1}{\sqrt{\|\hat{a}_n\|^2 + |\hat{b}_n|^2}} > 1$. We project $(\hat{a}_n, \hat{b}_n)$ onto the sphere by defining

$$(\tilde{a}_n, \tilde{b}_n) = \lambda_n (\hat{a}_n, \hat{b}_n).$$

Rescaling the outer weights by defining $\tilde{c}_n = \frac{1}{\lambda_n^k} \hat{c}_n$, by k -homogeneity of the activation the function value stays the same:

$$\begin{aligned} & \frac{1}{2M} \sum_{m=1}^M \left| \left(\sum_{n=1}^{\hat{N}} \tilde{c}_n \sigma(\tilde{a}_n \cdot x_m + \tilde{b}_n) \right) - V(x_m) \right|^2 \\ &= \frac{1}{2M} \sum_{m=1}^M \left| \left(\sum_{n=1}^{\hat{N}} \hat{c}_n \frac{1}{\lambda_n^k} \lambda_n^k \sigma(\hat{a}_n \cdot x_m + \hat{b}_n) \right) - V(x_m) \right|^2 \\ &= \frac{1}{2M} \sum_{m=1}^M \left| \left(\sum_{n=1}^{\hat{N}} \hat{c}_n \sigma(\hat{a}_n \cdot x_m + \hat{b}_n) \right) - V(x_m) \right|^2. \end{aligned}$$

The fidelity is invariant with respect to the scaling. A similar result holds for the second part of the fidelity term by linearity

$$\tilde{c}_n \nabla_x \sigma(\tilde{a}_n \cdot x + \tilde{b}_n) = \hat{c}_n \frac{1}{\lambda_n^k} \nabla_x \left(\lambda_n^k \sigma(\hat{a}_n \cdot x + \hat{b}_n) \right) = \hat{c}_n \nabla_x \sigma(\hat{a}_n \cdot x + \hat{b}_n).$$

On the other hand, the penalty term is nondecreasing by Assumption 3.11. Since $\lambda_n > 1$ and $k \geq 1$, it holds $|\tilde{c}_n| = |\hat{c}_n|/\lambda_n^k \leq |\hat{c}_n|$, hence

$$\phi_k(|\tilde{c}_n|) \leq \phi_k(|\hat{c}_n|).$$

The pair $\left(\tilde{N}, (\tilde{a}_n, \tilde{b}_n, \tilde{c}_n)_{n=1}^{\tilde{N}} \right)$ with

$$\tilde{N} = \hat{N} \in \mathbb{N}, \quad (\tilde{a}_n, \tilde{b}_n) \in \mathbb{S}^d, \quad \tilde{c}_n \in \mathbb{R}, \quad n = 1, \dots, \tilde{N},$$

is feasible for the sphere-constrained problem and yields a value no larger than the optimum of (10). Conversely, every sphere-feasible point is feasible for (10), so the two optimal values coincide; in particular the projected point is a global minimizer of (P_{ϕ_k}) , and the rescaling maps minimizers of one problem to minimizers of the other, which is the meaning of equivalence here. $\square$

The ReLU^k function. First we consider the ReLU function with the power k :

$$\sigma(x, \omega) = \max\{0, a \cdot x + b\}^k, \quad \omega = (a, b), \quad k \in [2, \infty).$$

The power k is a real number: the homogeneity analysis of this section only uses $k \geq 1$, while the integer-order statements of Section 2.2 ($B_1^k \subset H^k$) apply to integer k . Since these activations are not polynomials, the corresponding networks are dense in $C(K)$ on compact sets K [14]. In this case the gradient is

$$\nabla_x \sigma(x, \omega) = k \max\{0, a \cdot x + b\}^{k-1} a \in \mathbb{R}^d.$$

The activation function is non-decreasing and positive homogeneous of degree k . For the standard ReLU activation ($k = 1$) the network lies in H^1 but is not C^1 : the gradient-augmented loss is well defined — first derivatives exist weakly — but the derivatives of the atom with respect to its inner parameters, which the candidate search and the semismooth Newton step differentiate, are discontinuous, and second derivatives exist only as distributions. Choosing σ as a ReLU^k activation with $k > 1$ smooths the atom in x and in the parameters; for integer k , the Barron embedding of Section 2.2 in addition controls derivatives up to order k .

For $k > 1$, the activation function not only preserves the homogeneity of ReLU but also smooths the kink. The differentiability buys two things: smooth inner-parameter optimization for the candidate search and the Newton step, and a classical (rather than almost-everywhere) gradient in the pointwise evaluations of the gradient-augmented loss.

In the absence of Assumption 3.10, it is difficult to prove the existence of a global minimizer with the direct method of the calculus of variations. For example, for the penalty $\phi(c) = |c|^q$ with $q < 1$ we have $\phi'(0^+) = \infty$, and the weak-* lower semicontinuity of the penalized functional fails. However, assuming the local optimizer exists, we can still derive the necessary optimality condition.

Corollary 4.6 (Optimality conditions for $\phi'(0^+) = +\infty$). *Suppose ϕ satisfies Assumption 3.11 but with $\phi'(0^+) = +\infty$ in place of Assumption 3.10 (e.g., $\phi(z) = \frac{1}{q} z^q$ for $0 < q < 1$). Suppose $\sigma(\cdot, \omega) \in C^1(\overline{D})$ for every ω with $(x, \omega) \mapsto (\sigma, \nabla_x \sigma)$ continuous on $\overline{D} \times \mathbb{S}^d$ (satisfied by ReLU^k with $k \geq 2$), and let $V \in H^1(D)$. Let $(\bar{\omega}, \bar{c}) \in (\mathbb{S}^d)^N \times \mathbb{R}^N$, with $\bar{c}_n \neq 0$ for all n , be such that the outer weights are locally optimal for the population objective*

$$J(\bar{\omega}, \bar{c}) := \frac{1}{2} \|\mathcal{N}_{\bar{\omega}, \bar{c}} - V\|_{L^2(D)}^2 + \frac{1}{2} \|\nabla \mathcal{N}_{\bar{\omega}, \bar{c}} - \nabla V\|_{L^2(D)}^2 + \alpha \sum_{n=1}^N \phi(|\bar{c}_n|),$$

i.e. $c \mapsto J(\bar{\omega}, c)$ has a local minimum at $\bar{c}$ (this holds in particular at any global minimizer of (P_{ϕ_k}) after discarding zero-weight neurons). Write, as in Section 3, $\bar{p}(\omega) := \langle r, \sigma(\cdot, \omega) \rangle_{L^2(D)} + \langle \nabla r, \nabla_x \sigma(\cdot, \omega) \rangle_{L^2(D)}$ with $r := \mathcal{N}_{\bar{\omega}, \bar{c}} - V$. Then:

(1) Stationarity at each atom:

$$\bar{p}(\bar{\omega}_n) = -\alpha \phi'(|\bar{c}_n|) \operatorname{sign}(\bar{c}_n), \quad n = 1, \dots, N.$$

(2) Lower bound on atom weights: $\bar{p}$ is continuous on the compact sphere, so $\|\bar{p}\|_{C(\mathbb{S}^d)} < \infty$, and (1) gives $\phi'(|\bar{c}_n|) = |\bar{p}(\bar{\omega}_n)|/\alpha \leq \|\bar{p}\|_{C(\mathbb{S}^d)}/\alpha$ for every n . Since ϕ' is nonincreasing on $(0, \infty)$ (by concavity) with $\phi'(0^+) = +\infty$, the generalized inverse $(\phi')^{-1}(t) := \inf\{z > 0 : \phi'(z) \leq t\}$ is positive for every finite t , and

$$|\bar{c}_n| \geq (\phi')^{-1}\left(\frac{\|\bar{p}\|_{C(\mathbb{S}^d)}}{\alpha}\right) =: c_{\min} > 0, \quad n = 1, \dots, N.$$

In particular the width is controlled by the objective value: $N\phi(c_{\min}) \leq J(\bar{\omega}, \bar{c})/\alpha$.

Proof. (1) Perturb one outer weight: for $s = \pm 1$ and $\tau > 0$ small enough, local optimality gives $J(\bar{\omega}, \bar{c} + \tau s e_n) - J(\bar{\omega}, \bar{c}) \geq 0$. The fidelity term is smooth in c_n with partial derivative $\bar{p}(\bar{\omega}_n)$ at $\bar{c}$, so dividing by τ and letting $\tau \rightarrow 0^+$,

$$-s \bar{p}(\bar{\omega}_n) \leq \alpha \lim_{\tau \rightarrow 0^+} \frac{1}{\tau} [\phi(|\bar{c}_n + \tau s|) - \phi(|\bar{c}_n|)] = s \alpha \phi'(|\bar{c}_n|) \operatorname{sign}(\bar{c}_n);$$

the two signs together yield equality. Note $\phi'(|\bar{c}_n|) < \infty$ for $\bar{c}_n \neq 0$, so this step does not require $\phi'(0) < \infty$.

(2) Continuity of $\bar{p}$ on $\mathbb{S}^d$ follows from the continuity of $(\sigma, \nabla_x \sigma)$ on $\bar{D} \times \mathbb{S}^d$ by dominated convergence (the atoms are uniformly bounded there and $r \in H^1(D)$), hence $\|\bar{p}\|_{C(\mathbb{S}^d)} < \infty$. By (1), $\phi'(|\bar{c}_n|) \leq t^* := \|\bar{p}\|_{C(\mathbb{S}^d)}/\alpha$, i.e. $|\bar{c}_n|$ lies in $\{z > 0 : \phi'(z) \leq t^*\}$. Since ϕ' is nonincreasing, this set is an interval unbounded to the right, so $|\bar{c}_n| \geq \inf\{z > 0 : \phi'(z) \leq t^*\} = (\phi')^{-1}(t^*)$; and since $\phi'(0^+) = +\infty$ there is $z_0 > 0$ with $\phi'(z) > t^*$ on $(0, z_0)$, whence $c_{\min} = (\phi')^{-1}(t^*) \geq z_0 > 0$. Finally, $|\bar{c}_n| \geq c_{\min}$ and ϕ nondecreasing give $N\phi(c_{\min}) \leq \sum_n \phi(|\bar{c}_n|) \leq J(\bar{\omega}, \bar{c})/\alpha$. $\square$

Remark 4.7. Compared with Theorem 3.19:

- (1) The global dual bound $|\bar{p}(\omega)| \leq \alpha$ for all ω is *lost*: adding a fresh neuron at $\omega \in \mathbb{S}^d$ with outer weight $\pm\tau$ changes the objective by $\pm\tau \bar{p}(\omega) + \alpha \phi(\tau) + o(\tau)$, and since $\phi(\tau)/\tau \rightarrow \phi'(0^+) = +\infty$, the resulting inequality is vacuous. No constraint on $\bar{p}$ is obtained away from the support.
- (2) In exchange, the singularity $\phi'(0^+) = +\infty$ provides an *automatic minimum atom size* via (2), and with it an a priori width bound (Corollary 4.8) — without the merging argument of Theorem 3.20 or the local strong concavity of Assumption 3.13. In the finite-network class, finiteness of the support holds by definition; the content here is the quantitative bound.
- (3) The zero network ($c = 0$, at any width) is always locally optimal in the outer weights: giving any neuron a small weight $|c| \ll 1$ incurs a marginal penalty $\phi(|c|)/|c| \rightarrow +\infty$ that dominates the attainable decrease of the loss. This means the node insertion criterion $|\bar{p}(\omega)| > \alpha$ of the greedy algorithm no longer applies.

Corollary 4.8 (Existence of minimizer for (P_{ϕ_k}) with $\phi'_k(0^+) = +\infty$). *Suppose ϕ_k satisfies Assumption 3.11 on $(0, \infty)$ with $\phi'_k(0^+) = +\infty$; suppose $\sigma(\cdot, \omega) \in C^1(\bar{D})$ for every ω with $(x, \omega) \mapsto (\sigma, \nabla_x \sigma)$ continuous on $\bar{D} \times \mathbb{S}^d$ (satisfied by ReLU^k with $k \geq 2$); and let D be bounded and $V \in H^1(D)$ with $V \neq 0$ (for $V = 0$ the zero network is optimal). Then the discrete problem (P_{ϕ_k}) , posed with the population fidelity, admits a global minimizer of width at most $\lfloor N_{\max} \rfloor$, where*

$$N_{\max} = \frac{\|V\|_{H^1(D)}^2}{2\alpha\phi_k(c_{\min})}, \quad c_{\min} = (\phi'_k)^{-1}\left(\frac{\|V\|_{H^1(D)} \sup_{\omega \in \mathbb{S}^d} \|\sigma(\cdot, \omega)\|_{H^1(D)}}{\alpha}\right),$$

and $(\phi'_k)^{-1}$ is the generalized inverse introduced in Corollary 4.6(2).

Proof. 1. (Fixed N , existence by Weierstrass). For fixed N , the objective is a real-valued function on $(\mathbb{S}^d)^N \times \mathbb{R}^N$:

$$J(\omega, c) = \frac{1}{2} \int_D \left(\left| V(x) - \sum_{n=1}^N c_n \sigma(a_n \cdot x + b_n) \right|^2 + \sum_{i=1}^d \left| \partial_{x_i} V(x) - \sum_{n=1}^N c_n \partial_{x_i} \sigma(a_n \cdot x + b_n) \right|^2 \right) dx + \alpha \sum_{n=1}^N \phi_k(|c_n|).$$

We verify J is continuous in (ω, c) . By assumption both $\sigma(a \cdot x + b)$ and $\nabla_x \sigma(a \cdot x + b)$ are uniformly bounded on the compact set $\bar{D} \times \mathbb{S}^d$ (D bounded). For any convergent sequence $(\omega^{(j)}, c^{(j)}) \rightarrow (\omega, c)$, the integrand converges pointwise in x and is dominated by $C(|V(x)|^2 + |\nabla V(x)|^2 + 1) \in L^1(D)$ with some constant C . By dominated convergence, the integral is continuous in (ω, c) . Combined with the continuity of ϕ_k , J is continuous.

Since $\phi_k(z) \rightarrow \infty$ as $z \rightarrow \infty$ (Assumption 3.11) and $J \geq 0$, any minimizing sequence is bounded. Furthermore, since taking $c = 0$ is feasible and the penalty is non-negative, the optimal value is bounded by

$$J(\omega, 0) = \frac{1}{2} \|V\|_{H^1(D)}^2 \quad \text{for any } \omega,$$

so along a minimizing sequence eventually

$$\sum_{n=1}^N \phi_k(|c_n|) \leq \frac{1}{2\alpha} \|V\|_{H^1(D)}^2 + 1.$$

By the coercivity of ϕ_k , the minimizing sequence lies in $(\mathbb{S}^d)^N \times [-R, R]^N$ for some $R > 0$, which is compact. By Weierstrass, J attains its minimum for each N .

2. (Uniform bound on N). Let (ω^*, c^*) be a minimizer for N neurons with all $c_n^* \neq 0$. Corollary 4.6(1) gives $|p(\omega_n^*)| = \alpha \phi'_k(|c_n^*|)$, where

$$p(\omega) = \langle \mathcal{N}_{\omega^*, c^*} - V, \sigma(\cdot, \omega) \rangle_{H^1(D)}$$

is the dual variable of Section 3 restricted to the sphere, with the $H^1(D)$ inner product collecting the value term and the gradient term. By Cauchy–Schwarz and $\|\mathcal{N}_{\omega^*, c^*} - V\|_{H^1(D)}^2 \leq 2J(\omega^*, c^*) \leq \|V\|_{H^1(D)}^2$:

$$\alpha \phi'_k(|c_n^*|) = |p(\omega_n^*)| \leq \|\mathcal{N}_{\omega^*, c^*} - V\|_{H^1(D)} \|\sigma(\cdot, \omega_n^*)\|_{H^1(D)} \leq \|V\|_{H^1(D)} \sup_{\omega \in \mathbb{S}^d} \|\sigma(\cdot, \omega)\|_{H^1(D)}.$$

Since ϕ'_k is nonincreasing with $\phi'_k(0^+) = +\infty$, the generalized inverse gives $|c_n^*| \geq c_{\min} > 0$ with $c_{\min}$ as in the statement. Hence $N \phi_k(c_{\min}) \leq \sum_n \phi_k(|c_n^*|) \leq \frac{1}{2\alpha} \|V\|_{H^1(D)}^2$, giving $N \leq N_{\max}$.

Hence any width- N minimizer with all weights nonzero satisfies $N \leq N_{\max}$, i.e. $N \leq \lfloor N_{\max} \rfloor$ since N is an integer; a minimizer with a zero weight reduces, after discarding zero-weight neurons, to a smaller-width solution with the same objective value. Therefore $\inf_{N \in \mathbb{N}_0} J_N^* = \min_{0 \leq N \leq \lfloor N_{\max} \rfloor} J_N^*$, where $J_0^* = \frac{1}{2} \|V\|_{H^1(D)}^2$ is the zero-network value, and the minimum is attained by Step 1. $\square$

5. ALGORITHMS

This section describes the computational pipeline: the training data are generated via the open-loop problem, candidate neurons are inserted greedily, and the network is trained with the nonconvex penalty; insertion and training alternate for a fixed number of outer iterations.

5.1. Data generation via the open-loop problem. We come back to the original problem (P), the problem can be written as

$$\begin{aligned} & \inf_{u,y} J(y,u) \\ & \text{s.t.} \\ & E(y,u) = 0, \end{aligned}$$

with

$$E(y,u) = \begin{bmatrix} \frac{d}{dt}y - \left(f(y) + g(y)u \right) \\ y(0) - x \end{bmatrix}.$$

Assume that for every control the state equation admits a unique solution $y(u)$; setting $U := L^2(t_0, T; \mathbb{R}^{m_u})$, we obtain the unconstrained reduced problem

$$\inf_{u \in U} \hat{J}(u) := J(y(u), u).$$

The reduced gradient is computed by the standard adjoint (Lagrangian) approach — see, e.g., [12] — which avoids forming the implicit derivative of $u \mapsto y(u)$: it yields

$$\nabla \hat{J}(u) = 2\beta u + g(y)^T p \quad \text{in } U,$$

where the costate p solves the adjoint equation backward in time,

$$\begin{cases} \dot{p} + \partial_y(f(y) + g(y)u)^T p = -\partial_y l(y), \\ p(T) = 0, \end{cases}$$

which is exactly the costate system of (TPBVP). Each gradient evaluation therefore costs one forward solve of the state equation and one backward solve of the adjoint equation. For the data-driven approach it is essential that the initial dataset $\{x^j, V(0, x^j), \nabla V(0, x^j)\}_{j=1}^M$ is *causality-free*: the samples are generated without approximate values at neighboring points. [13] provides a comprehensive survey of generation methods. As in [1], we minimize the reduced objective with Barzilai–Borwein step sizes, using the reduced gradient above; the converged costate supplies the gradient sample ∇V .

5.2. Greedy insertion. In [2], a conditional gradient algorithm for finitely many samples is presented. To handle the sphere constraint we optimize an ambient vector $w \in \mathbb{R}^{d+1}$ and evaluate the candidate objective at its normalization $w/\|w\|$, which turns the constrained search into an unconstrained one; [23] use an inverse stereographic chart of $\mathbb{S}^d$ for the same purpose. For activations that are not positively homogeneous, the atom is not determined by its direction alone, so the search additionally optimizes, per candidate direction, a radial scale $r = e^s$ with s clamped to $[-3, 5]$ (an effective range $r \in [e^{-3}, e^5] \approx [0.05, 148]$); the computational dictionary is therefore a compact subset of the parameter domain even for the saturating activations of Example 3.1. For M data points, each carrying a value and d gradient components, the measurement dimension is $M(d+1)$, and an optimal representation of the target can be taken supported on at most $M(d+1)$ atoms.

The conditional gradient method minimizes, at the current iterate μ_t with network $f_t := \mathcal{N}\mu_t$, the linearization of the fidelity term over the total-variation ball $\{\|\mu\|_{\mathcal{M}(\Omega)} \leq M_0\}$; since the extreme points of the ball are the signed Dirac measures $\pm M_0 \delta_\omega$, this “Frank–Wolfe step” reduces to maximizing the absolute dual variable over the dictionary. In the empirical setting the dual variable carries both channels of the gradient-augmented loss — the value–gradient pairing of the atom with the current residuals $r_t := f_t - V$ and $R_t := \nabla f_t - \nabla V$ at the samples:

$$(11) \quad \Omega \ni \omega \mapsto p_t(\omega) = \frac{1}{N_x} \sum_{m=1}^M \left(w_1 \sigma(x^m, \omega) r_t(x^m) + w_2 \nabla_x \sigma(x^m, \omega) \cdot R_t(x^m) \right),$$

with the channel weights (w_1, w_2) and normalization $N_x = Md$ of the training loss $\mathcal{L}$ defined in the next subsection; p_t is the empirical counterpart of the dual variable $\bar{p}$ of Section 3. Finding the global maximum of this landscape is in general intractable (for ReLU-type dictionaries even NP-hard [2]); heuristic multi-start approaches are used for the greedy insertion instead.

5.3. The nonconvex penalized algorithms. We present two variants of the iterative node insertion and optimization algorithm, adapted from [23]. The first handles penalties ϕ satisfying Assumption 3.10. The implemented family is the ℓ^1 -log blend

$$\phi_{\gamma,\theta}(z) = \theta z + \frac{(1-\theta)^2}{\gamma} \log \left(1 + \frac{\gamma z}{1-\theta} \right), \quad \theta \in [0, 1), \gamma \geq 0,$$

which satisfies $\phi_{\gamma,\theta}(0) = 0$ and $\phi'_{\gamma,\theta}(0) = 1$; the limit $\gamma \rightarrow 0$ (or $\theta \rightarrow 1$) is the pure ℓ^1 penalty. All experiments fix $\theta = \frac{1}{2}$, and we abbreviate $\phi_\gamma := \phi_{\gamma,1/2}$ — every “log penalty” result below is this mixture, and rows with $\gamma = 0$ are pure ℓ^1 . The insertion criterion is the necessary optimality condition from Theorem 3.19: a candidate ω is accepted if $|p_t(\omega)| > \alpha$. The second handles the fractional-exponent penalty $\psi_k(z) = |z|^q$ with $q = 2/(k+1) < 1$ — the penalty ϕ_k of Section 4, written ψ_k in the numerical sections to keep it apart from the log blend ϕ_γ at a glance — where $\psi'_k(0^+) = +\infty$ renders the classical criterion vacuous (cf. the Remark after Corollary 4.6). Instead, a finite-step decrease criterion is used.

Both algorithms minimize the regularized loss

$$\mathcal{L}(\omega, c) = w_1 \cdot \frac{1}{2N_x} \sum_{m=1}^M |V(x^m) - \mathcal{N}_{\omega,c}(x^m)|^2 + w_2 \cdot \frac{1}{2N_x} \sum_{m=1}^M |\nabla V(x^m) - \nabla \mathcal{N}_{\omega,c}(x^m)|^2 + \alpha \sum_{n=1}^N \phi(|c_n|),$$

where $N_x = M \cdot d$ is the total number of gradient entries. Relative to the loss l_M of (3), which normalizes both channels by M , the implemented $\mathcal{L}$ differs by the constant factor $1/d$; this is immaterial for the fidelity alone but rescales the effective regularization strength — α in the present convention corresponds to $d\alpha$ in the convention of (3). All reported values of α use the present convention. We consider two choices of the weights (w_1, w_2) :

- L^2 loss: $(w_1, w_2) = (1, 0)$, fitting only the value function;
- H^1 loss: $(w_1, w_2) = (1, 1)$, fitting both the value function and its gradient.

The relative errors used to evaluate the approximation quality are

$$\begin{aligned} \text{Err}_{L^2} &= \frac{\|\mathcal{N} - V\|_{L^2}}{\|V\|_{L^2}}, \quad |\mathcal{N} - V|_{H^1} = \frac{\|\nabla \mathcal{N} - \nabla V\|_{L^2}}{\|\nabla V\|_{L^2}}, \\ \text{Err}_{H^1} &= \frac{(\|\mathcal{N} - V\|_{L^2}^2 + \|\nabla \mathcal{N} - \nabla V\|_{L^2}^2)^{1/2}}{(\|V\|_{L^2}^2 + \|\nabla V\|_{L^2}^2)^{1/2}}, \end{aligned}$$

where $|\cdot|_{H^1}$ denotes the H^1 semi-norm (gradient error only) and the norms are computed empirically over the data set.

For the q -penalty with $q < 1$, the zero network is always a local minimizer, and the criterion $|p_t(\omega)| > \alpha$ carries no information away from the support (Remark after Corollary 4.6). We replace it with a *finite-step decrease criterion*: given a candidate ω , we evaluate the objective decrease

$$\Delta J(c; \omega) = c p_t(\omega) + \frac{1}{2} c^2 S_t(\omega)^2 + \frac{\alpha}{q} |c|^q,$$

where $S_t(\omega)^2 = \frac{1}{N_x} \sum_{m=1}^M (w_1 \sigma(x^m, \omega)^2 + w_2 |\nabla_x \sigma(x^m, \omega)|^2)$ is the curvature term (the squared empirical value-gradient norm of the atom), and accept ω if $\min_c \Delta J(c; \omega) < 0$. The minimizer c^* is computed by Newton’s method on the stationarity equation $S_t(\omega)^2 c + \alpha |c|^{q-1} = |p_t(\omega)|$, and is used directly as the initial outer weight. The penalty coefficient in ΔJ is α/q , whereas the subsequent training step minimizes $\mathcal{L}$ with $\phi = \psi_k$, i.e. the penalty $\alpha \sum_n |c_n|^q$; since $\alpha/q > \alpha$ for $q < 1$, the finite-step test is *conservative* — every accepted atom also strictly decreases the

Algorithm 1 Iterative node insertion and optimization with ϕ_γ -penalty

- 1: Initial network $\omega^{(0)}, c^{(0)}$ of width $N(0)$.
 - 2: **while** $t < T_{\text{out}}$ **do**
 - 3: Sample N_{trial} random nodes $\omega \in \mathbb{S}^d$.
 - 4: Optimize (in parallel) the dual variable $|p_t(\omega)|$ by L-BFGS starting from the initial nodes; merge near-duplicate candidates.
 - 5: Select nodes with $|p_t(\omega)| > \alpha$ and add them to the network (of width $N(t+1/2)$). Initialize their outer weights by coordinate descent.
 - 6: Perform local training based on (9) by semismooth Newton method.
 - 7: Remove nodes with outer weight zero (resulting in width $N(t+1)$).
 - 8: $t = t + 1$.
 - 9: **end while**
-

training objective — and c^* serves as an initialization for the training step rather than an exact stationary point of it.

Algorithm 2 Iterative node insertion and optimization with $|c|^q$ -penalty

- 1: Initial network $\omega^{(0)}, c^{(0)}$ of width $N(0)$.
 - 2: **while** $t < T_{\text{out}}$ **do**
 - 3: Sample N_{trial} random nodes $\omega \in \mathbb{S}^d$.
 - 4: Optimize (in parallel) the dual variable $|p_t(\omega)|$ by L-BFGS starting from the initial nodes; merge near-duplicate candidates.
 - 5: For each candidate ω , solve $\min_c \Delta J(c; \omega)$ to obtain $c^*(\omega)$. Select nodes with $\Delta J(c^*; \omega) < 0$ and add them to the network with outer weight c^* (of width $N(t+1/2)$).
 - 6: Perform local training based on (9) with $\phi(z) = \psi_k(z) = |z|^q$ by semismooth Newton method.
 - 7: Remove nodes with outer weight zero (resulting in width $N(t+1)$).
 - 8: $t = t + 1$.
 - 9: **end while**
-

6. NUMERICAL EXAMPLES

Both benchmark problems are two-dimensional by design: the aim is to isolate the effect of the activation class and the penalty on the representation of smooth and nonsmooth value structure, with a reference solution available for every quantity we report. The curse-of-dimensionality motivation of Section 1 concerns the framework — causality-free data generation and sparse networks are built to scale — but scalability itself is not tested here.

All experiments share one computational protocol: a single random seed (42) drives sampling, initialization, and data splits; sweeps range over hyperparameters, not over repeated stochastic replications; validation sets are pointwise random splits of the sampled data, so nearby points of the same PMP characteristic can fall on both sides of the split, and input normalization is fitted on the full sample before splitting. Reported errors are therefore deterministic case studies under this protocol: they measure accuracy along the sampled characteristics, carry no uncertainty estimates, and comparative statements between activations and penalties are to be read within this design.

6.1. Van der Pol Oscillator. In this example we consider the Van der Pol equation. We consider the optimal control of the Van der Pol oscillator expressed as

$$\min_{u \in L^2(0, T; \mathbb{R})} J(u, y) = \int_0^T (y_1^2(t) + y_2^2(t) + \beta u^2(t)) dt$$

subject to

$$\begin{cases} \frac{dy_1}{dt} = y_2, \\ \frac{dy_2}{dt} = -y_1 + y_2(1 - y_1^2) + u, \\ (y_1(0), y_2(0)) = (x_1, x_2), \end{cases}$$

In this setting, $d = 2, k = 1, \beta = 0.1, T = 3$, and

$$f(y) = \begin{pmatrix} y_2 \\ -y_1 + y_2(1 - y_1^2) \end{pmatrix}, \quad g(y) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad l(y) = y_1^2 + y_2^2.$$

By the optimality condition, $u^* = -\frac{1}{2\beta}g(y^*)^T p^* = -\frac{1}{2\beta}p_2^*$. The two-point boundary value problem (TPBVP) becomes

$$(12) \quad \begin{cases} \dot{y}_1^* = y_2^*, \\ \dot{y}_2^* = -y_1^* + y_2^*(1 - y_1^{*2}) + u^*, \\ -p_1^* = 2y_1^* - p_2^*(1 + 2y_1^*y_2^*), \\ -p_2^* = 2y_2^* + p_1^* + p_2^*(1 - y_1^{*2}), \\ y^*(0) = x, \quad p^*(T) = 0. \end{cases}$$

For each initial condition $x = (x_1, x_2)$ sampled on the grid $[-3, 3]^2$, we solve (12) using a Barzilai–Borwein gradient method with backtracking line search. The control is represented in a Legendre polynomial basis $u(t) = \sum_{i=0}^{N_b-1} c_i P_i(t)$ with $N_b = 30$ coefficients. The optimization yields the dataset $\{x^m, V(x^m), \nabla V(x^m)\}_{m=1}^M$, where $V(x^m) = J(u^*; 0, x^m)$ and $\nabla V(x^m) = p^*(0)$.

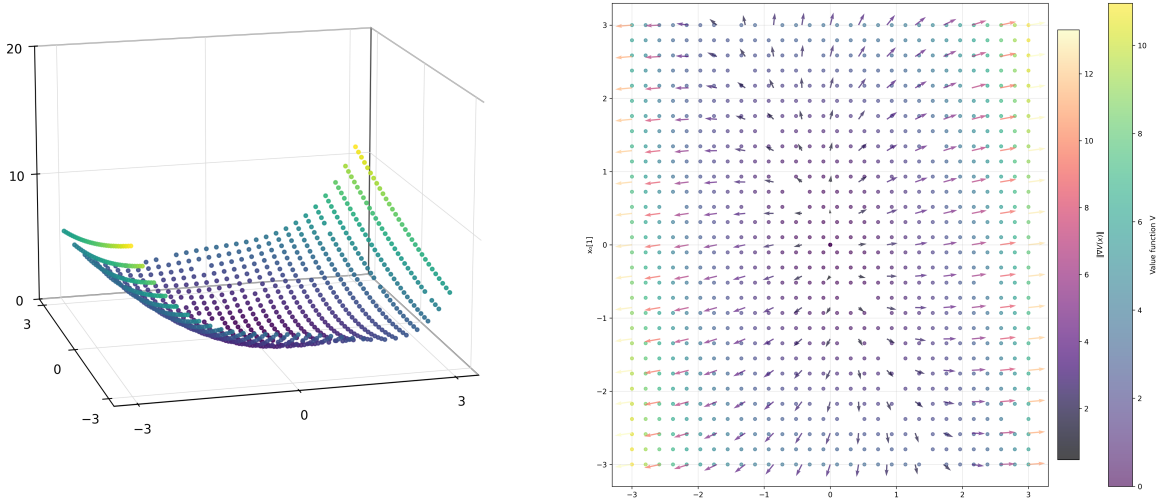


FIGURE 2. Left: value function $V(x)$ over $[-3, 3]^2$. Right: $V(x)$ (dot color) and $\nabla V(x)$ (arrows); arrow color indicates $\|\nabla V(x)\|$.

We solve the open-loop problem on a 30×30 grid over $[-3, 3]^2$ ($M = 900$ data points). All experiments run 10 outer iterations; each iteration samples $N_{\text{trial}} = 50$ candidate nodes, accepts at most 15 of them, and prunes neurons with $|c| \leq 10^{-8}$ after the local solve.

6.1.1. Algorithm 1: ϕ_γ -penalty with different activation functions. First we train the model with Algorithm 1, sweeping over activation functions and hyperparameter groups and reporting the effect on the validation error.

Table 1 isolates the effect of gradient augmentation (L^2 loss, value only, vs. H^1 loss, value + gradient) on the validation error.

TABLE 1. Effect of gradient training on validation error ($\alpha = 10^{-5}$, $\gamma = 1$).

Activation	L^2 trained	H^1 trained	Activation	L^2 trained	H^1 trained
softplus	8.61×10^{-2}	5.19×10^{-1}	softplus	5.63×10^{-1}	2.92×10^{-1}
tanh	3.59×10^{-2}	4.76×10^{-1}	tanh	4.68×10^{-1}	3.14×10^{-1}
Gaussian	3.05×10^{-2}	4.16×10^{-1}	Gaussian	4.64×10^{-1}	9.94×10^{-2}
(A) Effect of gradient training on Err_{L^2}			(B) Effect of gradient training on Err_{H^1}		

Choice of Activation Function in Gradient Training. We choose 3 classes of activation functions: ReLU-like (softplus), sigmoidal (tanh-form), and the Gaussian. Which shape suits the smooth target is a question about the derivative: because the gradient-augmented loss combines the value error with the gradient error, each neuron $\sigma(\cdot, \omega)$ contributes σ to the reconstruction of V and $\partial_x \sigma$ to that of ∇V , so the approximation efficiency is governed by the shape of $\partial_x \sigma$ rather than by that of σ itself. Figure 3 contrasts the three activations through their value, first derivative, and second derivative. A monotone, single-signed derivative that stays active over the whole axis (softplus) endows every neuron with a coherent gradient contribution of broad, bounded curvature, so that few atoms suffice to represent a smooth ∇V ; a derivative that saturates on both tails (tanh) contributes only on a narrow, localized range; and a localized bump (e^{-x^2}) concentrates its curvature near the origin and reverses the sign of its derivative, fitting a smooth field accurately only when densely tiled. The regularity of $\partial_x \sigma$ should therefore match that of the target gradient field: a smooth, broadly curved derivative suits a smooth value function.

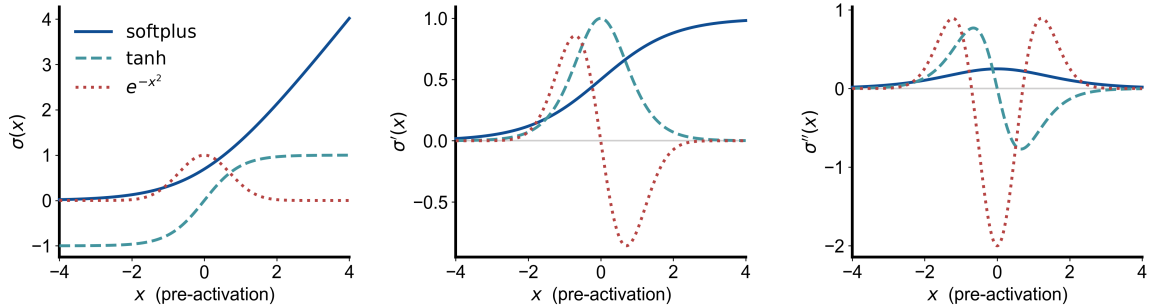


FIGURE 3. Three activations admissible under Assumption 3.9 — softplus, tanh, and the Gaussian bump e^{-x^2} — shown, from left to right, as the value σ , its first derivative σ' , and its second derivative (curvature) σ'' with respect to the pre-activation argument. Under the gradient-augmented loss the first derivative is the basis that reconstructs ∇V .

As observed in Table 1, the Gaussian activation achieves in general the best precision among the 3. All three capture the smooth bowl; they differ in how many neurons it takes.

Tanh saturates for large $|z|$, causing vanishing gradients. Softplus grows linearly and avoids saturation. The Gaussian is localized with exponential decay and has a sign-changing derivative $\sigma'(z) = -2ze^{-z^2}$ (a bump that is accurate where it sits but must be tiled densely to cover a smooth field).

With the H^1 loss, Err_{H^1} drops dramatically for the Gaussian (9.94×10^{-2}), whose localized sign-changing derivative reconstructs the smooth gradient field most accurately, while tanh (3.14×10^{-1}) and softplus (2.92×10^{-1}) improve more modestly. This accuracy comes at the largest neuron

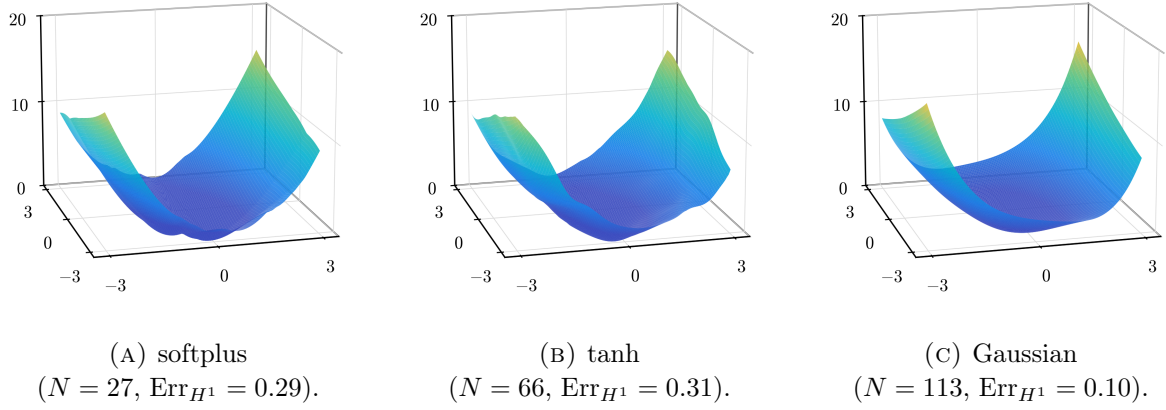


FIGURE 4. Learned value function under the H^1 loss for the signed model at the fixed operating point $\gamma = 1$, $\alpha = 10^{-5}$. All three capture the smooth bowl; they differ in how many neurons it takes.

TABLE 2. Network size (sparsity) for each activation, paired with the validation error natural to each loss (Err_{L^2} for the L^2 loss, Err_{H^1} for the H^1 loss; $\alpha = 10^{-5}$, $\gamma = 1$).

Activation	N	Err_{L^2}	Activation	N	Err_{H^1}
softplus	58	8.61×10^{-2}	softplus	27	2.92×10^{-1}
tanh	75	3.59×10^{-2}	tanh	66	3.14×10^{-1}
Gaussian	105	3.05×10^{-2}	Gaussian	113	9.94×10^{-2}
(A) L^2 trained			(B) H^1 trained		

count ($N = 113$), whereas softplus matches tanh’s Err_{H^1} (2.92×10^{-1} vs. 3.14×10^{-1}) with far fewer neurons (27 vs. 66). On this smooth target the choice of smooth activation trades sparsity against gradient accuracy rather than fit viability.

The performance differences across activations can be understood from two angles: approximation capability and sparsity.

Saturation of the derivative degrades the gradient fit. The gradient kernel column for neuron n at data point x_m is $\sigma'(a_n \cdot x_m + b_n) \cdot a_n$; the diversity of these columns determines how well the network can fit the gradient field. The distribution of $\sigma'(z)$ across all (neuron, data point) pairs indicates the mass saturation near zero (the gradient-kernel column scales with the inner weight a_n as well, so the share of $|\sigma'| < 0.05$ values at this arbitrary threshold is a heuristic indicator, not a direct redundancy measure):

The bottleneck for tanh is that two-sided saturation makes many gradient kernel columns redundant, so a given gradient accuracy costs more neurons rather than being unattainable; the Gaussian’s sign-changing derivative is the most expressive per neuron but the least sparse.

Many accepted candidates per iteration reduce sparsity. The Gaussian and softplus fits differ sharply in size: 113 neurons versus 27, a factor of 4.2, even though the Gaussian is the more accurate of the two ($\text{Err}_{H^1} = 9.9 \times 10^{-2}$ vs. 2.9×10^{-1}). Table 3 tracks the two fits neuron-by-neuron. The full objective $J = L(\mu) + \alpha \Phi(\mu)$ decreases at every iteration for both, confirming that the inserted neurons are genuine descent directions. But the Gaussian adds them in large batches (7–15 per iteration) whereas softplus adds only 1–5, and the descent per neuron

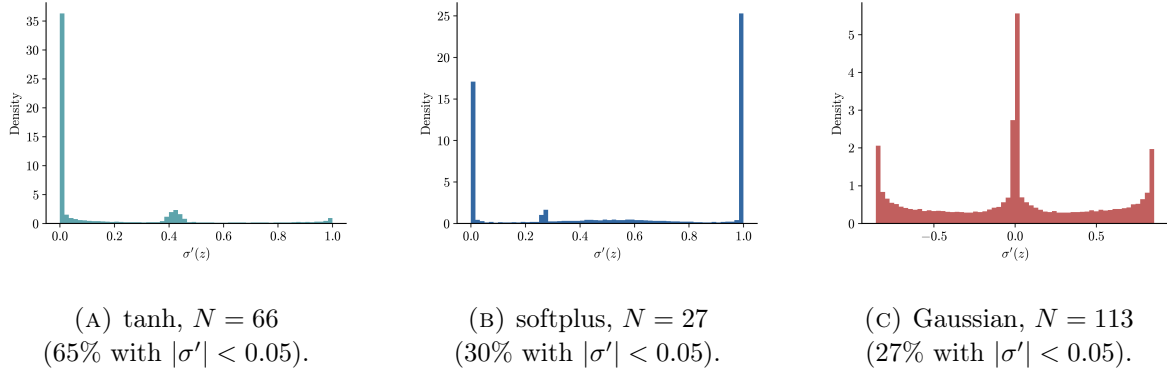


FIGURE 5. Distribution of $\sigma'(z)$ across all (neuron, data point) pairs for the signed H^1 model at the fixed operating point ($\gamma = 1$, $\alpha = 10^{-5}$). The percentage is the fraction of derivative values with $|\sigma'| < 0.05$.

differs accordingly: each Gaussian neuron lowers J by $\sim 10^{-4}$ (down to 2×10^{-5} by the final iteration), while a softplus neuron typically lowers it by 10^{-3} – 10^{-2} .

TABLE 3. Per-iteration insertion diagnostics for the Gaussian and softplus H^1 fits ($\alpha = 10^{-5}$, $\gamma = 1$).

Iter	Gaussian			Softplus		
	N	Ins	$\Delta J/n$	N	Ins	$\Delta J/n$
1	7	7	—	2	2	—
2	14	7	2.8×10^{-3}	3	1	1.7×10^{-1}
3	24	10	5.6×10^{-4}	6	3	5.2×10^{-3}
4	34	10	1.0×10^{-4}	8	2	3.6×10^{-3}
5	42	8	1.8×10^{-4}	11	3	1.9×10^{-2}
6	57	15	5.9×10^{-5}	13	2	1.1×10^{-3}
7	72	15	1.6×10^{-4}	15	2	1.2×10^{-3}
8	86	14	2.2×10^{-4}	18	3	1.4×10^{-3}
9	99	13	6.6×10^{-5}	22	4	2.8×10^{-4}
10	113	14	2.0×10^{-5}	27	5	9.3×10^{-4}
Total	113			27		

N = support size after SSN and pruning; Ins = neurons added that iteration; $\Delta J/n$ = decrease of the objective $J = L(\mu) + \alpha \Phi(\mu)$ per neuron added, relative to the previous iterate (iteration 1 has no predecessor).

The difference reflects how many descent directions the dictionary offers at each step, governed by the empirical dual variable $p_t(\omega)$ of (11) — the value–gradient pairing of the atom with the current residual. The Gaussian activation is localized: $\sigma(x, \omega)$ is a bump concentrated near the hyperplane $\{x : a \cdot x + b \approx 0\}$, so $p_t(\omega)$ is sensitive to local residual structure, and even small residual pockets push $|p_t(\omega)| > \alpha$ for many ω . The insertion step therefore finds a fresh batch of admissible atoms every iteration, each correcting only a small local portion of the residual. Softplus has global support, so $p_t(\omega)$ averages the residual over the entire domain—positive and

negative contributions cancel, fewer ω exceed the threshold α , but each admitted atom removes a global component of the residual.

In conclusion:

- Gaussian yields the best gradient accuracy ($\text{Err}_{H^1} = 9.9 \times 10^{-2}$) but uses 113 neurons. Its localized activation offers many shallow descent directions, so the algorithm keeps inserting neurons with diminishing per-neuron returns.
- Softplus reaches moderate gradient accuracy ($\text{Err}_{H^1} = 2.9 \times 10^{-1}$) with only 27 neurons—the sparsest fit. Each neuron removes a global component of the residual, so the dual variable drops below α after few insertions.
- Tanh is dominated: 66 neurons for $\text{Err}_{H^1} = 3.1 \times 10^{-1}$ —essentially softplus’s accuracy at more than twice the neuron count—because its two-sided saturation yields a narrow, mostly-dead derivative that wastes atoms without improving the gradient fit.

Effect of α and γ . Table 4 sweeps α for $\gamma = 0$ (pure ℓ^1) and $\gamma = 10$ (strong nonconvex correction) across the two activations (H^1 loss). Smaller α increases accuracy at the cost of more neurons for both, and the nonconvex correction helps the two activations differently. For softplus it roughly halves the H^1 error at a comparable neuron count ($3.35 \times 10^{-1} \rightarrow 1.76 \times 10^{-1}$ at $\alpha = 10^{-5}$; $3.24 \times 10^{-1} \rightarrow 1.68 \times 10^{-1}$ at $\alpha = 10^{-4}$). For the Gaussian it barely moves the already-low error ($\approx 1.0 \times 10^{-1}$) while the neuron count grows ($91 \rightarrow 119$ at $\alpha = 10^{-5}$): the localized activation is insensitive to the penalty’s nonconvexity, whereas the broad softplus ridge benefits from the sharper thresholding.

TABLE 4. Algorithm 1: effect of α with $\gamma = 0$ vs. $\gamma = 10$ (H^1 loss).

$\gamma = 0$					$\gamma = 10$				
α	N	Err_{H^1}	N	Err_{H^1}	α	N	Err_{H^1}	N	Err_{H^1}
10^{-2}	14	3.57×10^{-1}	10	3.78×10^{-1}	10^{-2}	12	2.31×10^{-1}	19	2.06×10^{-1}
10^{-3}	15	3.43×10^{-1}	14	2.79×10^{-1}	10^{-3}	26	1.64×10^{-1}	33	1.28×10^{-1}
10^{-4}	20	3.24×10^{-1}	25	1.68×10^{-1}	10^{-4}	81	1.08×10^{-1}	80	1.02×10^{-1}
10^{-5}	21	3.35×10^{-1}	23	1.76×10^{-1}	10^{-5}	91	1.03×10^{-1}	119	9.88×10^{-2}
(A) Softplus					(B) Gaussian				

6.1.2. *Algorithm 2: $|c|^q$ -penalty with ReLU^k activation.* Algorithm 2 uses the ReLU^k atom $\sigma(z) = \text{ReLU}(z)^k$ with the pure power penalty ψ_k , i.e. $\alpha \sum_i |c_i|^q$ with $q = 2/(k+1)$ and no log component. The swept parameter is the power k ; all fits below are at the fixed operating point $\alpha = 10^{-5}$.

Under the L^2 loss the gradient error stays large for every power ($\text{Err}_{H^1} \approx 4\text{--}5 \times 10^{-1}$): the value-only fit ignores ∇V , so the atoms are placed without regard to the gradient field. The H^1 loss drives the gradient error down to $\approx 1.0 \times 10^{-1}$ for *all* powers (Table 5). The effect of gradient augmentation is uniform in k : the fit becomes gradient-accurate regardless of the atom power.

Under the H^1 loss the gradient accuracy is essentially flat in k ($\text{Err}_{H^1} \approx 1.0 \times 10^{-1}$ for every power), while the neuron count falls sharply as the power grows: $k = 2$ needs 41 atoms, whereas $k = 3$, $k = 4$ and $k = 5$ need only about 20 (Table 5(b)). Raising the power increases sparsity at essentially no cost in accuracy: the exponent $q = 2/(k+1)$ shrinks, the coefficient penalty $|c|^q$ becomes more concave, and the more aggressive selection keeps the accuracy with fewer atoms—a smooth value function lies in the span of a few high-power atoms. The count drops sharply from $k = 2$ to $k = 3$ (41, 42, 20, 22, 21) and then plateaus, while the error is flat up to a mild degradation at $k = 5$ ($0.098 \rightarrow 0.104$): no intermediate optimum appears, and the sparsity

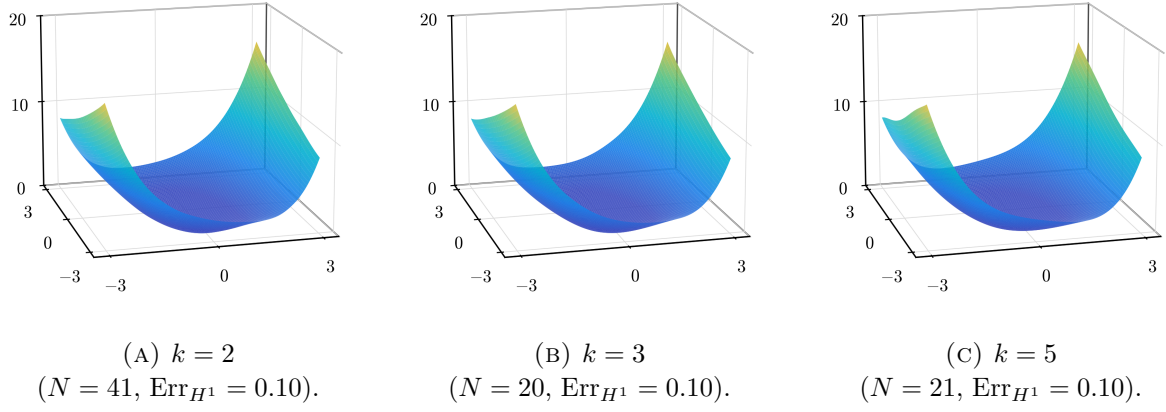


FIGURE 6. H^1 -trained model with $\alpha = 10^{-5}$, for the representative powers $k \in \{2, 3, 5\}$. All reconstruct the smooth bowl; they differ only in neuron count.

TABLE 5. Algorithm 2: network size and validation error vs. the atom power k ($\alpha = 10^{-5}$).

Power	N	Err_{L^2}	Err_{H^1}	Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	14	5.05×10^{-2}	4.83×10^{-1}	$k = 2$	41	4.16×10^{-1}	9.84×10^{-2}
$k = 2.01$	14	4.83×10^{-2}	4.81×10^{-1}	$k = 2.01$	42	4.17×10^{-1}	9.93×10^{-2}
$k = 3$	12	5.98×10^{-2}	4.94×10^{-1}	$k = 3$	20	4.17×10^{-1}	1.00×10^{-1}
$k = 4$	12	5.48×10^{-2}	4.61×10^{-1}	$k = 4$	22	4.16×10^{-1}	9.97×10^{-2}
$k = 5$	10	4.77×10^{-2}	4.04×10^{-1}	$k = 5$	21	4.14×10^{-1}	1.04×10^{-1}
(A) L^2 trained				(B) H^1 trained			

gain saturates rather than growing monotonically. Under the L^2 loss the fits are uniformly sparse (10–14 atoms) but gradient-blind ($\text{Err}_{H^1} \approx 4\text{--}5 \times 10^{-1}$, Table 5(a)).

6.1.3. *Summary.* Figure 7 plots insertion growth trajectory — the best relative H^1 error reached with a given number of neurons. Algorithm 2 dominates: ReLU^k at $k = 5$ and $k = 2$ reach the best gradient accuracy ($\text{Err}_{H^1} \approx 1.0 \times 10^{-1}$) with only ≈ 20 –40 neurons, whereas the most accurate Algorithm-1 activation, the Gaussian, needs 113 neurons for the same accuracy, and softplus/tanh plateau at a coarser error. At equal gradient accuracy Algorithm 2 uses a fraction of the atoms.

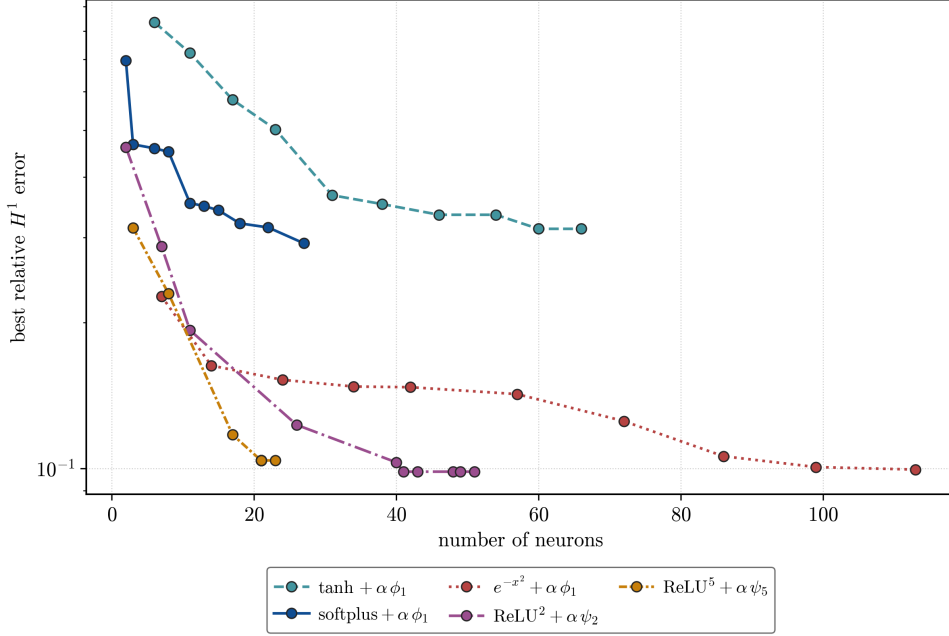


FIGURE 7. Running best relative H^1 validation error vs. network size ($\alpha = 10^{-5}$).

We synthesize feedback laws from $y_0 = (2, 1)$ and roll them out over the horizon $T = 12$ with fixed-step RK4, zero-order-hold controls clipped to $|u| \leq 50$ as a numerical guard. The reference is the *PMP-derived frozen-time feedback*: the time-zero costate component of the dataset, interpolated by a smooth C^1 (Clough–Tocher) interpolant and applied as a stationary law. The finite-horizon optimal feedback is time-dependent and the rollout horizon exceeds the data horizon $T = 3$, so this reference is itself a surrogate — its rollout cost 6.48 is not $V(0, y_0)$ — and the comparison is a stationary-surrogate diagnostic rather than a finite-horizon optimality test. Figure 8 shows $\|y(t)\|$ and $|u(t)|$ for softplus, the Gaussian, and ReLU^k ($k = 5$), beside this reference. All three drive the state to the origin and track the reference control, at essentially its closed-loop cost (6.68, 6.51, 6.49 vs. 6.48).

Figure 9 shows where the inner weights sit for the Gaussian, softplus, and ReLU^k ($k = 5$). The two algorithms produce structurally different representations: Algorithm 2 constrains its atoms to the unit sphere $\mathbb{S}^2$, while Algorithm 1 leaves them free — the Gaussian atoms in particular span a wide range of norms.

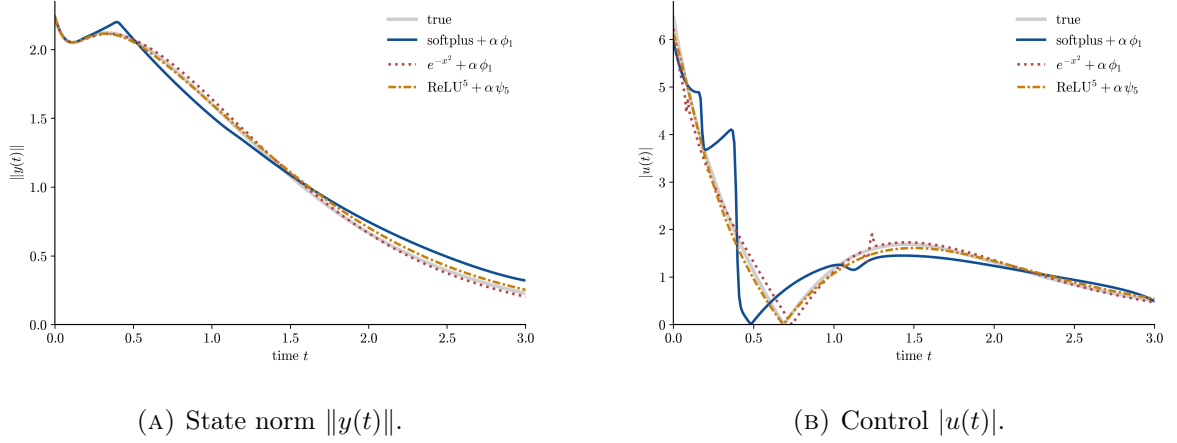


FIGURE 8. Evolution of the state and control from $y_0 = (2, 1)$ under the feedback synthesized from each fitted value function, beside the PMP-derived frozen-time reference control.

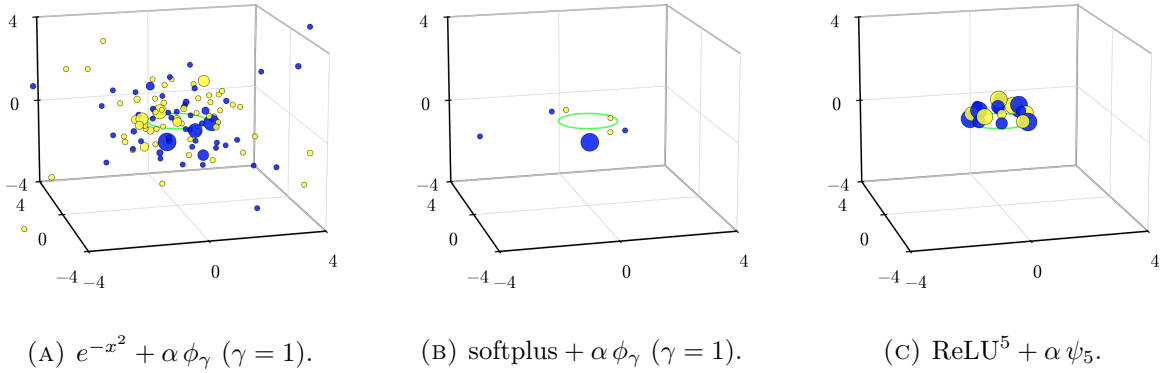


FIGURE 9. Inner-weight portrait: raw (a_1, a_2, b) with the unit equator circle (light green, at $b = 0$) for reference ($\alpha = 10^{-5}$). Dot color indicates the sign of the outer weight, dot size its magnitude.

6.2. Pendulum Swing-Up. We have established that our networks approximate value functions in the Barron space $B_1^k(D)$, and that this space embeds into $H^k(D)$: at any fixed variation-norm budget, the achievable H^1 error at a gradient discontinuity is bounded away from zero (Section 2.2) — and the budget is exactly what the sparsity regularizer controls. We now exhibit this finite-budget limit on the pendulum example and determine which atom class spends its budget most effectively at the discontinuity.

When the value function possesses a switching set — a manifold across which ∇V is discontinuous, as in the pendulum swing-up problem — a smooth derivative can no longer reproduce the target gradient field economically. Representing a jump in ∇V requires a derivative $\partial_x \sigma$ that is itself able to break across a hyperplane, which relaxes the differentiability of Assumption 3.9. Figure 10 contrasts three such shapes. The rectified linear unit ReLU has a step derivative, so a single neuron seats a finite gradient jump exactly on its hyperplane; its squared variant ReLU^2 has a kinked but continuous derivative — a jump in the curvature — that builds a sharp gradient transition while its quadratic growth still fits the smooth far field; softplus, whose derivative is smooth, can only smear the jump and consequently plateaus near the switching set.

The positive homogeneity of the rectified families also makes them the natural setting for the fractional-exponent penalty of Section 4.

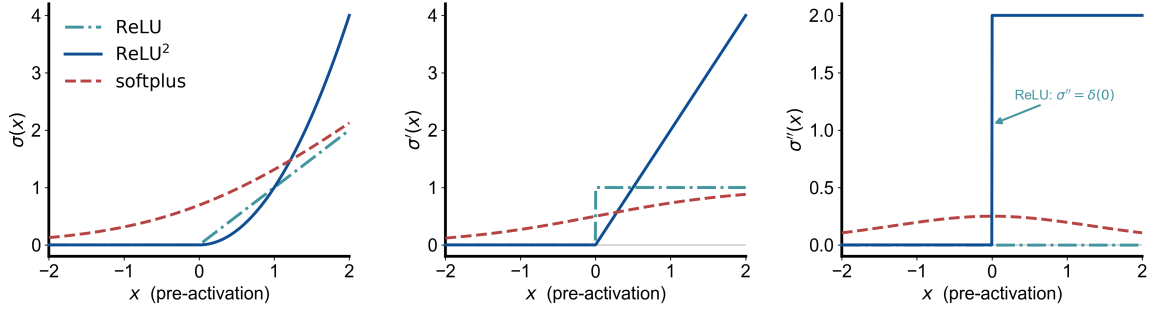


FIGURE 10. Three activations for a target gradient that jumps across a switching set — ReLU, ReLU^2 , and softplus (smooth reference) — shown, from left to right, as the value σ , its first derivative σ' , and its second derivative σ'' with respect to the pre-activation argument. Only the rectified derivatives can break at the origin: ReLU' jumps and $(\text{ReLU}^2)'$ kinks, whereas the softplus derivative is smooth. The curvature of ReLU is a Dirac mass at the origin (marked).

The example we consider follows from [10]. We write the state as

$$x = (x_1, x_2)^T = (\theta, \dot{\theta})^T,$$

the pendulum angle and angular velocity. (The letter ω remains reserved for the network parameters.) We consider the optimal control problem

$$\min_{u \in L^2(0, \infty; \mathbb{R})} \int_0^\infty (q_1 \sin^2 \theta(t) + q_1 (\cos \theta(t) - 1)^2 + q_2 \dot{\theta}^2(t) + r u^2(t)) dt$$

subject to

$$\begin{cases} ml^2 \ddot{\theta} = -b\dot{\theta} + mgl \sin \theta + u, \\ (\theta(0), \dot{\theta}(0)) = (x_1, x_2). \end{cases}$$

The target equilibrium is $x_U = (2j\pi, 0)$, $j \in \mathbb{Z}$.

In this setting the state dimension is $d = 2$ and the control is scalar. The physical and cost parameters are $m = l = 1$, $b = 0.1$, $g = 9.8$, $q_1 = q_2 = 1$, and $r = 1$. To initialize the backward integration, we use the local linear-quadratic regulator (LQR) approximation near x_U . Let $\xi = x - x_U$ be the local coordinate around the upright equilibrium. Linearizing the dynamics at $(x_U, 0)$ gives

$$\dot{\xi} = A\xi + Bu, \quad A = \begin{pmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{b}{ml^2} \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ \frac{1}{ml^2} \end{pmatrix}.$$

The running cost is locally approximated by

$$\xi^T Q \xi + r u^2, \quad Q = \begin{pmatrix} q_1 & 0 \\ 0 & q_2 \end{pmatrix}.$$

The local infinite-horizon LQR value function is

$$V^\infty(\xi) = \xi^T P \xi,$$

where P is the positive definite solution of the algebraic Riccati equation

$$A^T P + P A - \frac{1}{r} P B B^T P + Q = 0.$$

The superscript ∞ refers to the infinite-horizon LQR problem. For $\varepsilon > 0$, we define the local LQR sublevel set and boundary by

$$L_\varepsilon = \{\xi : V^\infty(\xi) \leq \varepsilon\}, \quad \partial L_\varepsilon = \{\xi : V^\infty(\xi) = \varepsilon\}.$$

The corresponding state-costate system becomes

$$(13) \quad \begin{cases} ml^2 \ddot{\theta}^* = -b\dot{\theta}^* + mgl \sin \theta^* + u^*, \\ -\dot{p}_1^* = 2q_1 \sin \theta^* + \frac{g}{l} \cos \theta^* p_2^*, \\ -\dot{p}_2^* = 2q_2 \dot{\theta}^* + p_1^* - \frac{b}{ml^2} p_2^*, \\ x^*(0) = x, \quad x^*(T_x) - x_U \in \partial L_\varepsilon, \quad p^*(T_x) = \nabla V^\infty(x^*(T_x) - x_U). \end{cases}$$

In the experiment we use $\varepsilon = 2 \times 10^{-4}$, and on ∂L_ε the terminal costate is

$$p^*(T_x) = \nabla V^\infty(\xi^*(T_x)) = 2P\xi^*(T_x).$$

For the data generation, we solve the open-loop PMP system by integrating 2000 characteristics backward from the local LQR boundary ∂L_ε . Along each PMP trajectory, the accumulated running cost *plus the boundary value* ε gives the value sample $V(x)$, and the costate gives the gradient sample $\nabla V(x) = p$. The backward integration of each raw trajectory stops when the accumulated value reaches the cap $V \leq 100$; the in-basin training samples are additionally restricted to values below 50 (the resolved part of the basin). This yields the dataset

$$\{x^m, V(x^m), \nabla V(x^m)\}_{m=1}^M.$$

This construction records the nonsmooth geometry of the pendulum value function. Because the clockwise and counter-clockwise swing-up branches have equal cost at the bottom position $((2k+1)\pi, 0)$, the optimal value function is not globally C^1 . The generator detects the switching curve by intersecting equal-value contours with their 2π -shifted copies and restricts the PMP trajectories to a consistent smooth branch. Restriction alone, however, produces one-sided data — every sample stops at the switching curve, so the gradient jump itself would never appear in the training set. The generator therefore harvests a two-sided band: the raw trajectories are tiled by $\pm 2\pi$ in θ and gated to within 0.5 of the (tiled) switching arms, and a tiled point is kept only if its branch value beats the competing branch's locally extrapolated value *and* matches its own branch's local lower envelope — a numerical envelope screen (the competing branch is extrapolated to first order from its 32 nearest raw samples within radius 0.1 and must lose by a margin of 0.3), so the kept points carry the branch-optimal $(V, \nabla V)$ up to this screen's tolerance. The band splits into a near-side pad (300 samples: the central branch between the basin's conservative trim and the true arm) and a far-side collar (600 samples: the neighboring upright's branch across the arm). We fix the terminology: the *switching set* (or curve) is the geometric locus across which ∇V jumps; the *switching band* is the two-sided slice of training data straddling it; the *switching tube* is the evaluation region within distance 0.3 of the (tiled) curve. The final (x, V, dV) dataset (3,900 samples: a 3,000-sample in-basin body plus the 900-sample two-sided band) thus contains the gradient discontinuity as an interior feature. This makes the pendulum benchmark a stress test for the gradient-augmented sparse neural approximation: the value is continuous while its gradient jumps across a switching set that the models see from both sides. Figure 11 shows the dataset generated.

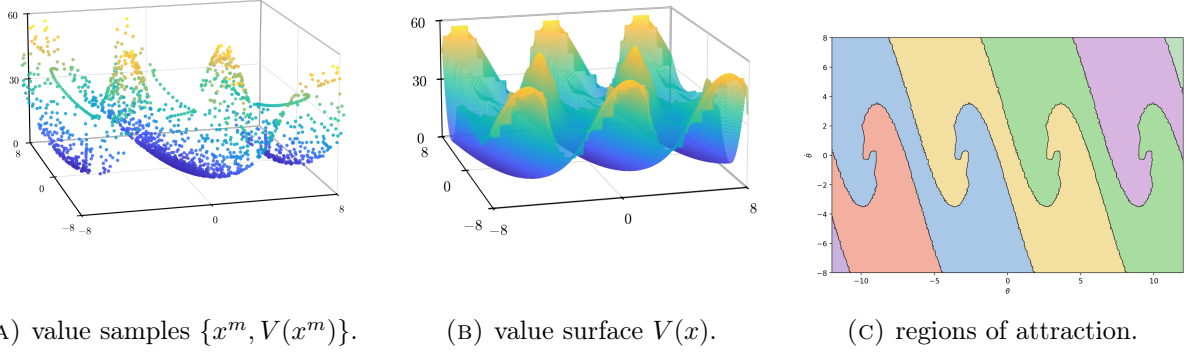


FIGURE 11. Optimal value function $V(x)$ over the $(\theta, \dot{\theta})$ plane, 2π -periodic in θ . (a) The raw (x, V) samples produced by the PMP generator. (b) The reconstructed value surface. (c) Regions of attraction of the periodic uprights; their boundaries are the switching-set spirals across which ∇V is discontinuous.

6.2.1. *The representation limit at the switching set.* As in the last example we fit the training data with the models we selected. Figure 12 shows the fitted surfaces on the same physical window and value range as the true surface of Figure 11b. The switching profiles are reproduced only approximately — the quantitative limitation appears on the transect below.

Figure 13 shows a normal cross-section of the switching curve through the densest data region. The true value (left) is the continuous lower envelope of the two PMP branches, with a concave kink at the crossing; the true normal gradient $n \cdot \nabla V$ (right) therefore jumps at the switching set $s = 0$. The training data here carry samples on both sides of the curve, so the jump is in-sample. No model reproduces the magnitude of the jump: all undershoot the steep pre-jump gradient. The two kink-capable atoms come closest, each in the shape its atom geometry dictates. ReLU² develops a visible kink at $s \approx 0$ — its piecewise-quadratic atoms can break the derivative across a line — and tracks the true value level best. The leaky-ReLU fit is a staircase: a piecewise-linear network has zero curvature between its atom lines, so its normal gradient is piecewise *constant*, jumping where an atom line crosses the section and holding a plateau of ≈ -30 to -42 in between. The smooth activations necessarily interpolate a gentle slope through the discontinuity. The discontinuous gradient is thus a genuine representation cost.

Since the method is data-driven, we investigate next to what extent the representation limit can be mitigated by oversampling the switching region. The hypothesis is that a model with good local approximation power should improve near the switching set when the sample density there is increased. We test this with stratified sampling: From the same generator we build four training sets that differ only in the fraction of samples placed around the switching curve: 6,000 samples with the reference fraction of 23%, the same 6,000 budget reallocated to 40% and to 60%, and the reference set with 2,000 additional samples (8,000 total). On each set we train the Gaussian with ϕ_γ and ReLU² with ψ_2 , each for three regularization weights α , and keep the best run per set. All models are evaluated on one common out-of-sample evaluation pool of $\approx 9.4 \times 10^5$ points — a dense re-evaluation of the same 2,000 PMP characteristics with every training row excluded, so it is disjoint from the training sets but not an independent test set over new characteristics — with the relative H^1 error reported separately on the switching tube (distance ≤ 0.3 to the switching curve) and on the rest of the domain.

Neither model improves at the switching set (Table 6, Figure 14). The Gaussian is flat across all four sets: a smooth atom cannot represent the kink, and more samples of it do not change that. ReLU² *degrades monotonically* as the fraction grows ($0.25 \rightarrow 0.29 \rightarrow 0.35$, with the rest

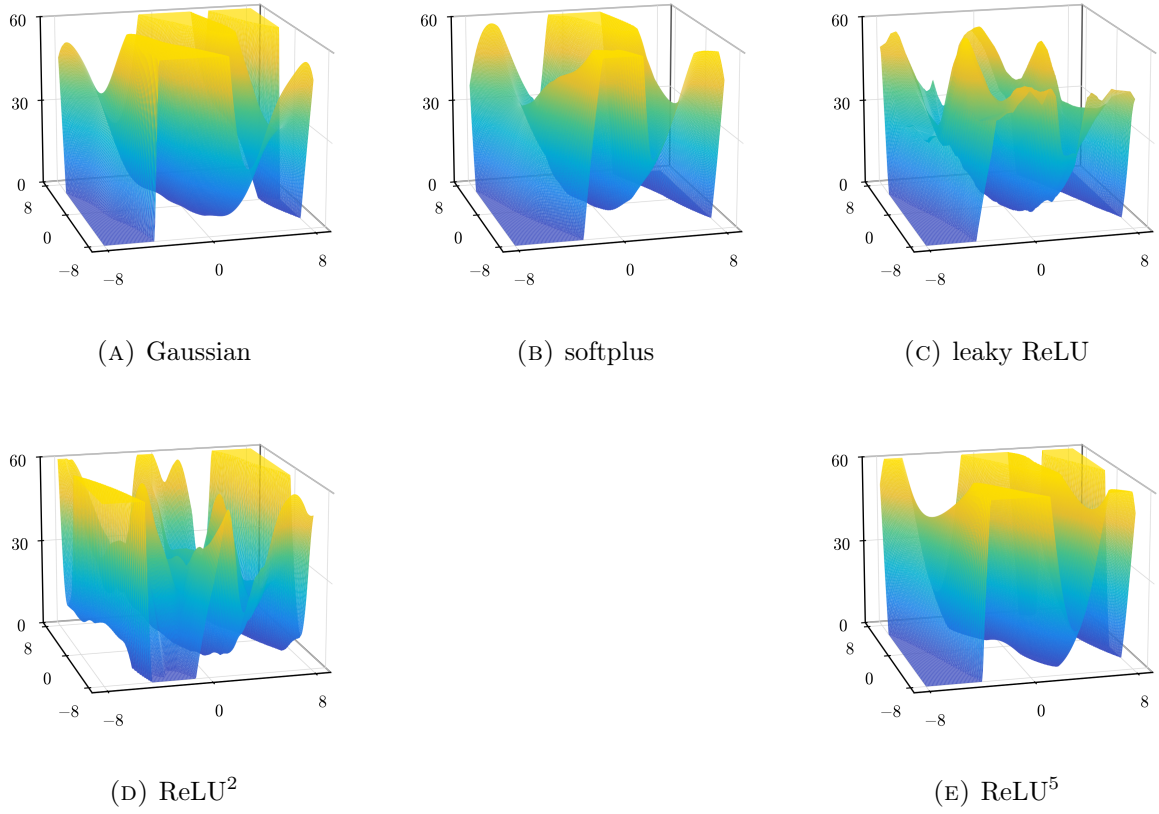


FIGURE 12. Learned value surfaces $\hat{V}(x)$ over the $(\theta, \dot{\theta})$ window $[-8, 8]^2$, displayed on the common range $\hat{V} \in [0, 60]$. The ReLU^k models are trained with the fractional-exponent penalty ψ_k , the others with the log penalty ϕ_γ ; the weights α and γ are selected per model.

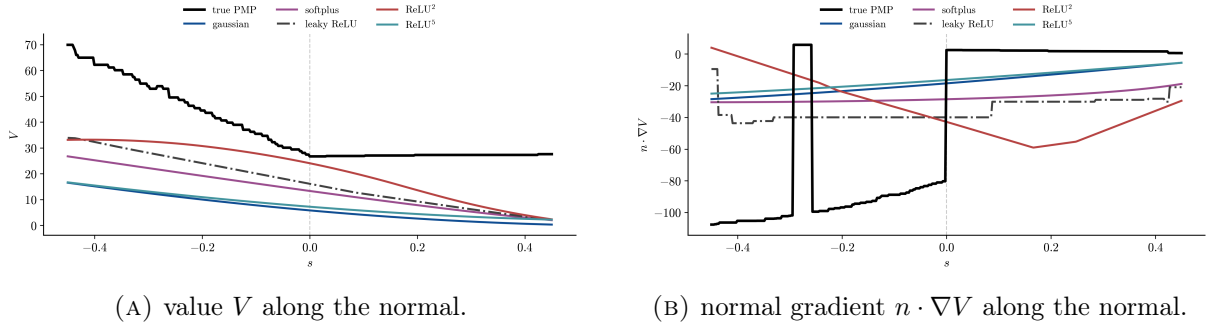


FIGURE 13. Normal cross-section of the switching curve, parametrized by the normal distance s . Left: the true value (dashed) is a continuous lower envelope with a kink at $s = 0$. Right: the true normal gradient jumps at $s = 0$; ReLU^2 develops a kink near $s = 0$ and leaky ReLU a piecewise-constant staircase, while the smooth models interpolate through the discontinuity. The rectangular excursion of the reference curve near $s \approx -0.3$ is a nearest-neighbor artifact of the envelope reference, not a feature of V .

Model	Training set	Switching tube	Rest
Gaussian + ϕ_γ	6k, 23% band (reference)	5.81×10^{-1}	5.89×10^{-1}
	6k, 40% band	5.72×10^{-1}	5.30×10^{-1}
	6k, 60% band	6.02×10^{-1}	6.68×10^{-1}
	6k + 2k band	5.97×10^{-1}	5.68×10^{-1}
ReLU ² + ψ_2	6k, 23% band (reference)	2.46×10^{-1}	1.56×10^{-1}
	6k, 40% band	2.89×10^{-1}	1.72×10^{-1}
	6k, 60% band	3.46×10^{-1}	2.20×10^{-1}
	6k + 2k band	2.88×10^{-1}	1.84×10^{-1}

TABLE 6. Relative H^1 error on the common evaluation pool, inside the tube of distance ≤ 0.3 to the switching curve and on the rest of the domain; each cell reports the best of the three regularization weights for that region, so the two cells of a row need not come from the same run.

error rising in step): samples moved into the switching set are samples removed from the interior, and the fit loses more there than it gains at the curve. Adding 2,000 band samples on top of the budget is less harmful than reallocation but still does not beat the reference. Under the tested architecture, regularization grid, optimizer budget, and seed, increasing the switching-band share did not reduce the error in either region — the limit is not a sampling deficit within this protocol.

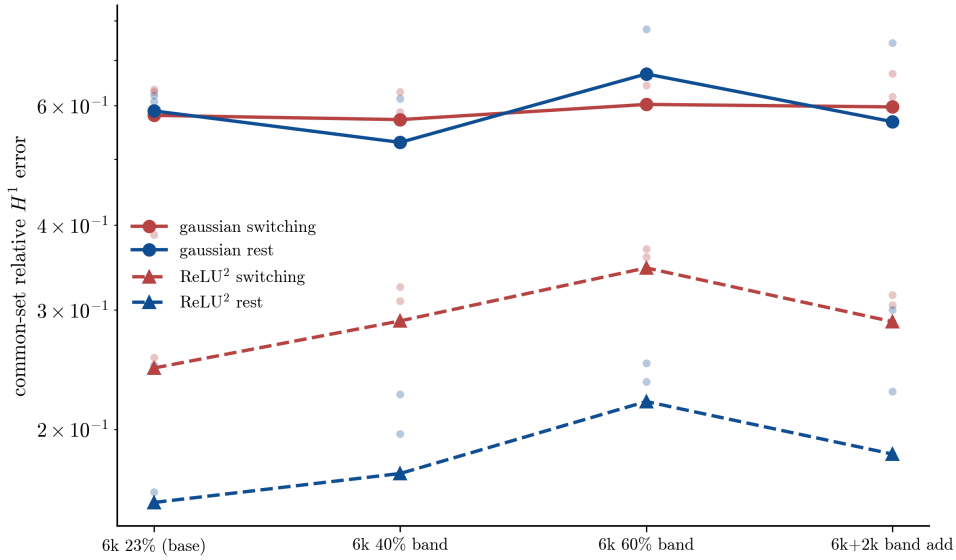


FIGURE 14. Relative H^1 error on the common evaluation pool against the four training sets, inside the switching tube (red) and on the rest (blue), for the Gaussian (solid) and ReLU² (dashed). Faint dots: the three regularization weights; markers: the best run per set. Increasing the band fraction improves neither region for either model.

6.2.2. *ReLU² fits the discontinuous gradient best.* Despite the limitation in representing the discontinuous gradient, ReLU² fits the switching profile markedly better than the other activation functions.

Figure 15 tracks the best relative H^1 validation error reached as neurons are inserted, for each activation paired with its penalty. (Per-activation hyperparameters are selected by validation error; the selected log-penalty representatives are softplus at $\gamma = 10$, while for the Gaussian and leaky ReLU the selection lands at $\gamma = 0$, i.e. the pure ℓ^1 member of the blend.) ReLU² (fractional penalty ψ_2) separates from the other models within the first insertions and reaches the lowest error of all ($\approx 3.1 \times 10^{-1}$ at 131 neurons); leaky ReLU — the other kink-capable atom — is the runner-up ($\approx 4.4 \times 10^{-1}$); softplus plateaus near 5.2×10^{-1} , and the Gaussian and ReLU⁵ (ψ_5) stall around 6.1×10^{-1} . The per-neuron accuracy advantage of the ReLU² atom is unambiguous; raising the power to 5 forfeits it, and even the piecewise-linear leaky ReLU trails ReLU² by roughly 40%.

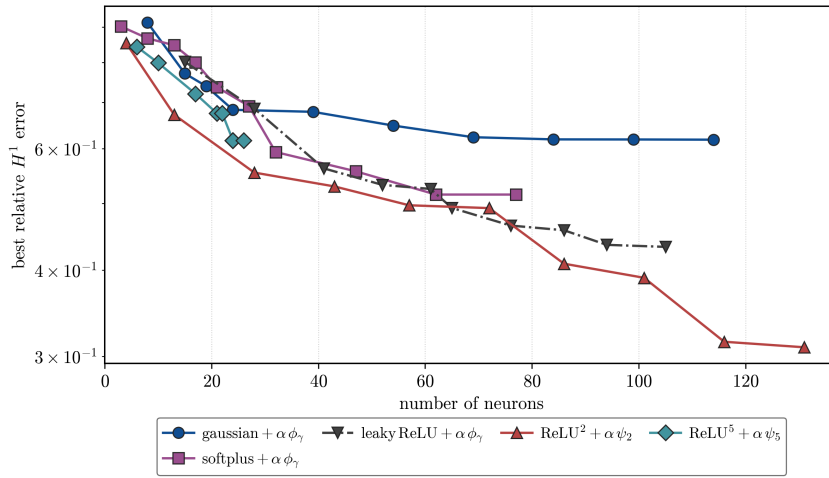


FIGURE 15. Best relative H^1 validation error against neuron count, one curve per activation–penalty pair (ϕ_γ the log penalty, ψ_k the fractional-exponent penalty). ReLU² reaches the lowest error overall; leaky ReLU follows at $\approx 1.4\times$ its error, softplus at $\approx 1.7\times$, and the Gaussian and ReLU⁵ plateau at $\approx 2\times$.

The same ordering holds region by region. Figure 16 reports the relative H^1 error — scored out of sample on the dense two-sided evaluation pool ($\approx 9.6 \times 10^5$ points reconstructed from the same characteristics, training rows excluded), from each run’s recorded best-objective iterate (selection is by training objective; the live final model, which may contain atoms inserted after the last record, is not used) — separately inside the switching tube and on the rest of the domain: ReLU² leads in *both* regions, leaky ReLU is the runner-up in both, and the two kink-capable atoms sit $1.5\text{--}3\times$ ahead of the smooth activations. ReLU⁵ is the one model that is better inside the tube than outside: its stiff high-degree ridges seat the switching band but pay for it everywhere else.

Figure 17 draws each atom’s active line $\{a \cdot x + b = 0\}$ in the physical $(\theta, \dot{\theta})$ plane, with line strength proportional to the outer weight $|c_n|$, for ReLU² (left) and Gaussian (right); the switching curve is shown in black. ReLU² concentrates its strongest ridges parallel to the diagonal switching arms — piecewise low-degree ridges whose derivative kinks exactly along the line where the target’s gradient jumps — whereas the Gaussian spends its capacity on localized slabs (each atom $e^{-(a \cdot x + b)^2}$ is constant along its line, not an isotropic bump) that superpose to fill the wells but cannot seat a sharp gradient transition. This alignment is consistent with the frontier advantage above and with the transect kink of Figure 13.

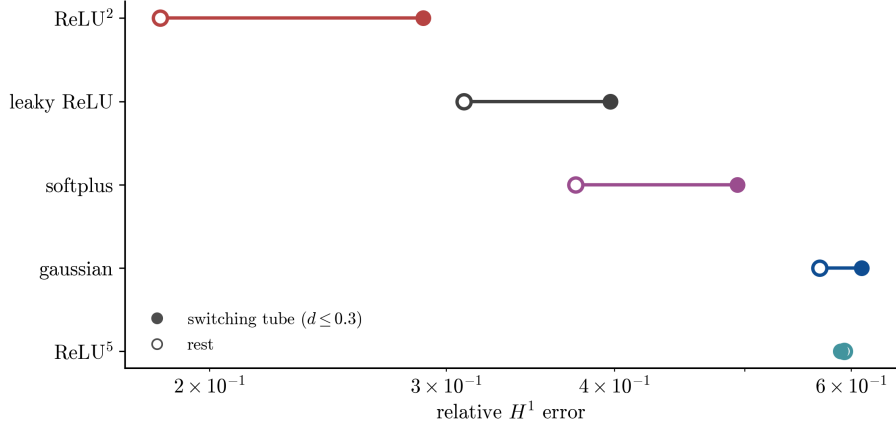


FIGURE 16. Relative H^1 error (log scale) inside the switching tube (distance ≤ 0.3 to the switching curve; filled markers) and on the rest of the domain (open markers), per representative model, rows ordered by the rest error.

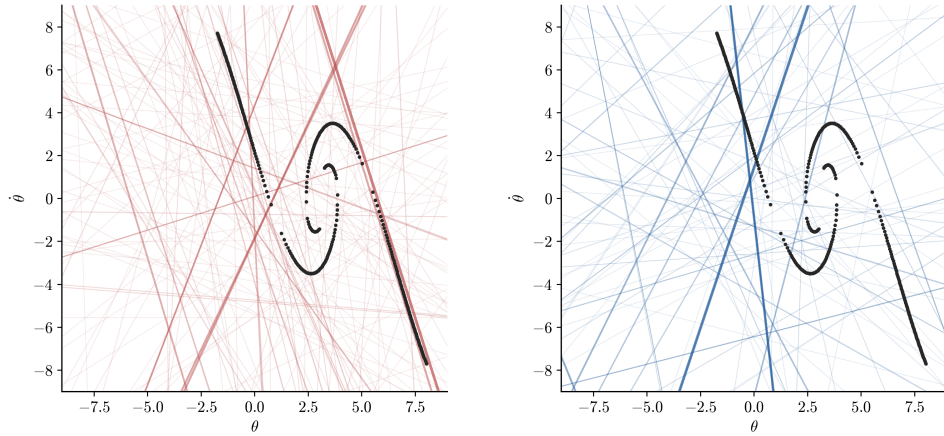


FIGURE 17. Active atom lines $\{a \cdot x + b = 0\}$ in the $(\theta, \dot{\theta})$ plane (strength $\propto |c_n|$) for ReLU^2 (left) and Gaussian (right); the switching curve is black. ReLU^2 's strongest ridges run parallel to the switching arms; the Gaussian slab atoms spread across orientations with no alignment to the switching curve.

6.2.3. *Feedback synthesis across the switching set.* Minimizing the Hamiltonian in u gives the optimal control in feedback form,

$$u(x) = -\frac{1}{2rml^2} \partial_{\dot{\theta}} V(x);$$

replacing V by a fitted $\hat{V}$ yields a candidate feedback law. We test each law in closed loop from two initial states placed on either side of the switching curve, $A = (0.71, 0.68)$ and $B = (0.23, 0.53)$, integrating the dynamics over the horizon $T = 10$ with fixed-step RK4 ($\Delta t = 5 \times 10^{-3}$), zero-order-hold controls, and a numerical clamp $|u| \leq 30$ (the OCP itself is unconstrained). The two starts demand different optimal behavior: from A the optimal law swings the pendulum over the top to the upright at $\theta = 2\pi$, while from B it brakes directly to the upright at $\theta = 0$ — so a correct law must change branch across the curve. The reference law is a nearest-neighbour

envelope controller on the raw PMP data: at each state it takes the lowest-value branch among the 40 nearest raw samples and applies the feedback formula to that branch’s costate.

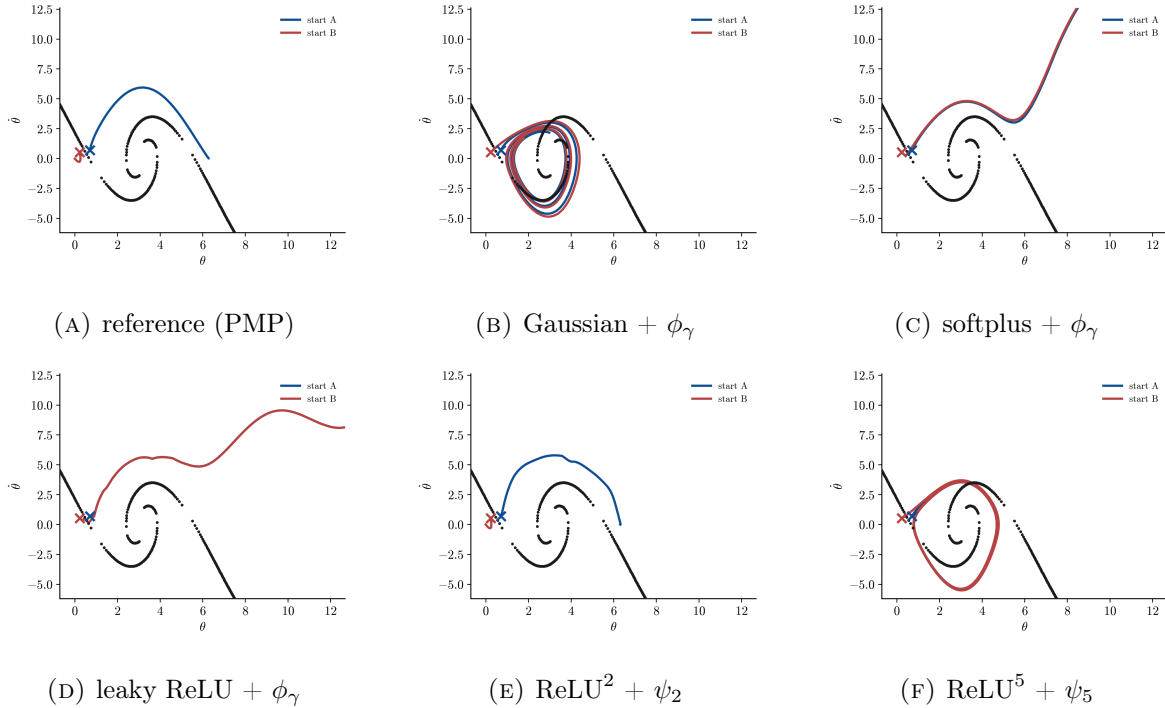


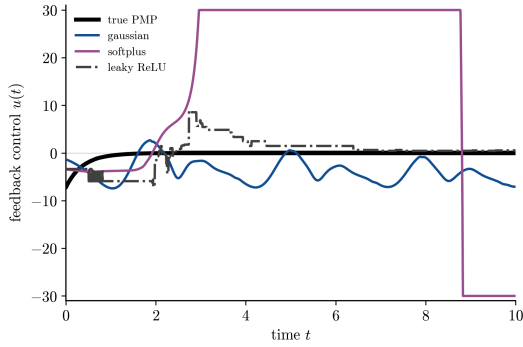
FIGURE 18. Closed-loop trajectories in the $(\theta, \dot{\theta})$ plane from the two starts A (blue) and B (red) on either side of the switching curve (black), one panel per feedback law; all panels share the same axes. The reference law swings over the top from A and brakes to $\theta = 0$ from B ; ReLU^2 reproduces both behaviors.

Only ReLU^2 reaches the upright neighbourhood from both starts (Table 7). From B it brakes to the upright at essentially the reference cost (10.3 vs 10.2); from A it takes the correct branch — swinging over to the upright at 2π — though with an over-energetic arc that roughly doubles the cost (57.9 vs 26.2). Every other law fails from both starts: softplus and leaky ReLU over-accelerate, pass the uprights and never brake; the Gaussian settles into a periodic orbit around the wells without reaching an upright; ReLU^5 swings over from A but arrives at 2π too slowly to be captured, and from B stalls short of the upright. Figure 19 shows the control signal from B : the ReLU^2 law tracks the reference almost exactly, while the others oscillate or saturate. The failures track the global fit quality of the value approximation: a feedback law is only as good as the gradient field.

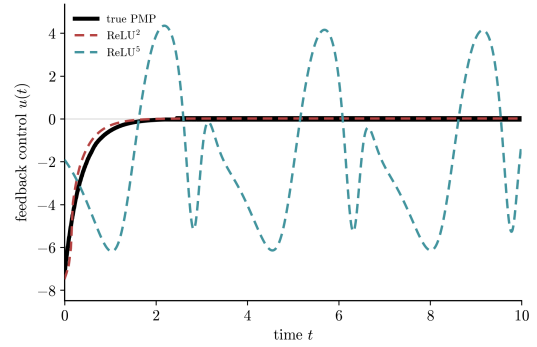
In summary, ReLU^2 with the fractional-exponent penalty is the best atom class for this nonsmooth target. With the switching set in the training data, only ReLU^2 — whose derivative can kink across a hyperplane — builds a sharp continuous transition at the jump (Figure 13). In closed loop the branch decision at the curve becomes learnable, and ReLU^2 alone makes it correctly from both sides of the switching set — at the reference rollout cost from B , and along the correct branch at roughly twice the reference cost from A ; every other law fails from both starts. The bottleneck has moved from data coverage across the curve to the atoms’ ability to represent the kink — which the sparse rectified atoms address directly.

Feedback law	start $A = (0.71, 0.68)$		start $B = (0.23, 0.53)$	
	cost	reaches upright	cost	reaches upright
reference (PMP)	26.2	yes	10.2	yes
Gaussian + ϕ_γ	298.5	no	278.9	no
softplus + ϕ_γ	73013.3	no	66996.7	no
leaky ReLU + ϕ_γ	869.1	no	776.9	no
$\text{ReLU}^2 + \psi_2$	57.9	yes	10.3	yes
$\text{ReLU}^5 + \psi_5$	235.9	no	232.4	no

TABLE 7. Closed-loop rollout cost over the horizon $T = 10$ and terminal outcome for each feedback law from the two starts on either side of the switching curve; “reaches upright” means the final state satisfies $\text{dist}(\theta, 2\pi\mathbb{Z}) < 0.4$ and $|\dot{\theta}| < 0.4$ at $T = 10$ — a terminal-neighbourhood criterion over one finite rollout, not an asymptotic-stability statement; the reference row is the nearest-neighbour PMP envelope controller.



(A) log-penalty models.



(B) ReLU^k models.

FIGURE 19. Feedback control $u(t)$ from start B , per law, beside the reference (black); the axis is clipped to the informative band — softplus’s actuator-saturation excursion of ± 30 leaves the frame. The reference brakes to $\theta = 0$ with u rising from ≈ -7 to 0; ReLU^2 tracks it almost exactly.

7. CONCLUSION AND OUTLOOK

This thesis studied the approximation of value functions of nonlinear optimal control problems from PMP-generated data by sparse shallow networks, combining a gradient-augmented (H^1) loss with nonconvex sparsity penalties over a measure-space formulation.

On the theoretical side, the analysis of [23] was extended in three directions. For localized ridge activations, the existence of global minimizers, the first-order optimality conditions, and the finite support of local minimizers were established on the *unbounded* parameter domain $\mathbb{R}^{d+1}$ (Theorems 3.18, 3.19, 3.20, 3.23), with the no-escape condition (H) making the behavior of the dual variable at infinity explicit. The localization was shown to be sharp: for saturating or growing activations fit mass can escape to infinity — for tanh this destroys the weak-* lower semicontinuity of the fidelity and with it the direct method (Example 3.1); for softplus nonattainment is genuine and the regularization even degenerates entirely, since every ReLU network is reached at vanishing

penalty. For positively k -homogeneous activations, the problem was reformulated on the sphere with the fractional-exponent penalty $|c|^{2/(k+1)}$, and the optimality and existence theory was extended to this singular penalty class despite the failure of the classical assumptions at zero.

On the numerical side, the two benchmarks isolate two regimes. On the smooth Van der Pol value function, gradient augmentation is decisive for gradient accuracy, and the nonconvex penalties act as a pure sparsity lever: the ReLU^k atoms with the fractional-exponent penalty reach the accuracy of the best log-penalty activation with a fraction of the neurons, and every resulting feedback law stabilizes at essentially the optimal cost. On the pendulum swing-up, whose value function carries a gradient discontinuity along the switching set, the picture inverts: the switching set is a representation limit at the trained budgets — every fitted model leaves its dominant error in the switching tube, and oversampling the switching region does not pay it down under the tested protocol — and the choice of atom decides control viability rather than merely sparsity: only ReLU^2 , whose derivative can kink across a hyperplane, develops a sharp transition at the switching set and yields a feedback law that makes the correct branch decision from both sides in the tested rollouts.

Several directions remain open. The experiments are two-dimensional; the causality-free data generation and the sparse networks are built to scale, but high-dimensional deployment is untested. In the oversampling study, per-sample objective weighting is the natural untried lever against the interior cost of two-sided switching data. On the theory side, the counterexamples suggest a well-posed completion for the saturating class: an impact-normalized penalty $\phi(|c| \|\sigma(\cdot, \omega)\|_{H^1})$ closes the zero-cost escape directions, and the compactified dictionary — whose boundary atoms are exactly the ReLU ridges and constants reached by the dilation limits, cf. the embedding of Section 2.2 — would carry the sparse-measure theory for softplus-type activations. Finally, an explicit affine term in the model would absorb the constant and affine escape directions directly and plausibly restore existence on the unbounded domain for the saturating class.

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